Usability Engineering

Design Phase: Design Principles & Patterns, Sketching for Ideation| Wintersemester 2026/27

FH-Prof. Dr. Mandy Keck

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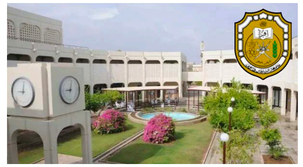
mandy.keck@fh-hagenberg.at - Room FH 1.217

About me

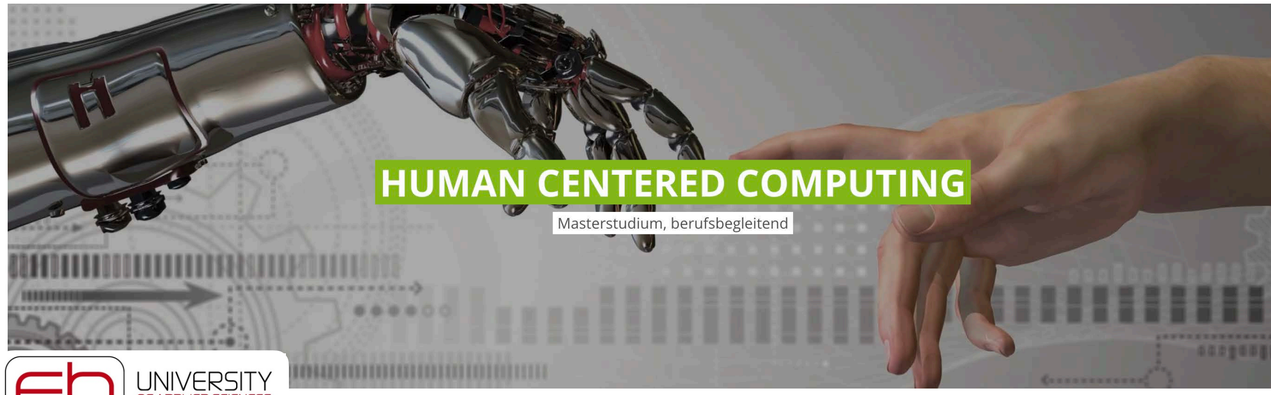
FH-Prof. Dr.-Ing. Mandy Keck
Professorin für User Experience
und Interaktionsdesign



TECHNISCHE UNIVERSITÄT DRESDEN 2009 - 2020

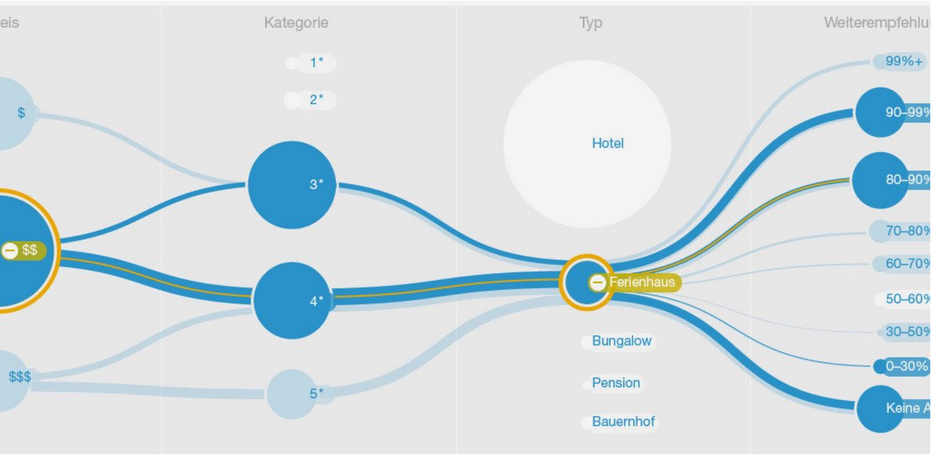


2015 - 2016
Sultan Qaboos
University

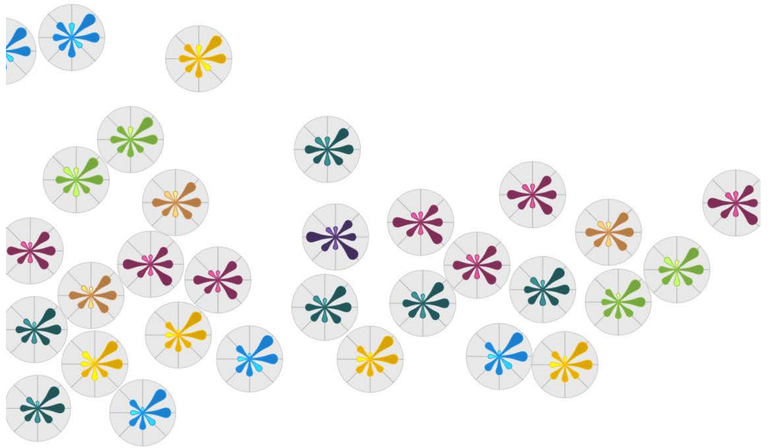


FH OÖ UNIVERSITY OF APPLIED SCIENCES UPPER AUSTRIA seit 2021

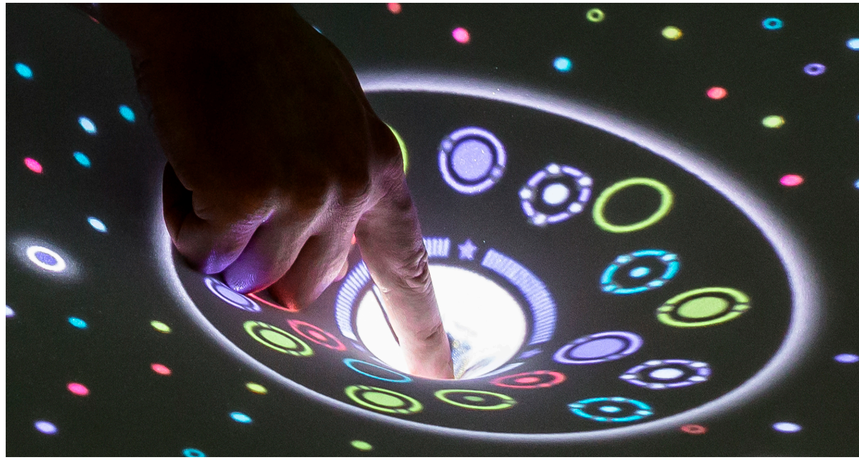
Schwerpunkte in Lehre und Forschung:



Informationsvisualisierung



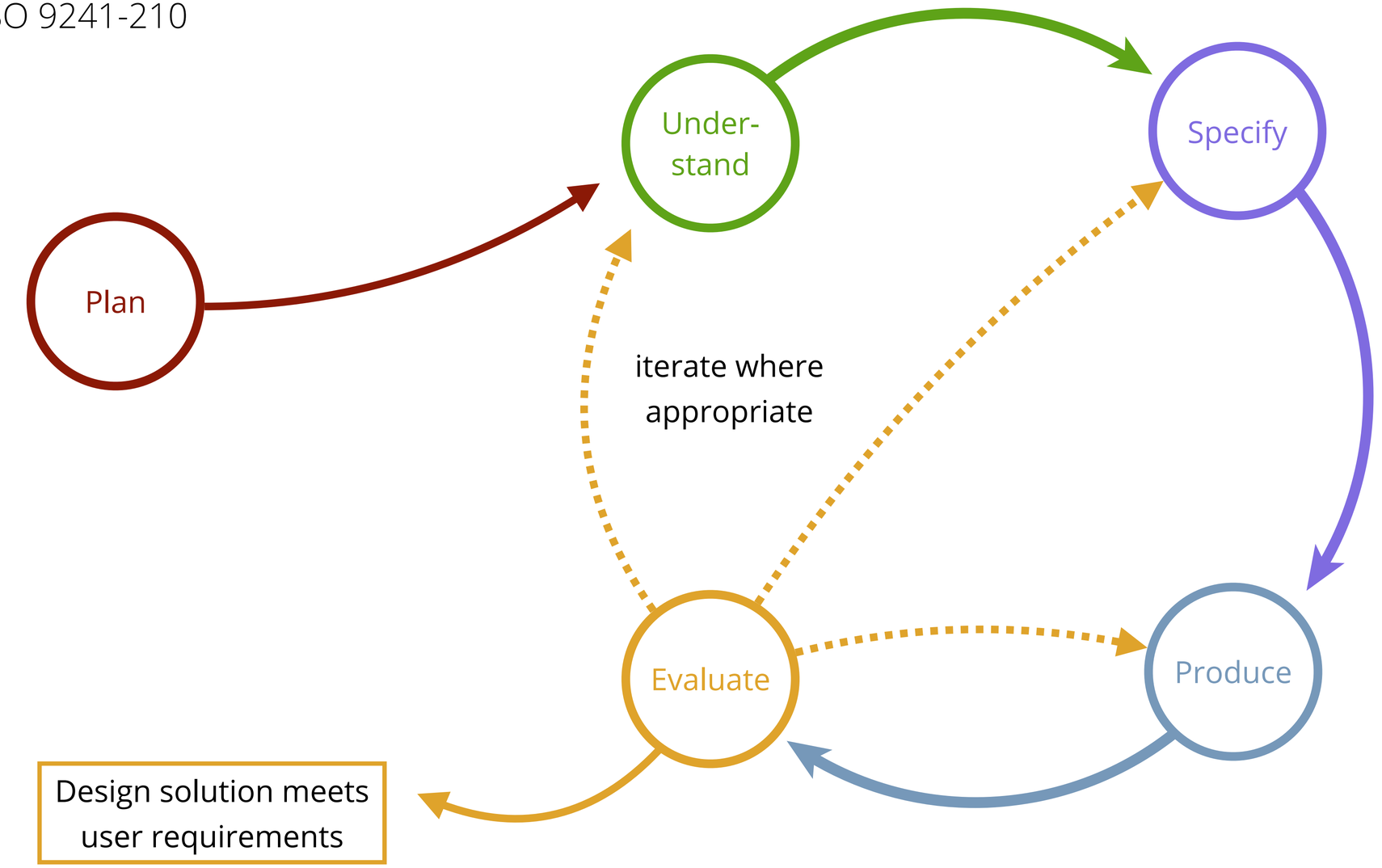
Visuelle Exploration



Interaktionsdesign

Human-Centered Design Process

ISO 9241-210



Simplified Design Process

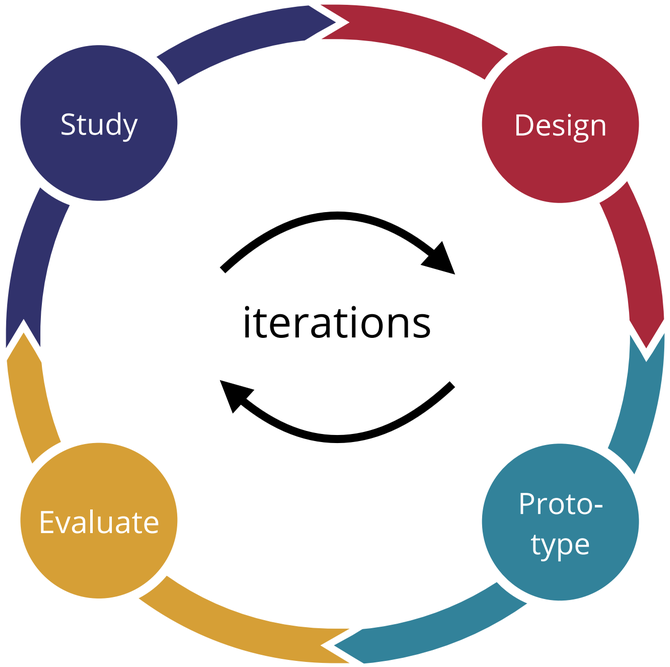
Derived from the UX Lifecycle Wheel [Hartson & Pyla 2019]

Requirement Analysis

Scenarios, Personas, ...







Sketching

Scribbles, Storyboards, ...





(GUI) Design

GUI Sketches, Wireframes, ...



Evaluation

Heuristic Evaluation,
Usability Tests, ...



Develop

(not covered here)

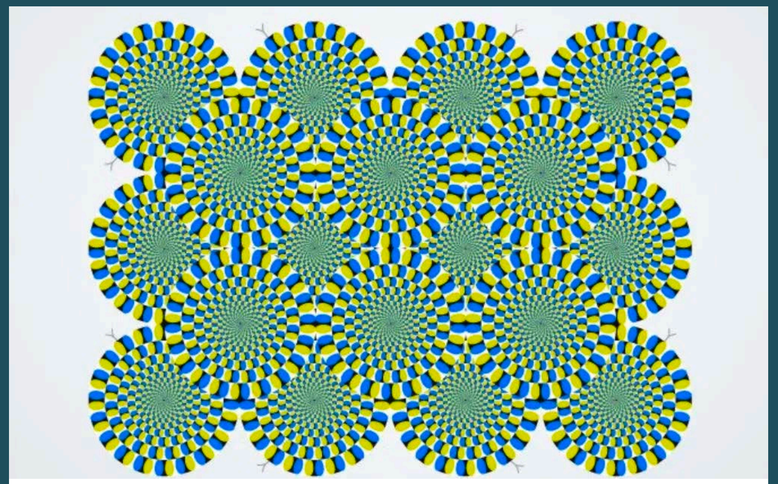
Building & Prototyping

Mockups, Prototypes, ...



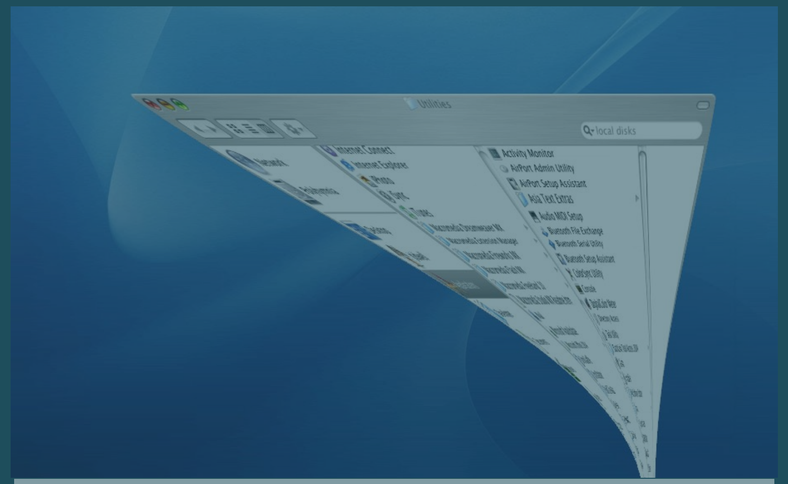
What can help us in creating good user interfaces?

Psychology & cognitive science



Getting a better understanding of how users perceive & think about UIs

Design principles & patterns



Following fundamental principles & guidelines of good interaction design

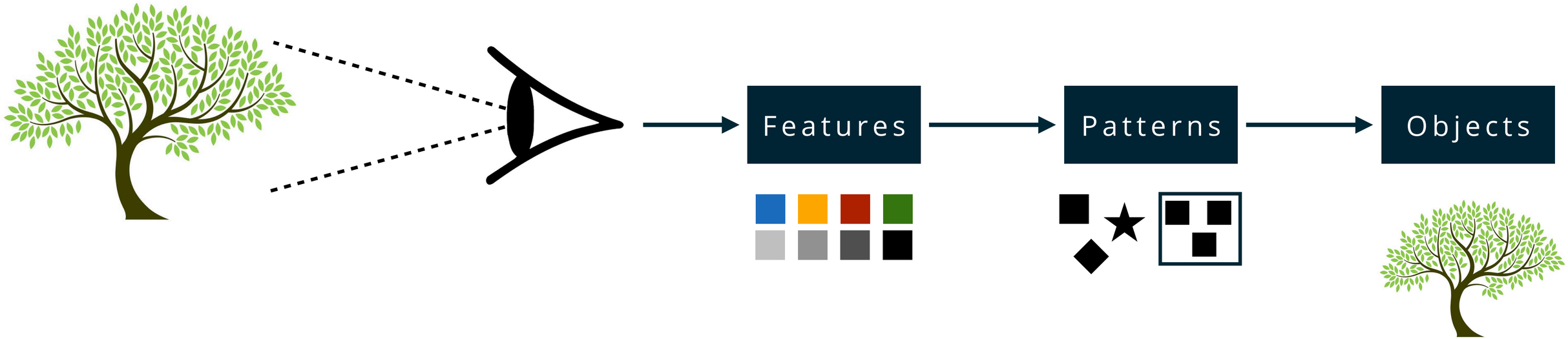
Human-centered Design



Following a human-centered design process

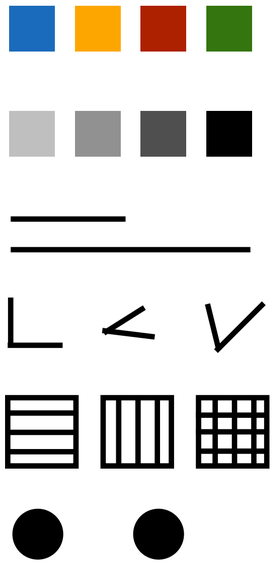
Perception

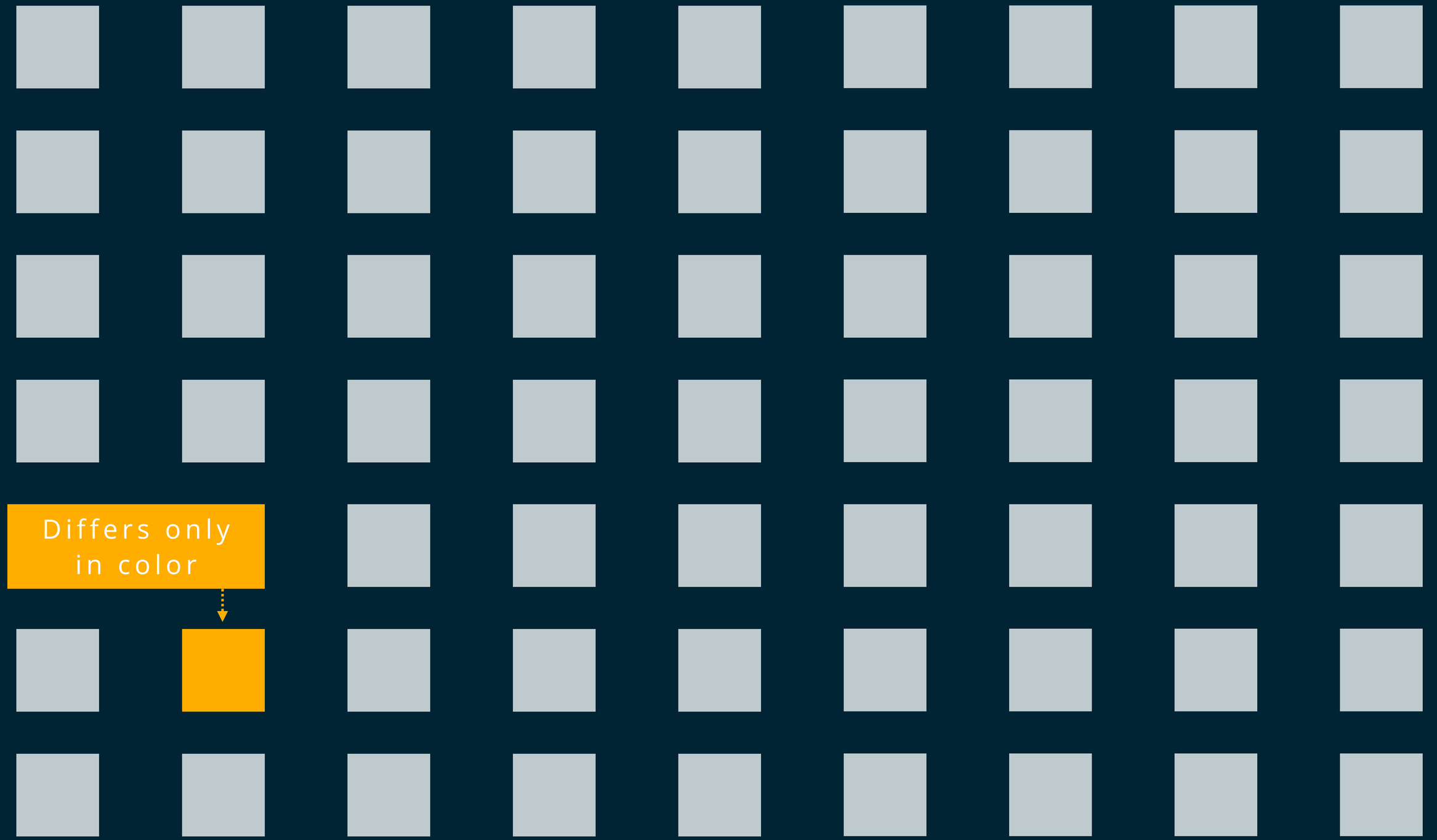
- the interaction with a user interface is based on the perception of visual, acoustic and tactile stimuli
- we will focus on **visual perception**: most information is conveyed visually

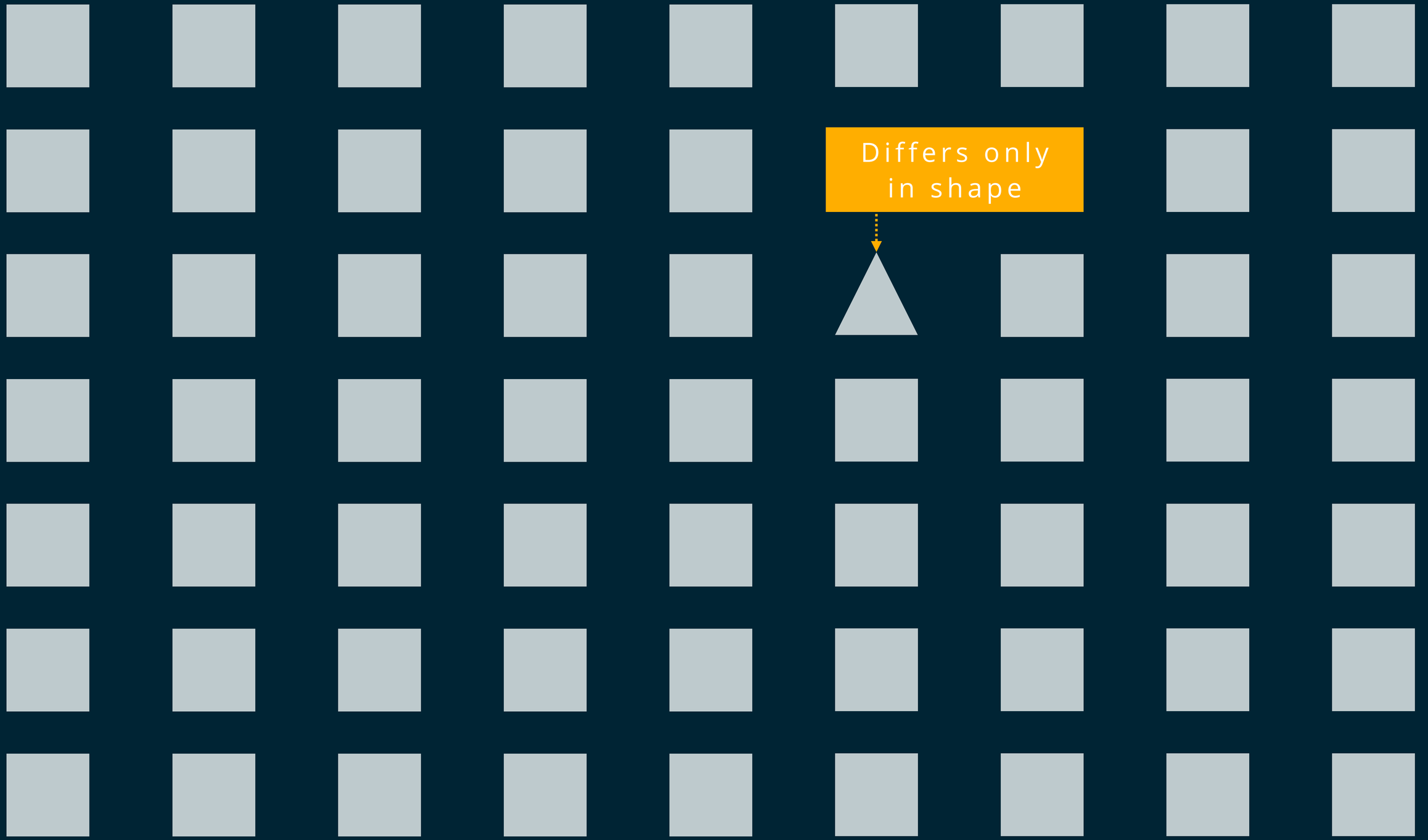


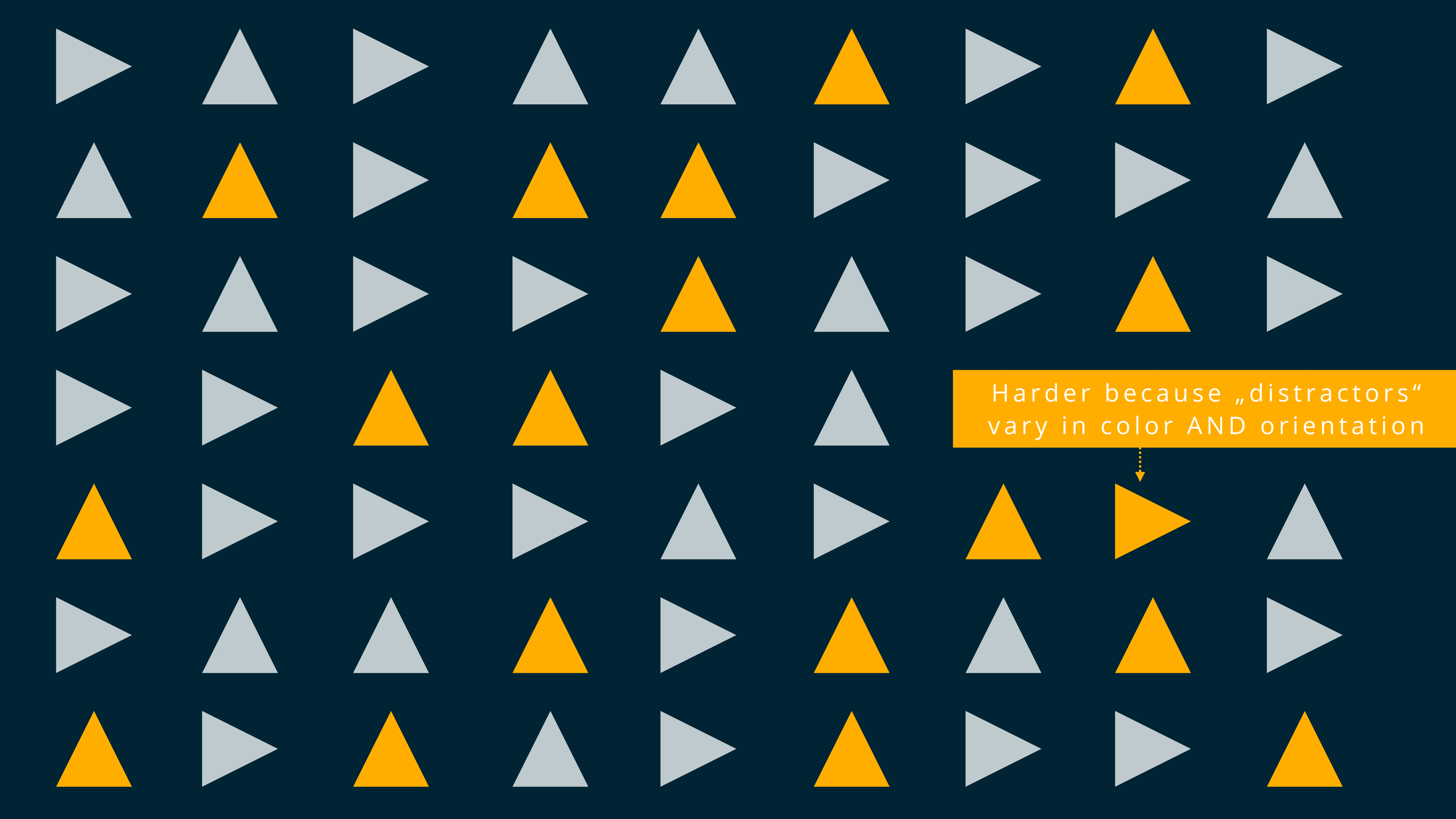
Feature Detection

- Color
- Value
- Length
- Angle
- Texture
- Motion



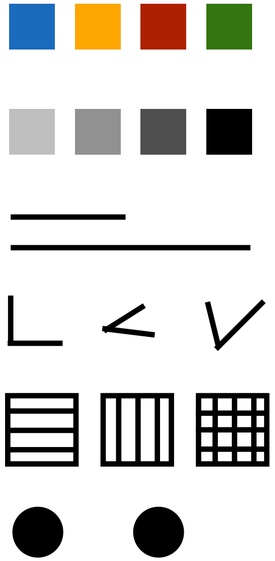

Differs only
in color


Differs only
in shape


Harder because „distractors“
vary in color AND orientation

Feature Detection

- Color
- Value
- Length
- Angle
- Texture
- Motion



- very fast
- allows „pop-out“
- support subsequent stages like recognition of patterns & objects
- pop-out should be used to attract attention

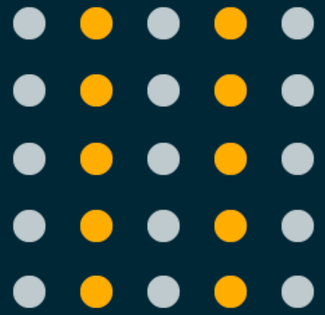
Feature Detection



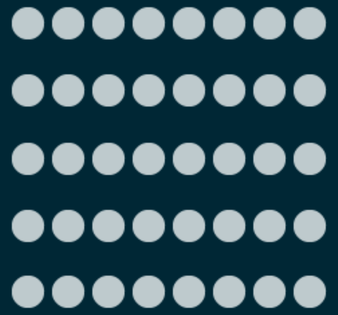
source: https://www.lingscars.com/

- use „pop-out“ wisely!

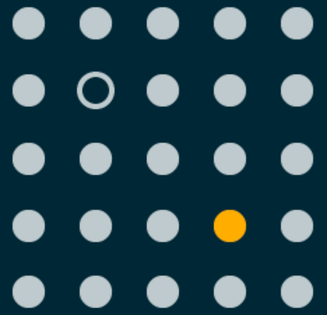
Gestalt Laws



Similarity



Proximity



Prägnanz



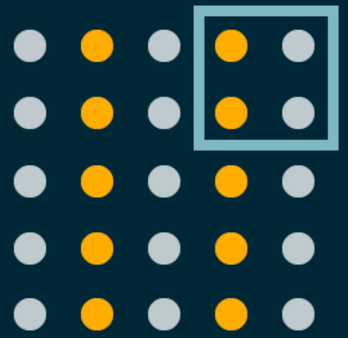
Closure



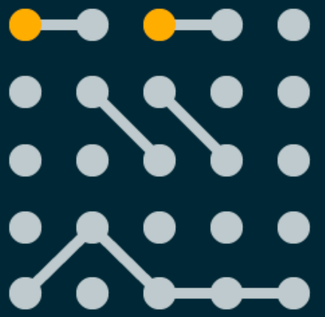
Continuation



Common fate



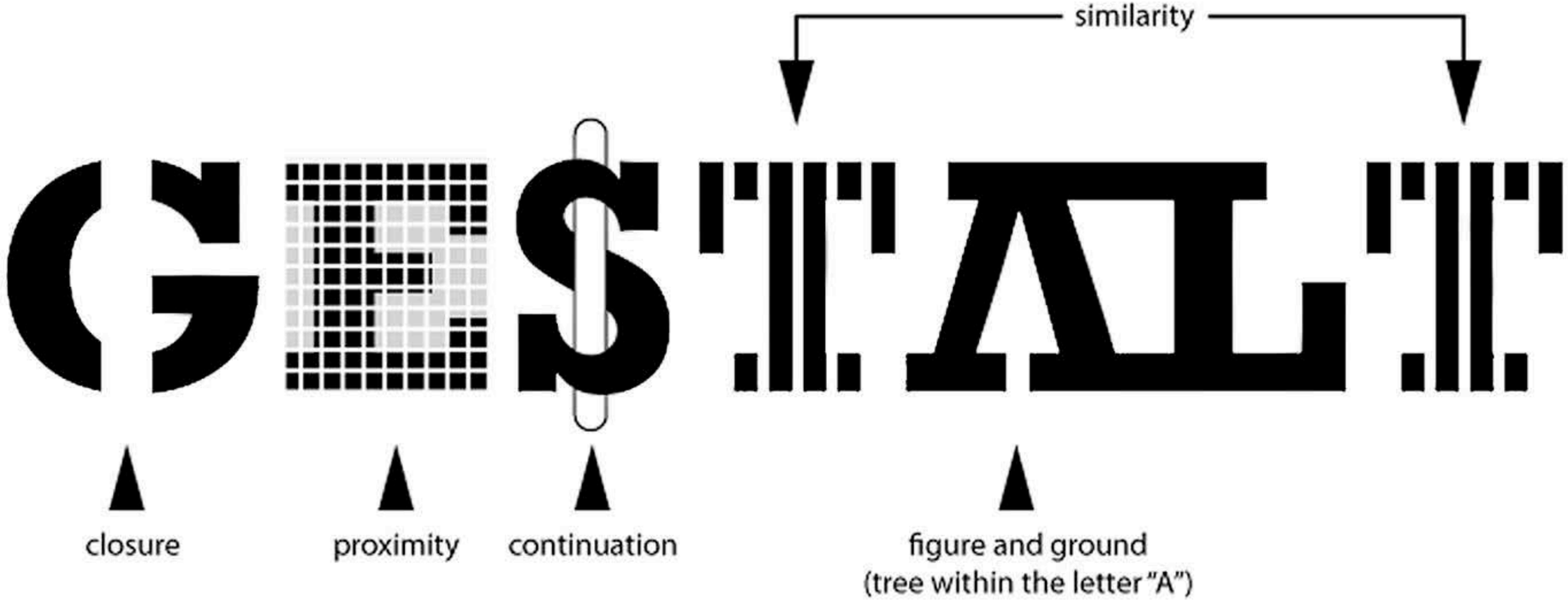
Common region



Connectedness

Gestalt Laws / Gestalt Principles of Visual Perception

- **Gestalt psychology:** we strive to find the simplest solution to incomplete visual information
- human perception selects the interpretation that results in the smallest cognitive effort



source: https://medium.com/ringcentral-ux/gestalt-principles-learn-how-to-influence-perception-83112932d0bc

Law of Figure & Ground

- people instinctively perceive objects as either being in the foreground or the background



Law of Figure & Ground

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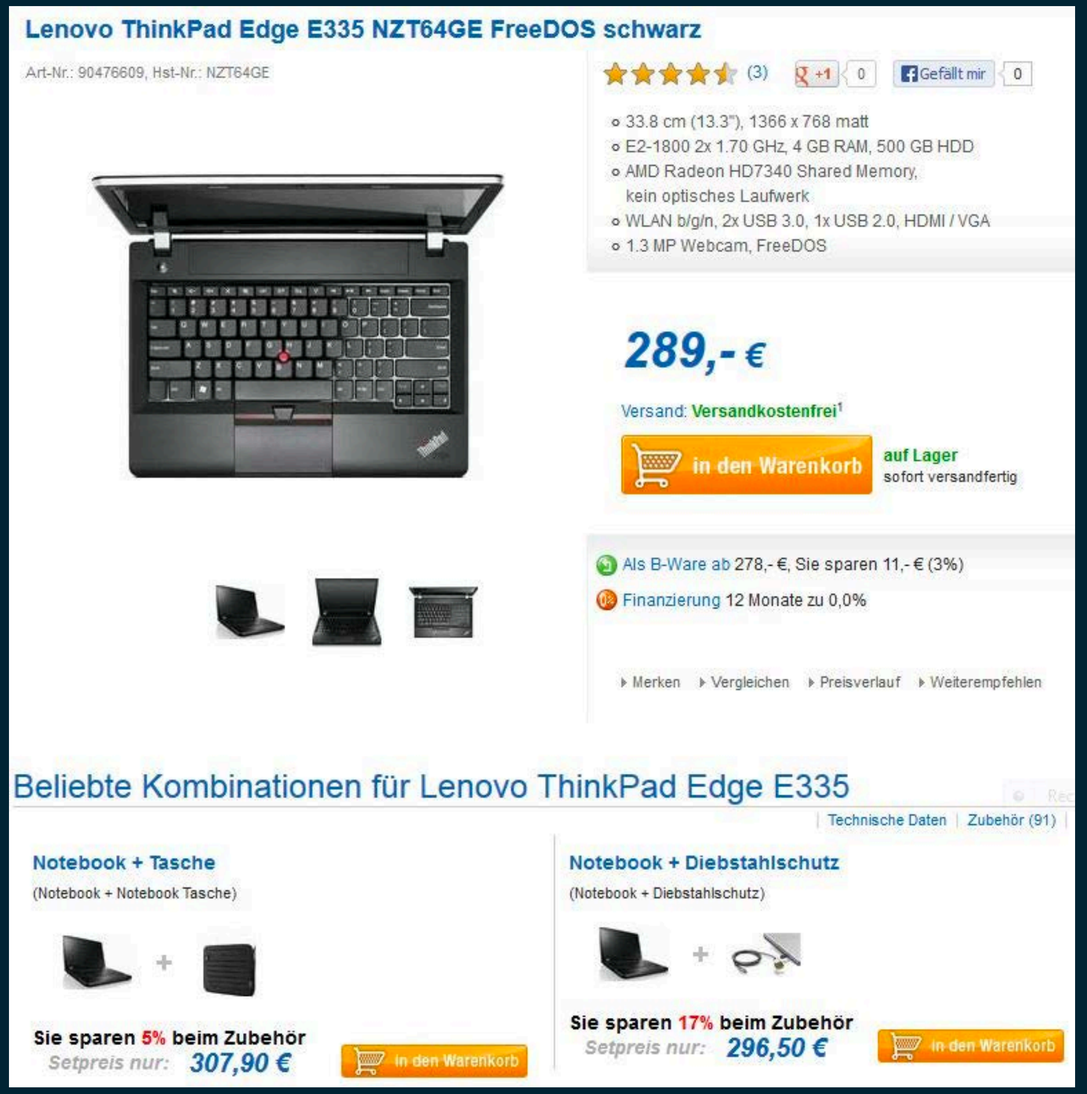
Law of Similarity

- similar elements are visually grouped (e.g. by color, size, shape)



Law of Similarity

- similar elements are visually grouped (e.g. by color, size, shape)
- similarity can be used to tie together elements that might not be right next to each other in a design



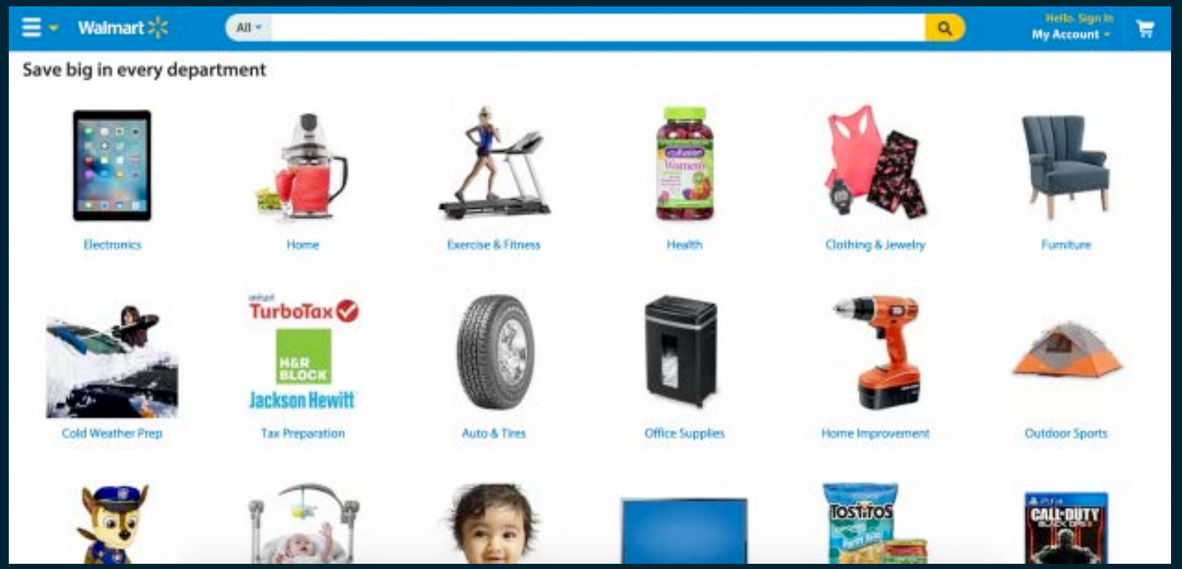
Law of Proximity

- things that are close together appear to be more related than things that are spaced farther apart



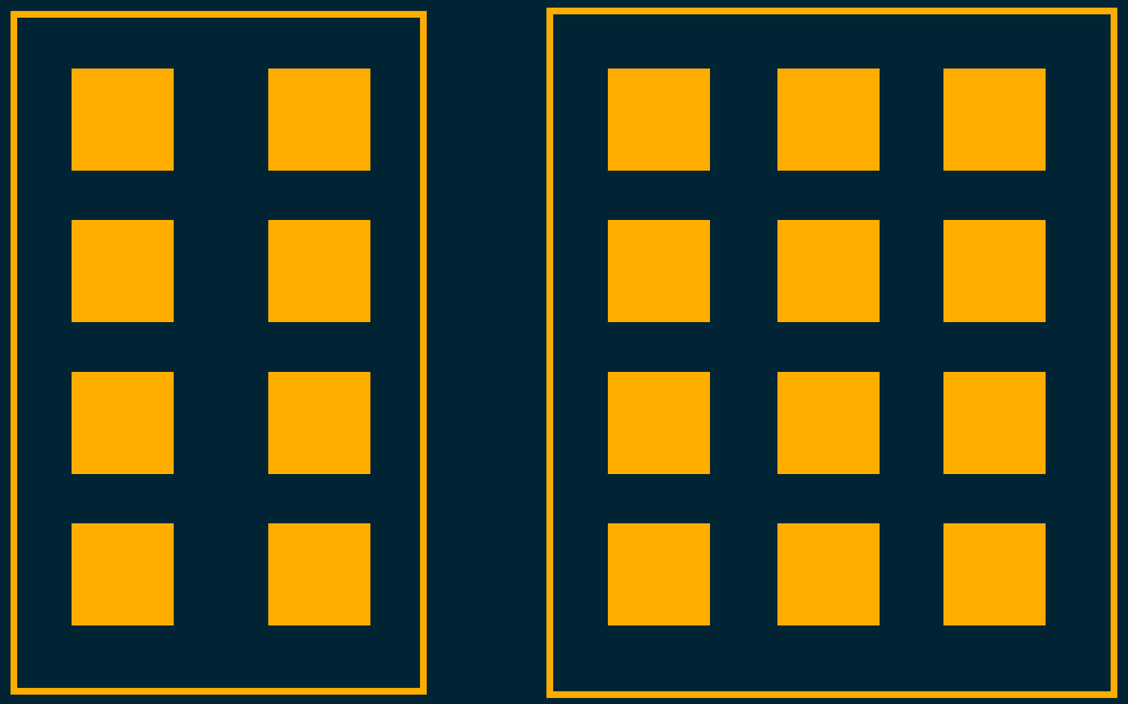
Law of Proximity

- things that are close together appear to be more related than things that are spaced farther apart
- The nearness of each image and its corresponding text communicates that they’re related to one another



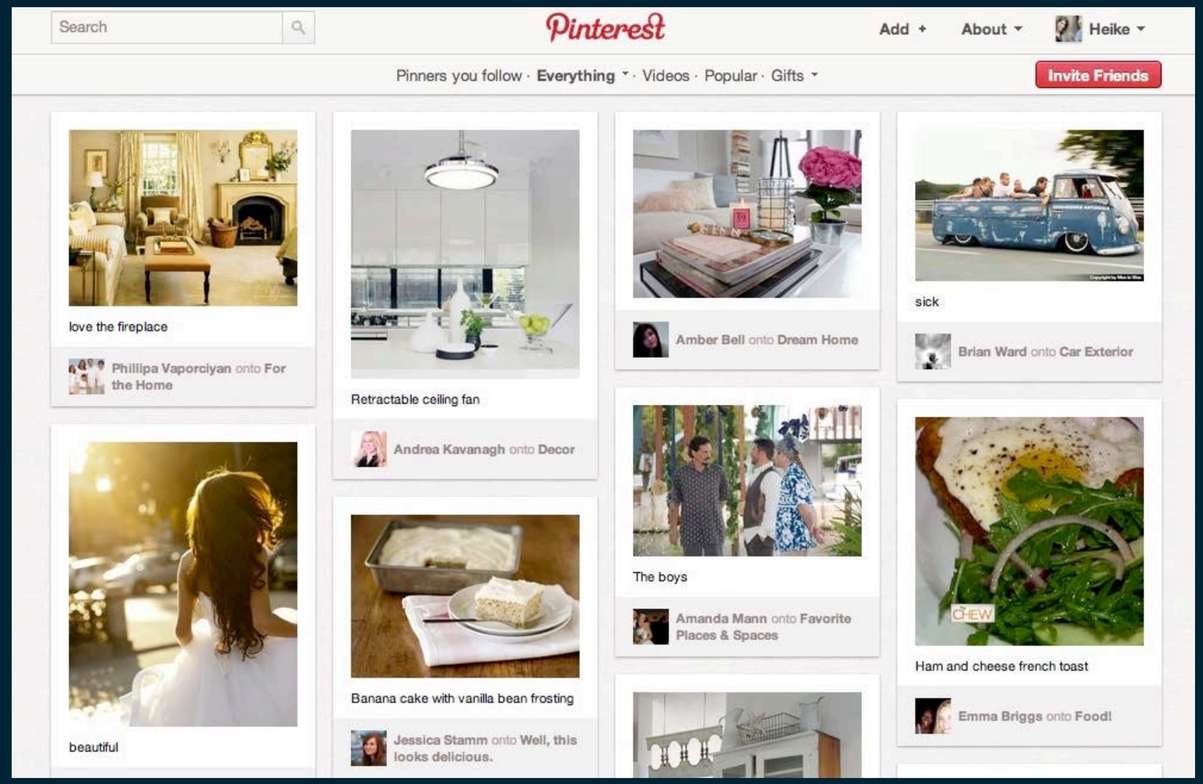
Law of Common Region

- when objects are located within the same closed region, we perceive them as being grouped together



Law of Common Region

- when objects are located within the same closed region, we perceive them as being grouped together
- e.g.in Pinterest, the common region principle is used to separate each pin



Law of Closure

- when we look at a complex arrangement of visual elements, we tend to look for a single, recognizable pattern
- is often used in logo design







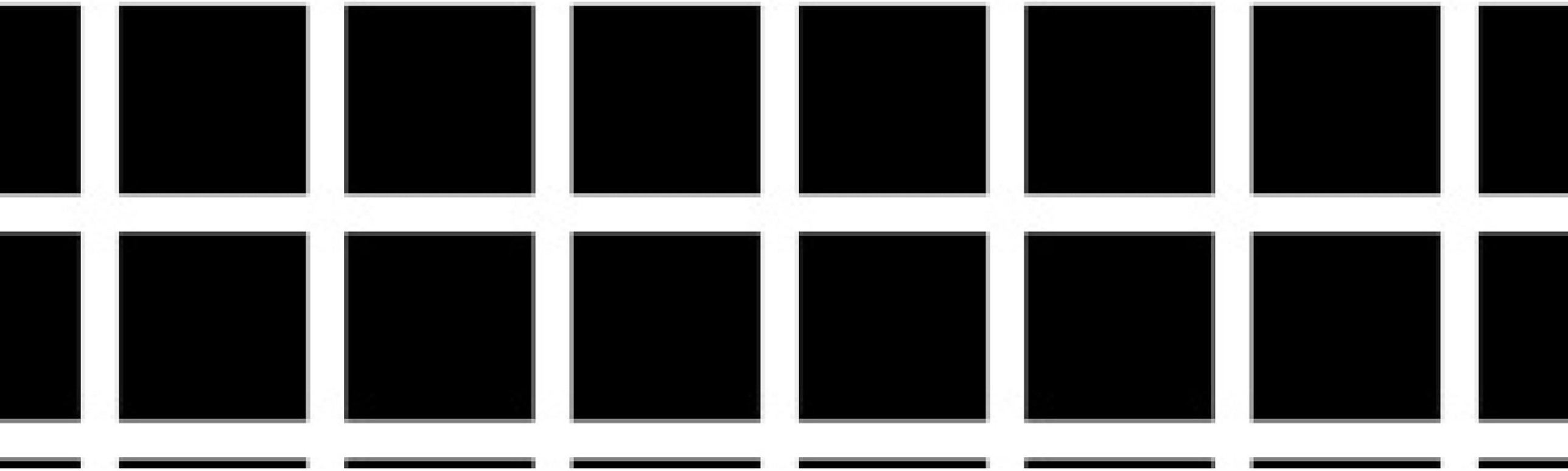


Gestalt Laws

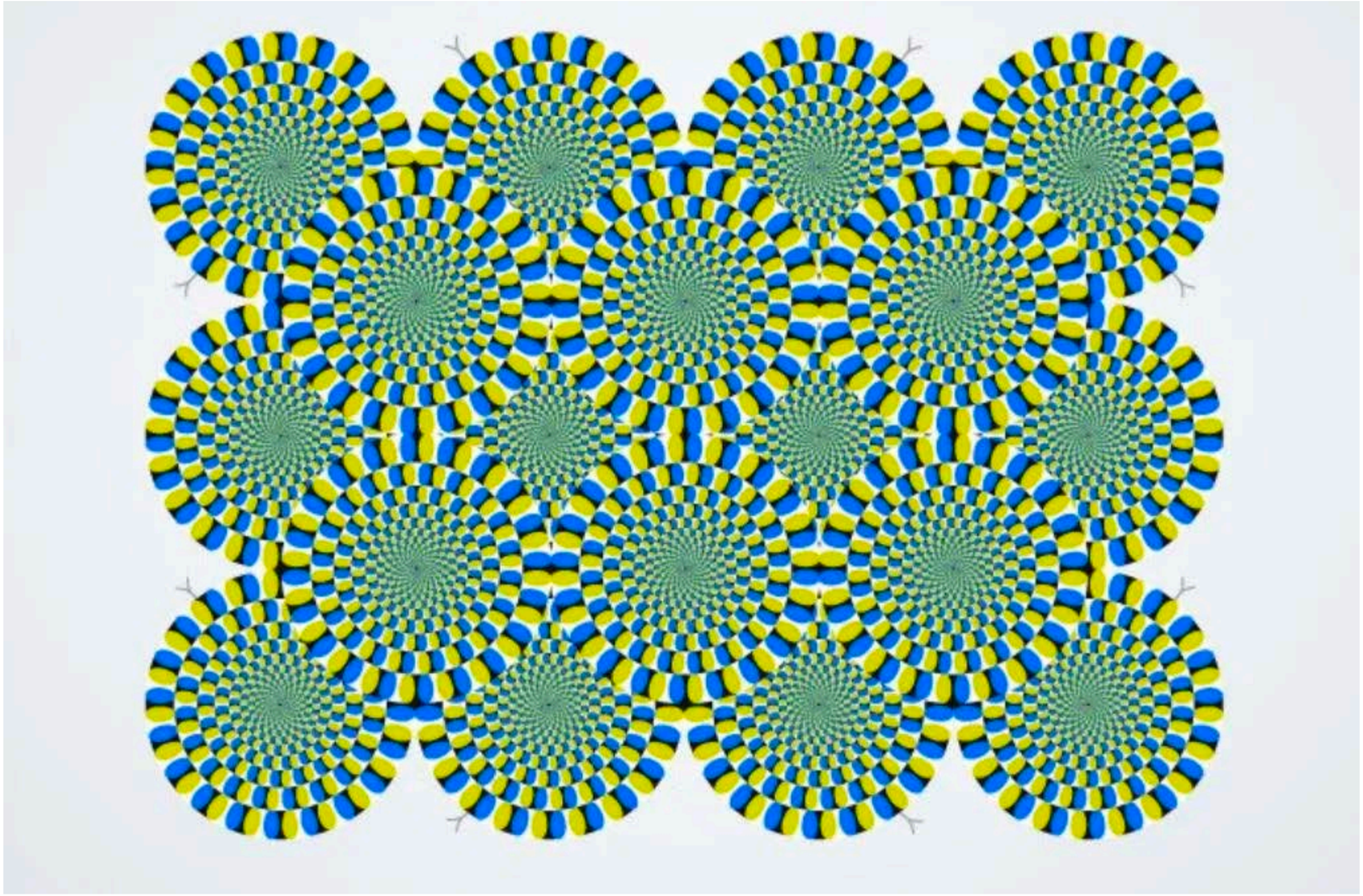


source: https://www.reddit.com/r/keming/comments/dndz8r/no_safety_smoking_first/

Limits of Visual Perception



Limits of Visual Perception



From senses to memory

What happens to information after we perceived it?

- **Working memory** has a duration of about 30 seconds [Hartson & Pyla 2019]
  - other intervening activities will cause the contents of working memory to fade even faster
- **Long-term memory:** information stored in working memory can be transferred to long-term memory by „learning“



Working Memory

Limited capacity:

- the **„magic number“ 7 ± 2** (Miller’s Law, 1956):
  7 ± 2 chunks of information can be held in working memory
- Miller’s magical number 7 ± 2 was later revised to a more conservative 4 ± 2
- information that is not retained is lost

Chunking:

- a chunk is a basic human memory unit containing one piece of data that is recognizable as a single gestalt, e.g.
  a chunk = one word for spoken expressions [Hartson & Pyla 2019]

0110101111000111

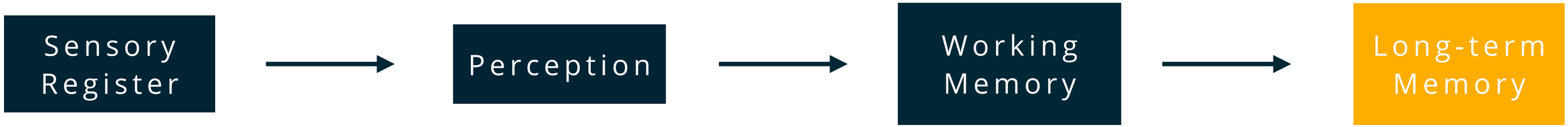
0110 1011 1100 0111

004350233233

0043 50 233 233

From senses to memory

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- requires attention
- a small amount of what is available is actually perceived

whatever is perceived is available for a few seconds

a small amount of information is „learned“

Long-term Memory

- Unlimited capacity
- **Learning:** information must be „copied“ from working memory to long-term memory

Transfer to long-term memory:

- **Association:** a piece of information is connected with an information that is already known



horse

+



donkey

=



mule

Long-term Memory

- Unlimited capacity
- **Learning:** information must be „copied“ from working memory to long-term memory

Transfer to long-term memory:

- **Association:** a piece of information is connected with an information that is already known
- **Repetition:** expending effort to commit something to long-time memory

Which is more likely to be remembered?

Associative

Phone number: 0043 50 233 233

IBAN: AT03 0100 0000 0552 4464

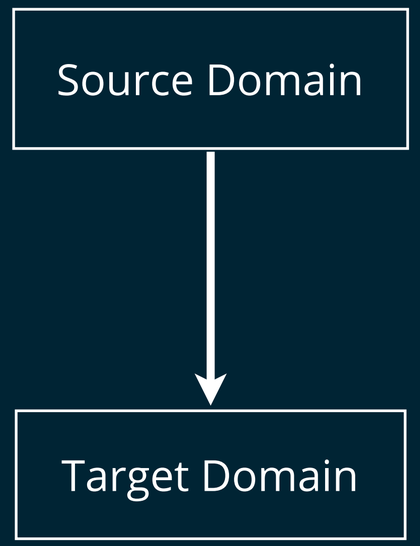
```
C:\Users\kevin\Documents\dev\android-praktikum>git push
warning: push.default is unset; its implicit value has changed in
Git 2.0 from 'matching' to 'simple'. To squelch this message
and maintain the traditional behavior, use:
```

Principles

- support association
  - use metaphors

„The essence of metaphor is understanding and experiencing one kind of thing in terms of another“

[Lakoff & Johnson 2003]



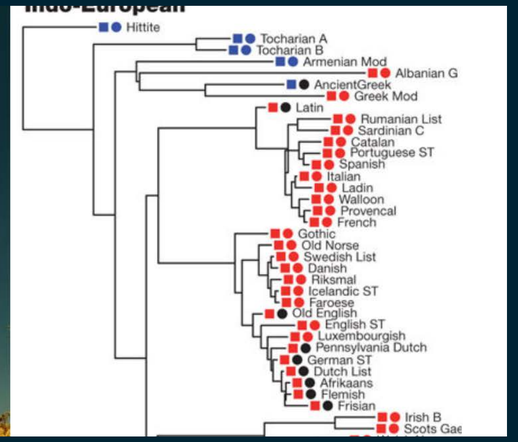
familiar situation

unknown problem

Principles

- support association
  - use metaphors

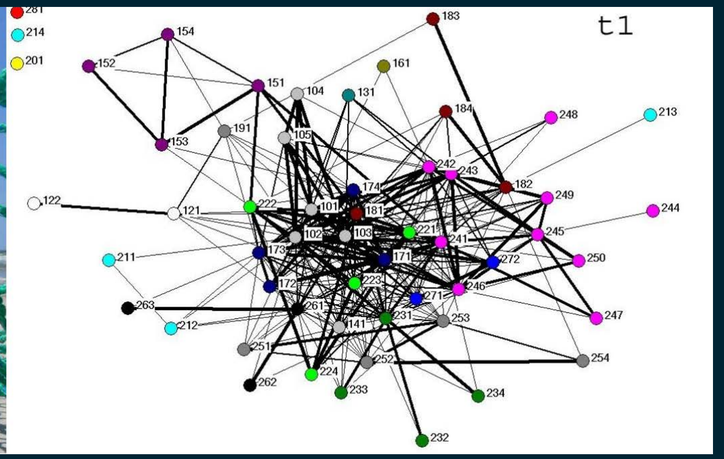




Source Domain

Target Domain





Principles

- support association
  - use metaphors

Shopping card at amazon:



Advantages:

- actions and goals are anchored in concepts that are already understood
- powerful tool that can facilitate learning and understanding of complex content

Principles

- support association
  - use metaphors

metaphor should be abstract and not too close to the source domain

Operating system "Bob" from Microsoft



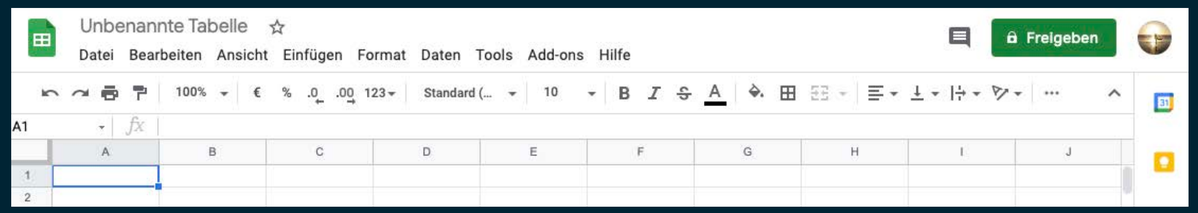
Disadvantages:

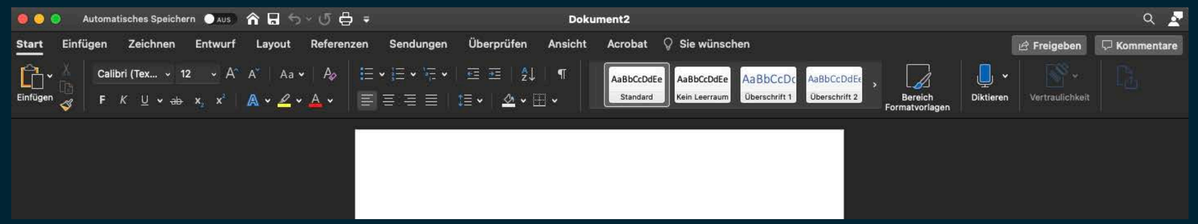
- Scalability: metaphor that works for a simple process may fail as systems become more complex
- Risk that metaphors lead to misunderstandings or assumptions about the target domain are not valid

Principles

- support association
  - use metaphors
  - leverage standards and consistency



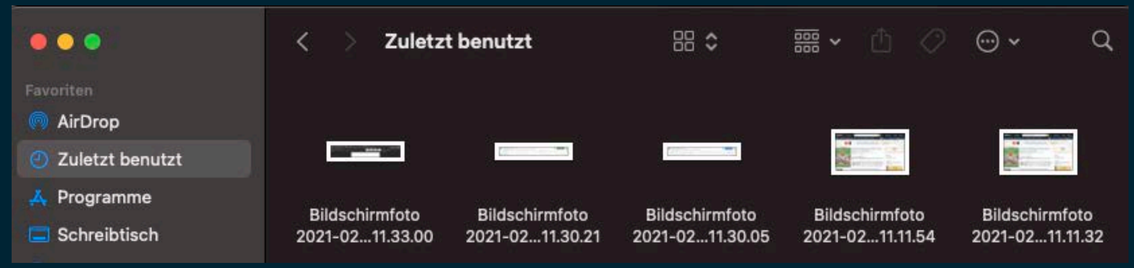


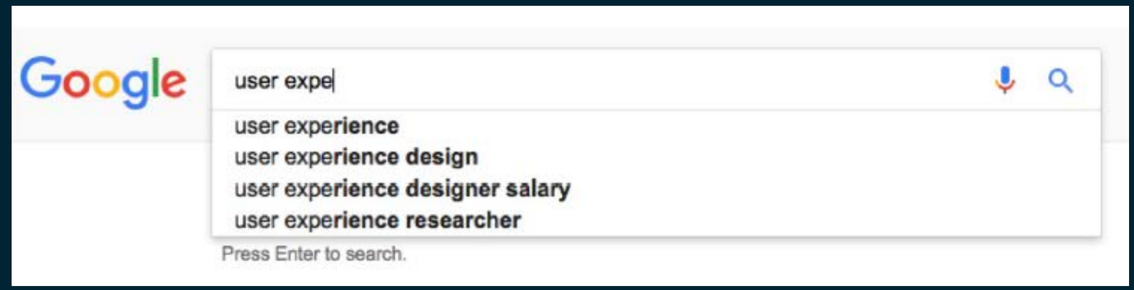


Principles

- support association
  - use metaphors
  - leverage standards and consistency
- prioritize recognition over recall

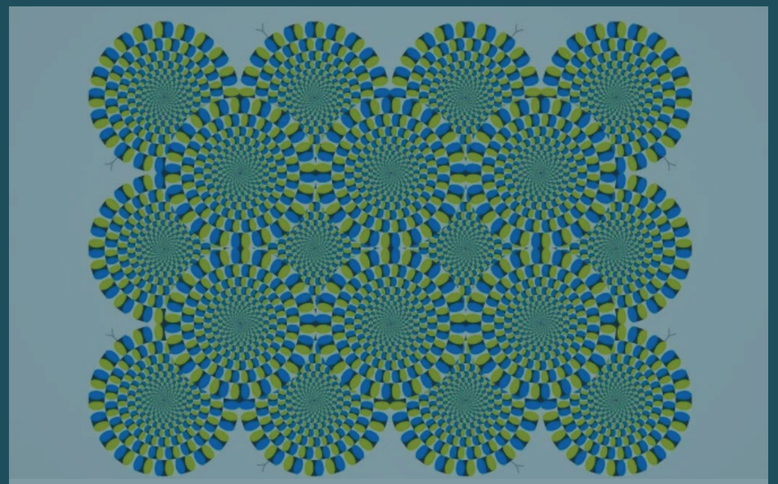
„I cannot remember it exactly, but I will recognize it when I see it“





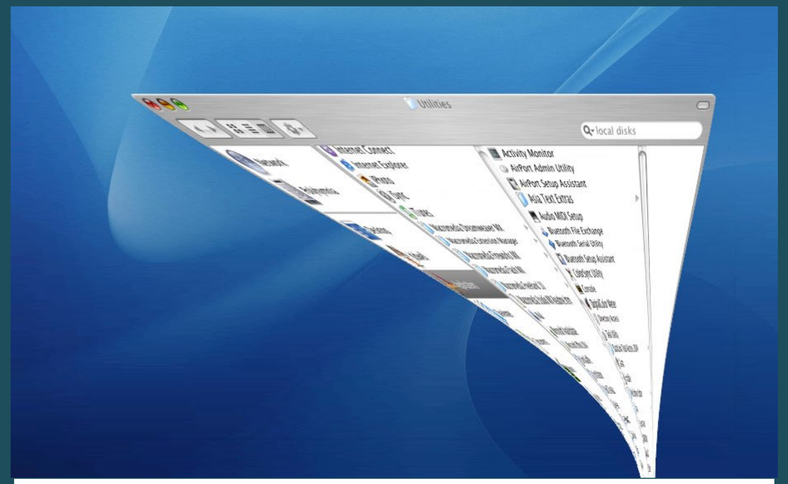
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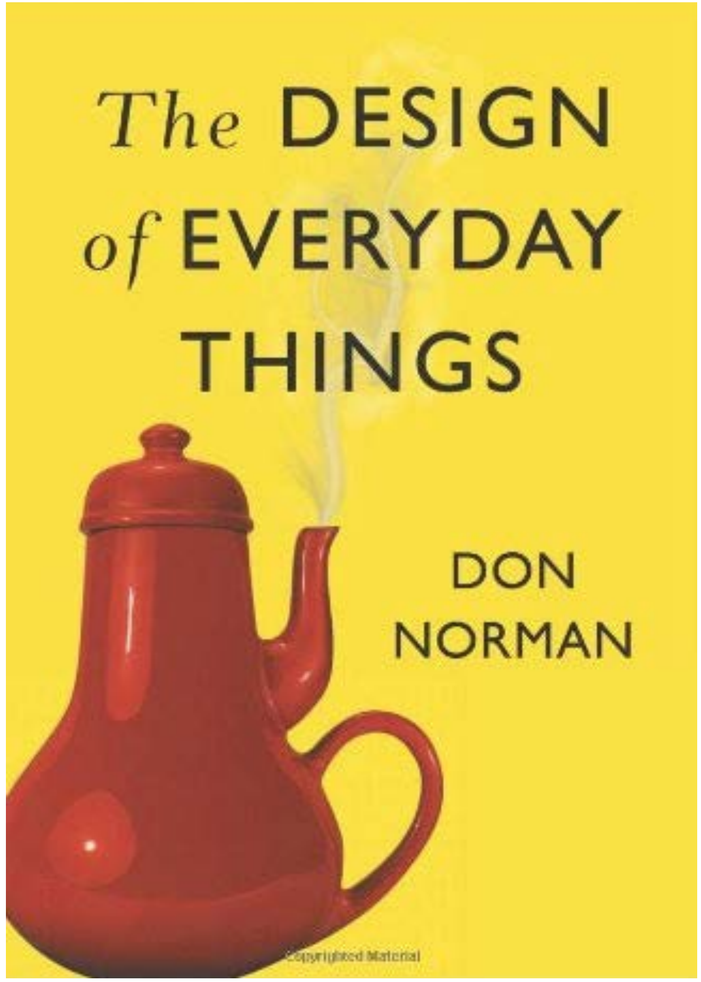
Human-centered Design



Following a human-centered design process

Design Principles

- generalizable abstractions that deal with the aspects of behavior, form and content [Cooper 2014]
- used by designers to analyze, critique, and discuss designs



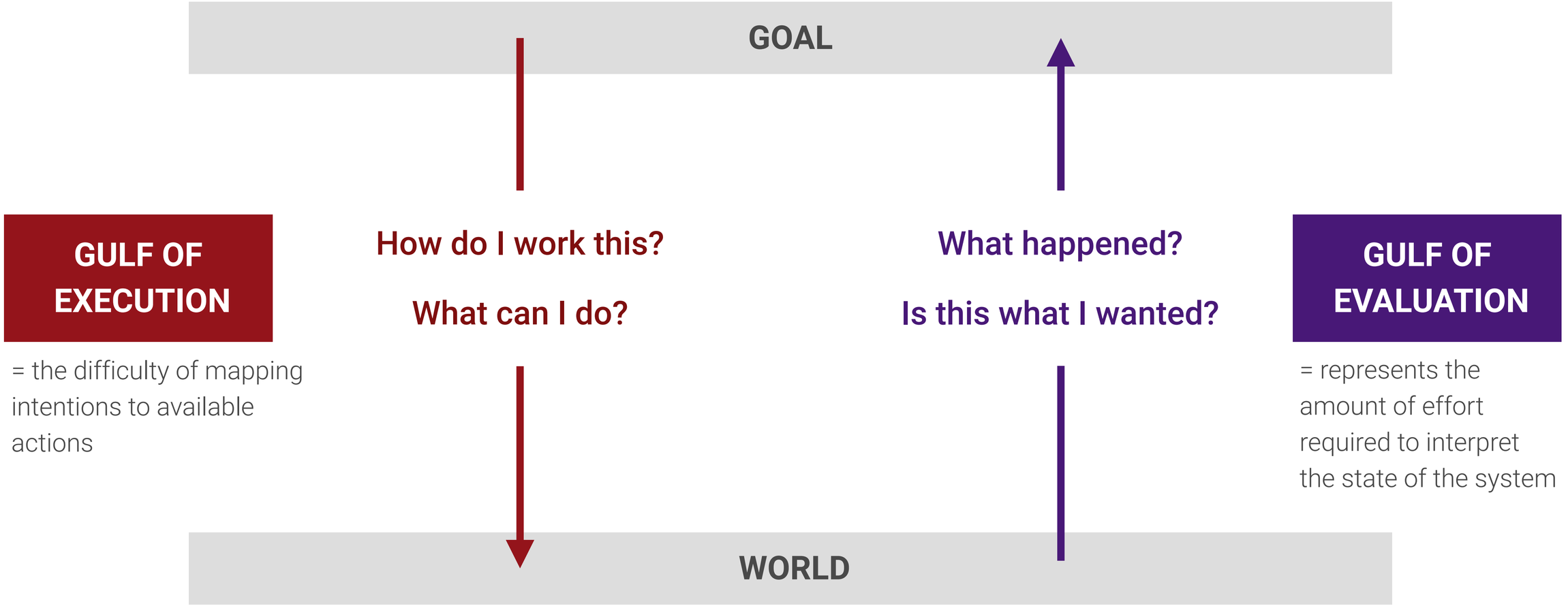


Vox

Video: https://youtu.be/yY96hTb8WgI?si=25L2A6EqWYF5MkYC

Gulf of Execution and Evaluation

[Norman 2014]



GULF OF EXECUTION

= the difficulty of mapping intentions to available actions

GULF OF EVALUATION

= represents the amount of effort required to interpret the state of the system

The role of the designer is to help people bridge the two gulfs!

Gulfs of Execution and Evaluation



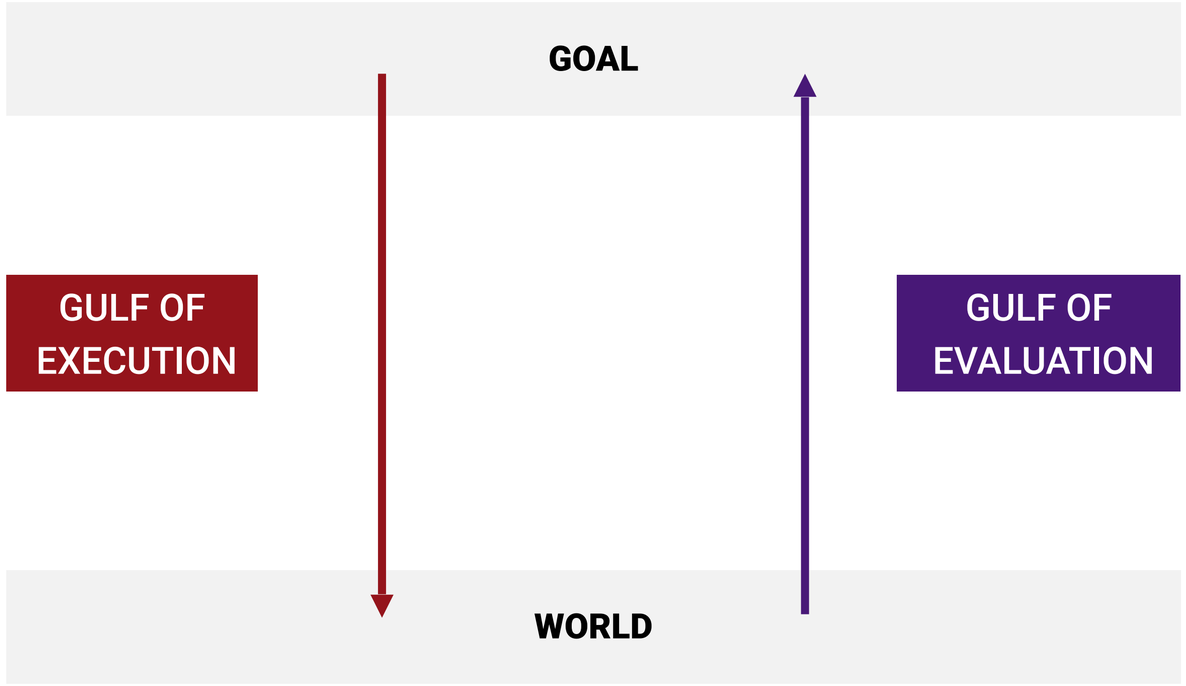
https://www.youtube.com/watch?v=bg4UwyPPZ6U

Fundamental Principles of Design

[Norman 2014]

Seven key design principles that help minimizing the gulf of execution and evaluation:

1. Discoverability
2. Feedback
3. Affordances
4. Signifiers
5. Mappings
6. Constraints
7. Conceptual Models



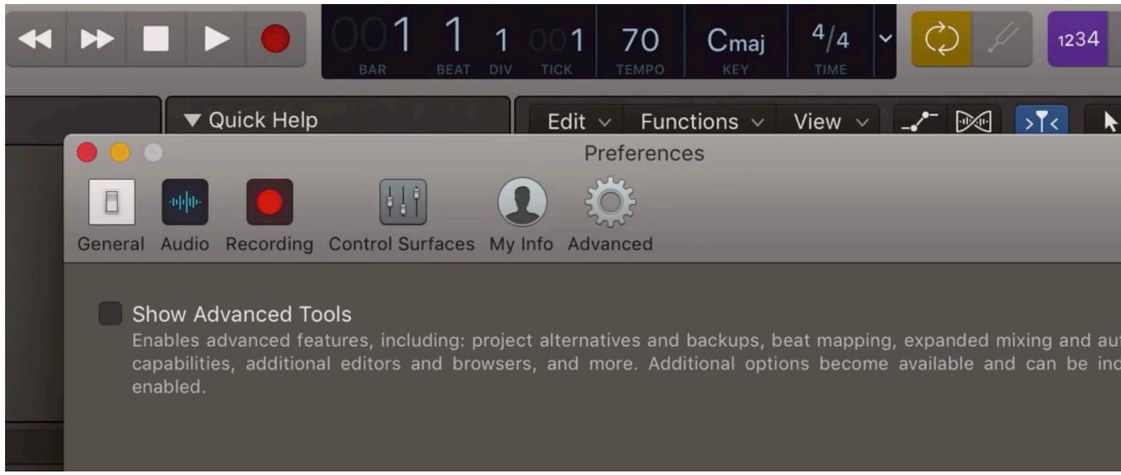
Discoverability

Discoverability increases understanding of the available options and where to perform them.

- the most important/frequently used functions should be easy to find/access
- as systems have become more complex it has become necessary to hide more functionality

  Do not overwhelm users!



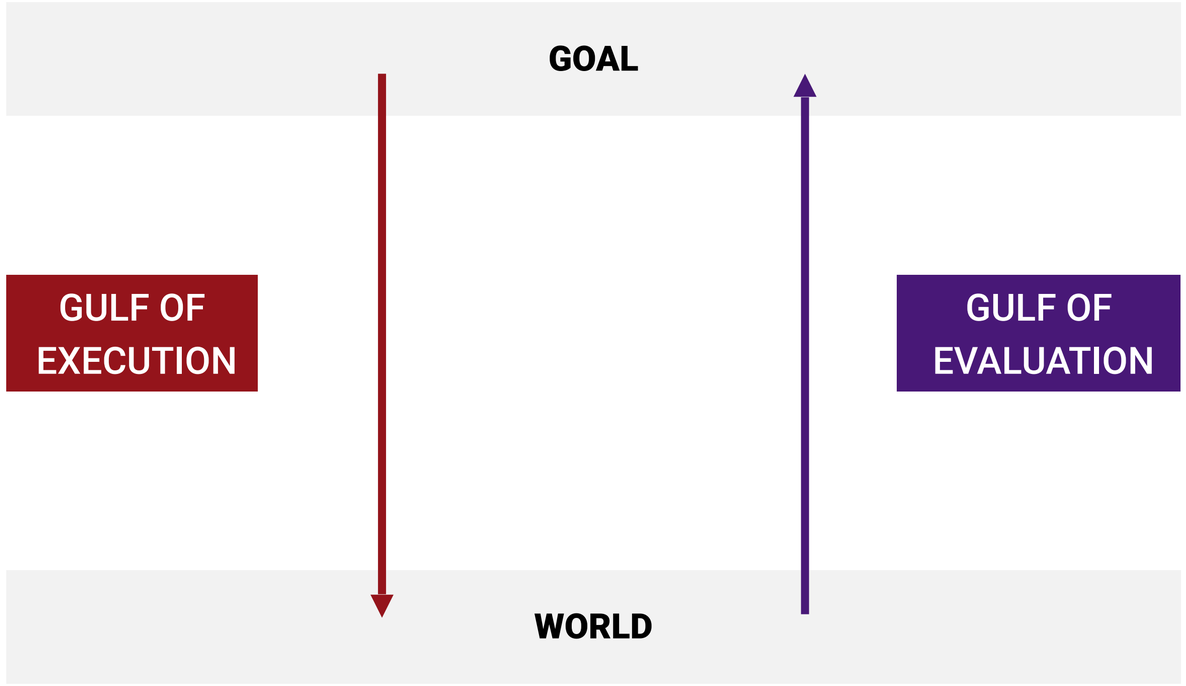


Fundamental Principles of Design

[Norman 2014]

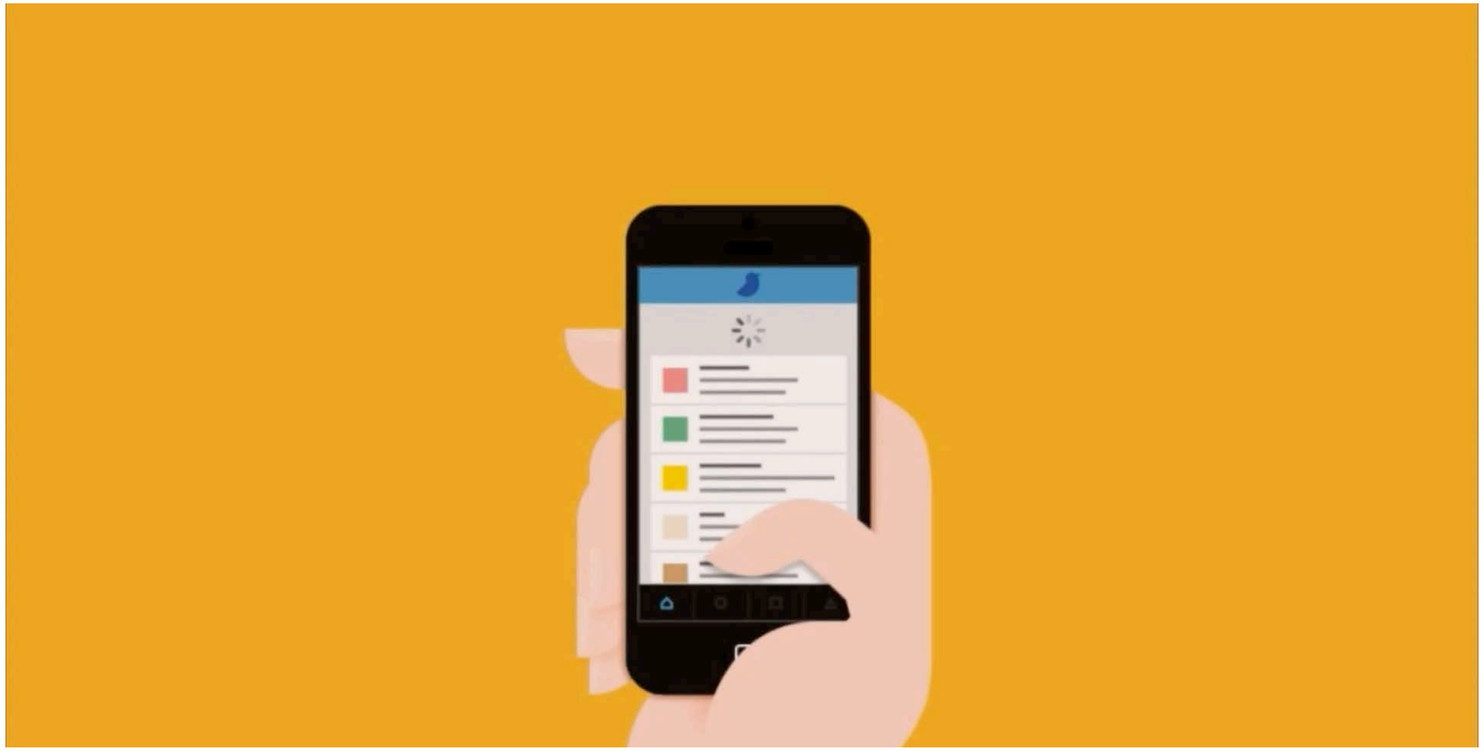
Seven key design principles that help minimizing the gulf of execution and evaluation:

1. Discoverability
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3. Affordances
4. Signifiers
5. Mappings
6. Constraints
7. Conceptual Models



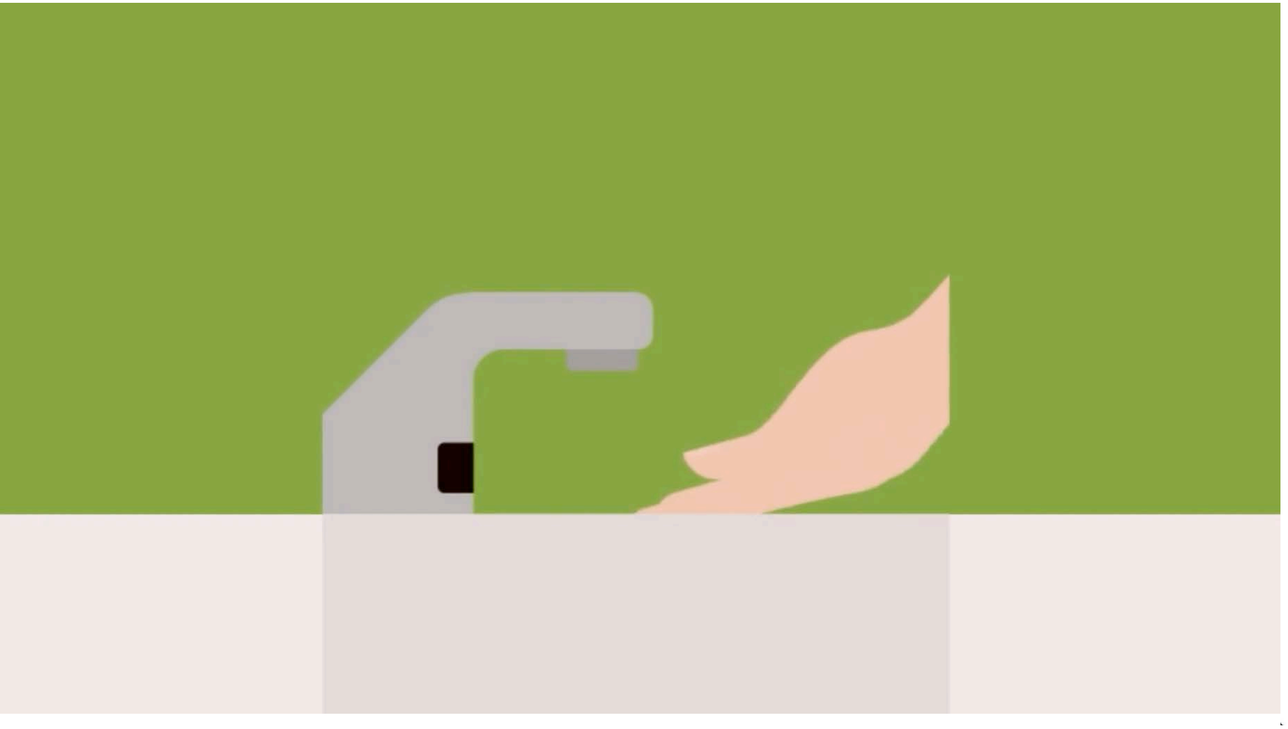
Feedback

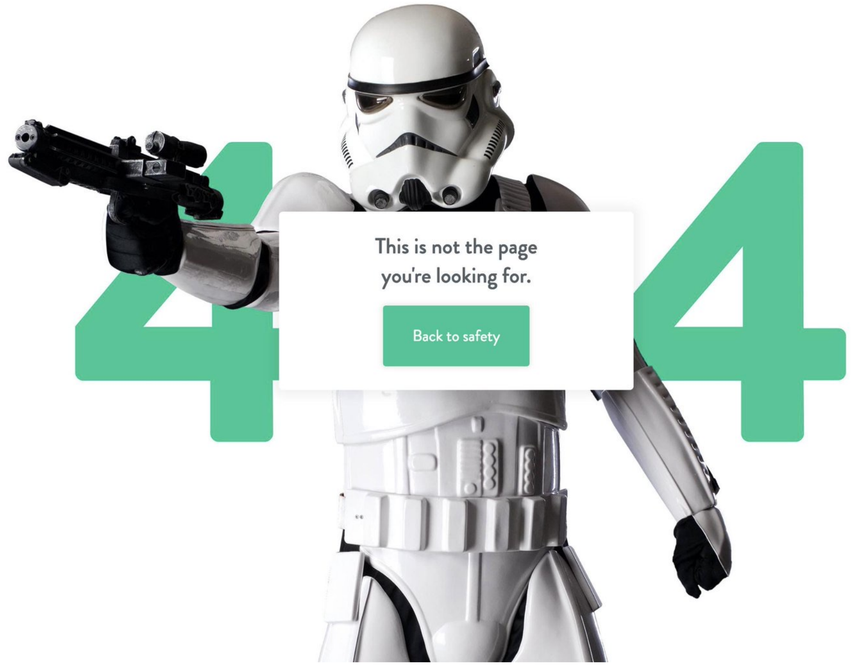
- always provide feedback to let users know the effect of their actions



Feedback

- feedback involves sending back information about what action has been done and what has been accomplished



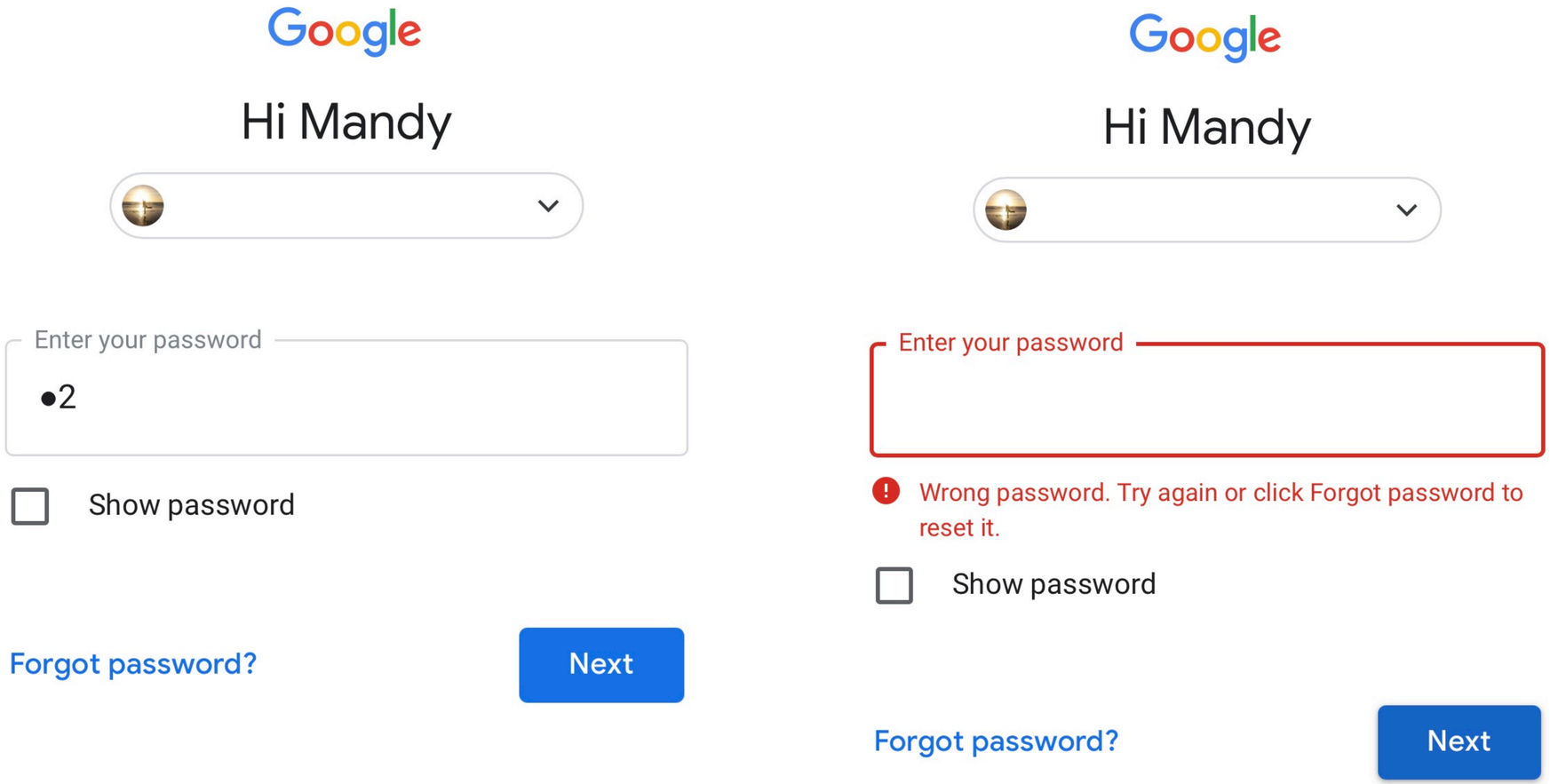


Zestful

Also provide feedback when the action cannot be completed

Feedback

- feedback involves sending back information about what action has been done and what has been accomplished



Feedback

- the system should always inform the user what is going on by providing appropriate feedback within a reasonable time
- the user should always receive feedback, both for successful and unsuccessful actions

DURATION OF AN ACTION	FEEDBACK
< 0.1 s	not noticed
0.1 - 1 s	no feedback necessary
1 - 5 s	"Busy"
> 5 s	Progress Bar

[MOSER 2012]

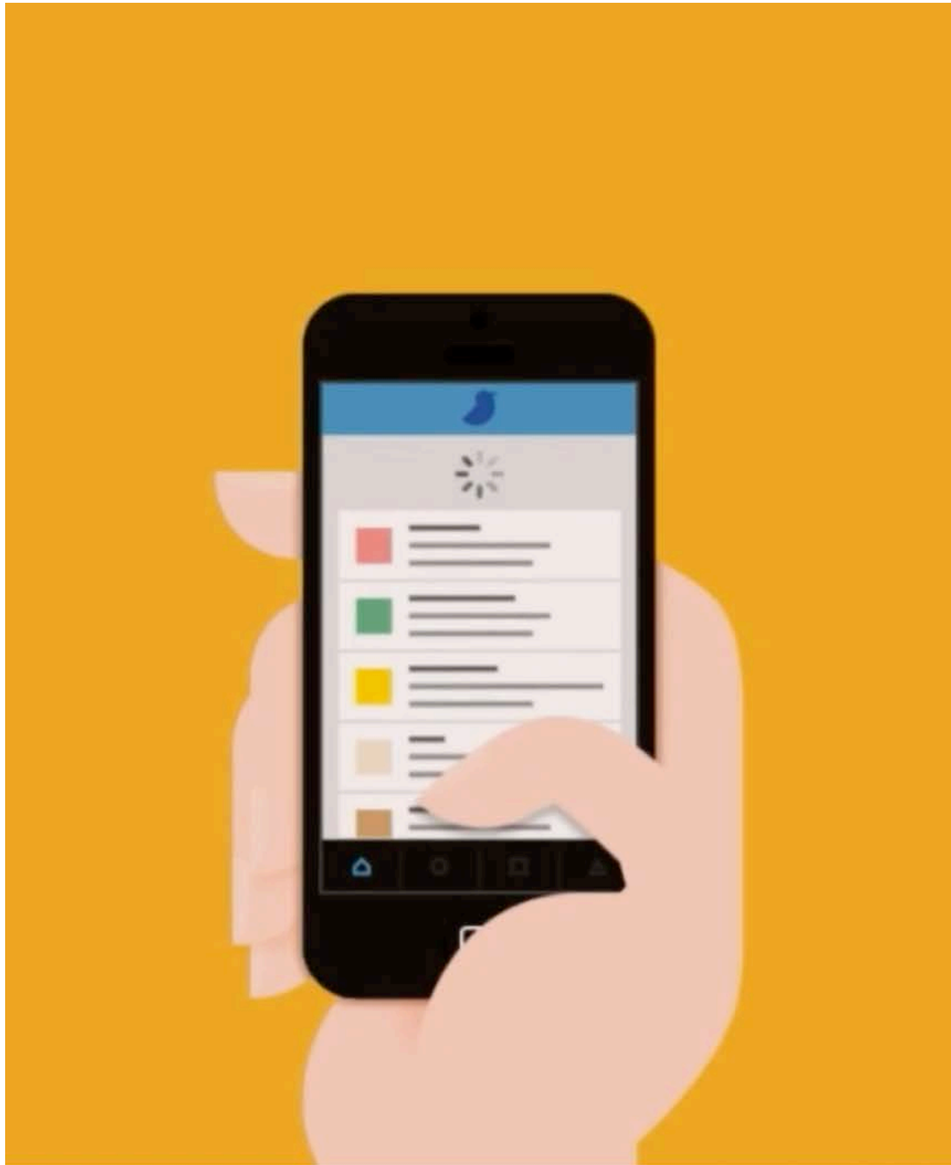
Feedback

- for actions shorter than 1 second, the results can be presented immediately



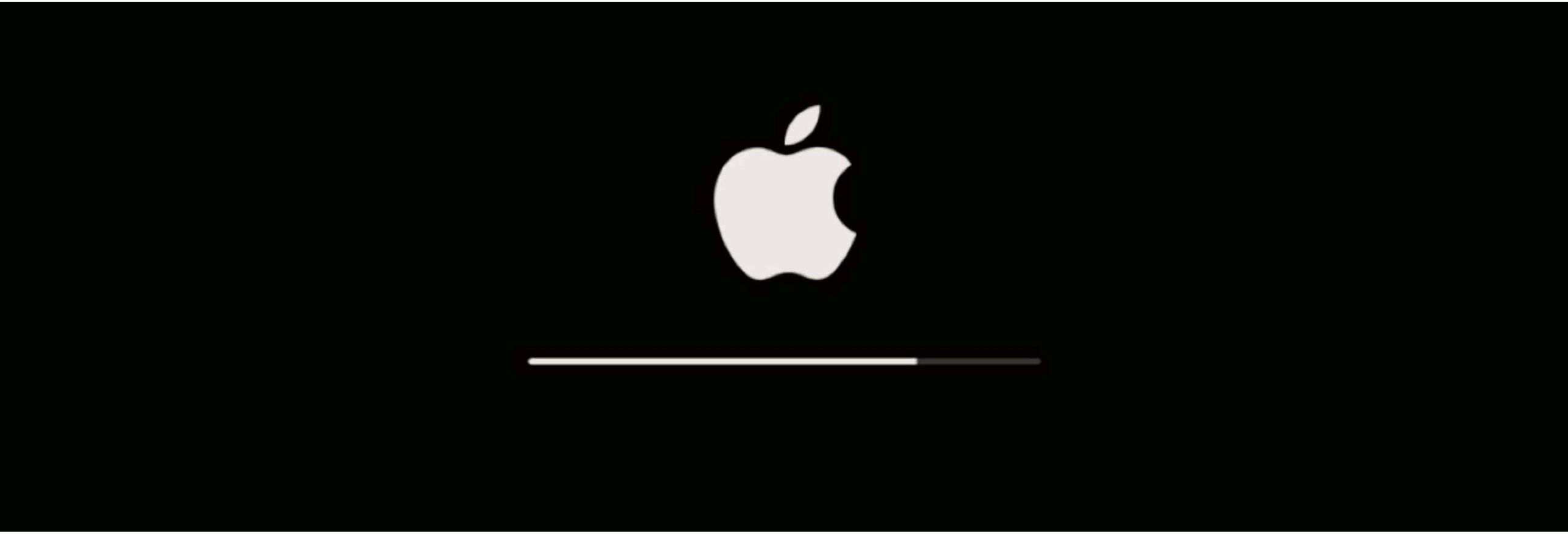
Feedback

- 1 - 5 seconds: feedback should be given so the user knows that the computer is still working



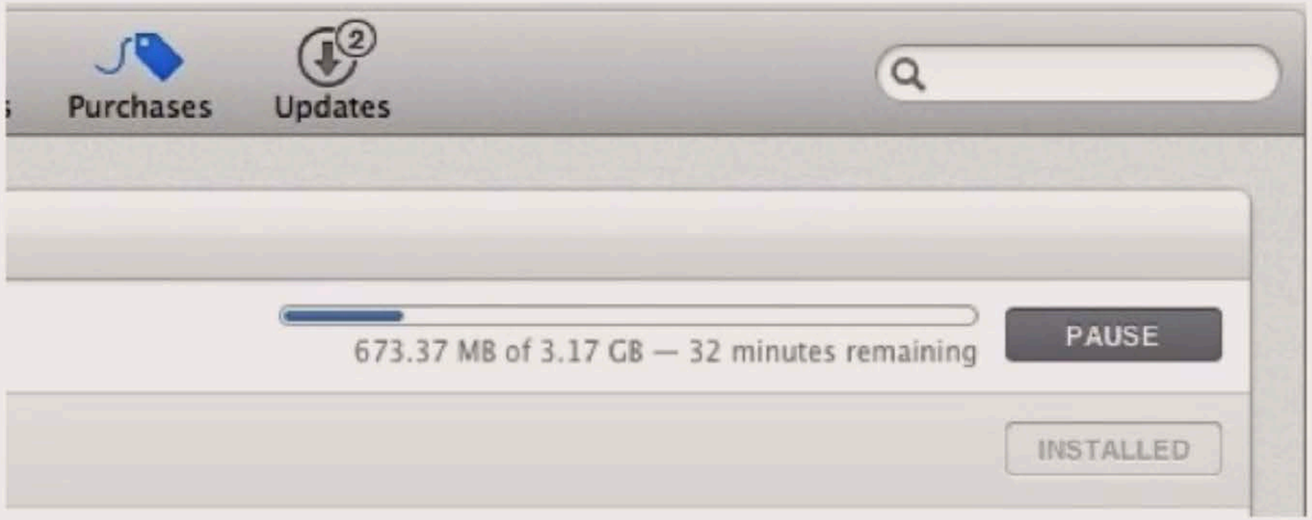
Feedback

- more than 5 seconds: use of a progress bar



Feedback

- if it takes longer than 1 minute, a timer should be offered

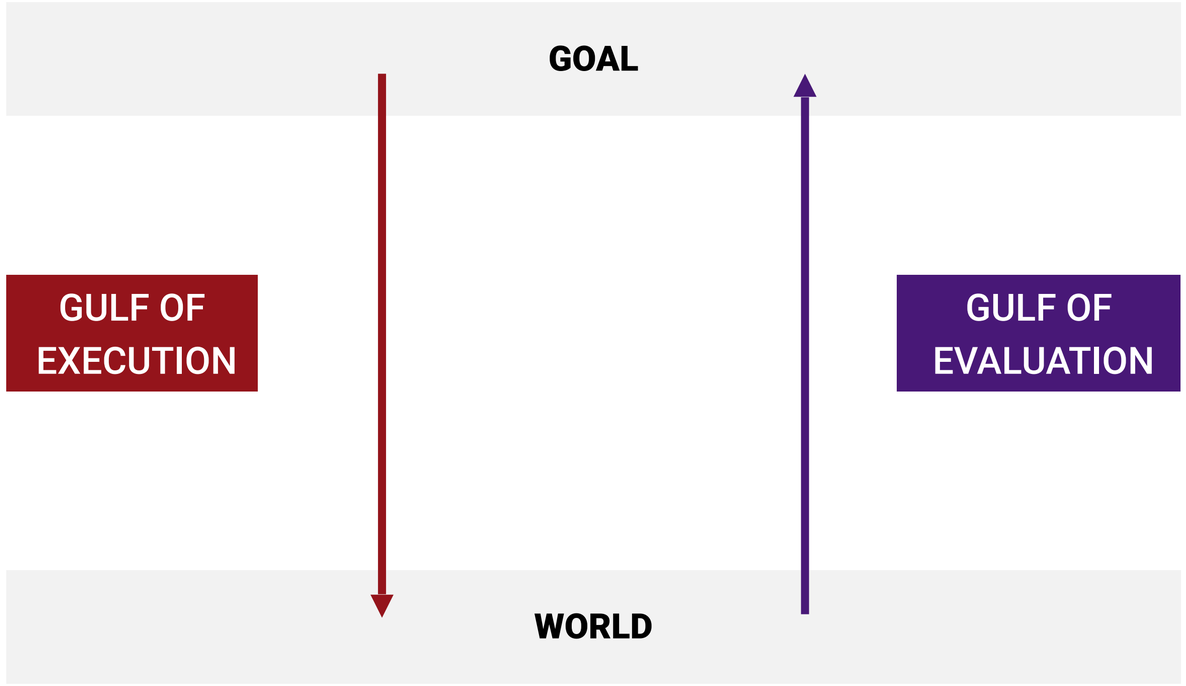


Fundamental Principles of Design

[Norman 2014]

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Affordance

- perceived and actual properties of an object that give clues to its operation
- when the affordances of a physical object are perceptually obvious, it is easy to know how to interact with it








PUSH
PUSH


PULL
PULL

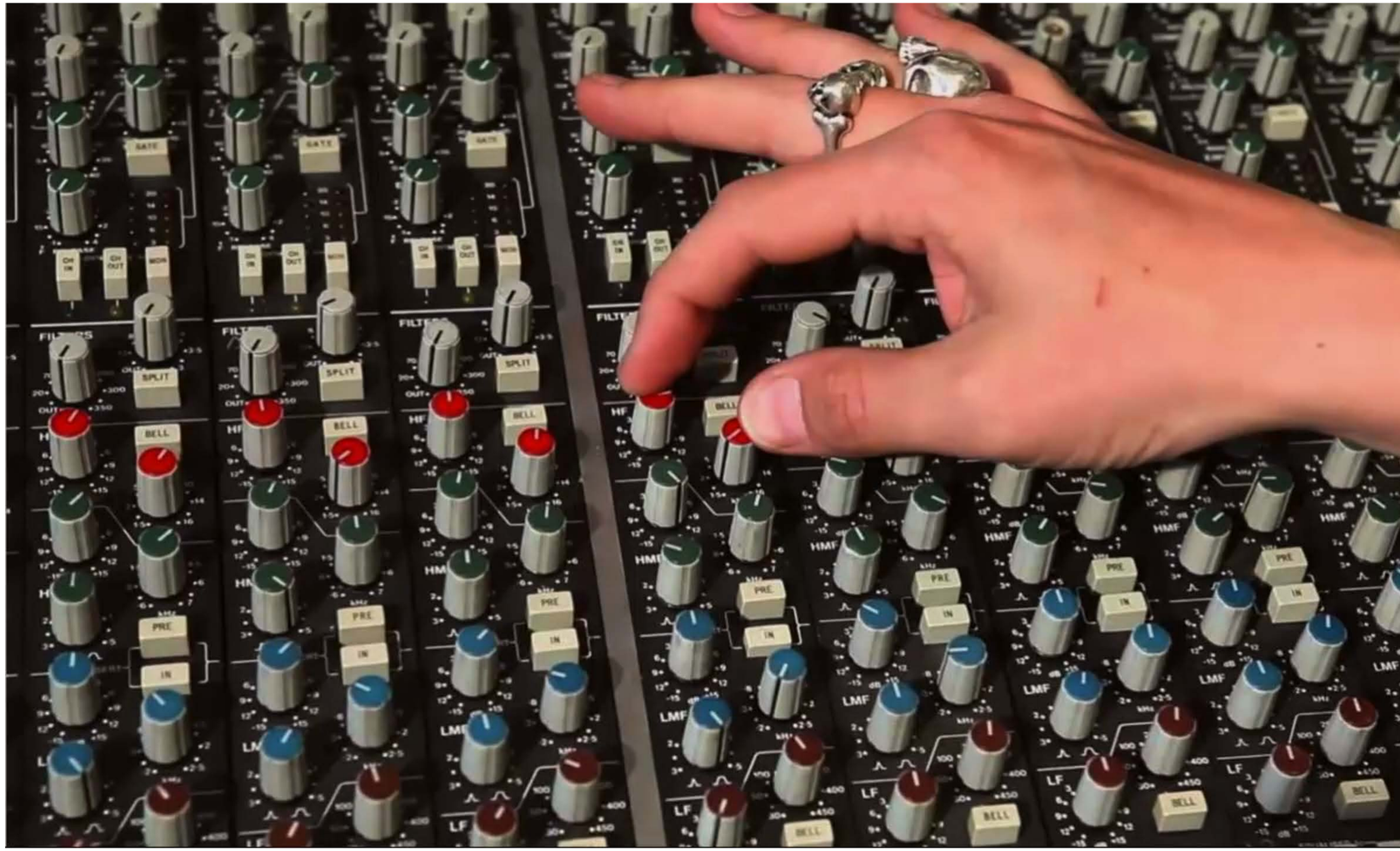
Affordance

- e.g. a mouse button invites pushing

  by the way it is physically

  constrained in its plastic shell





Affordance





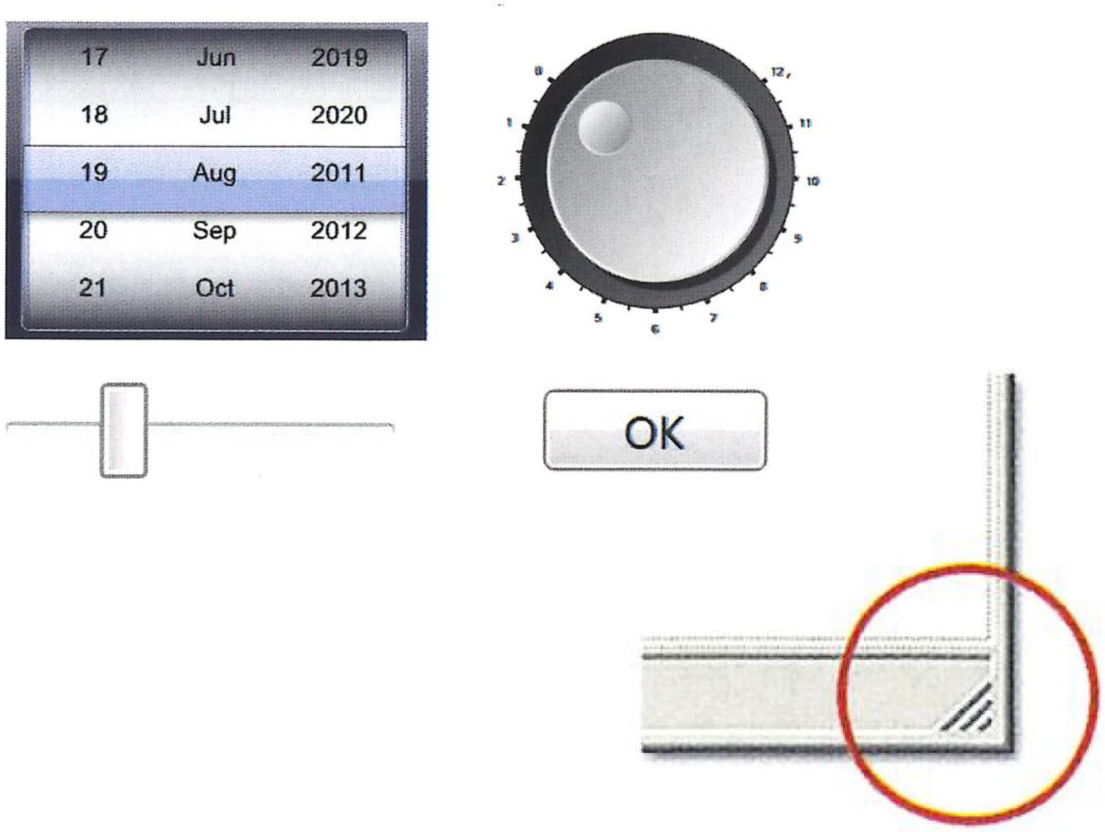


Affordances used incorrectly

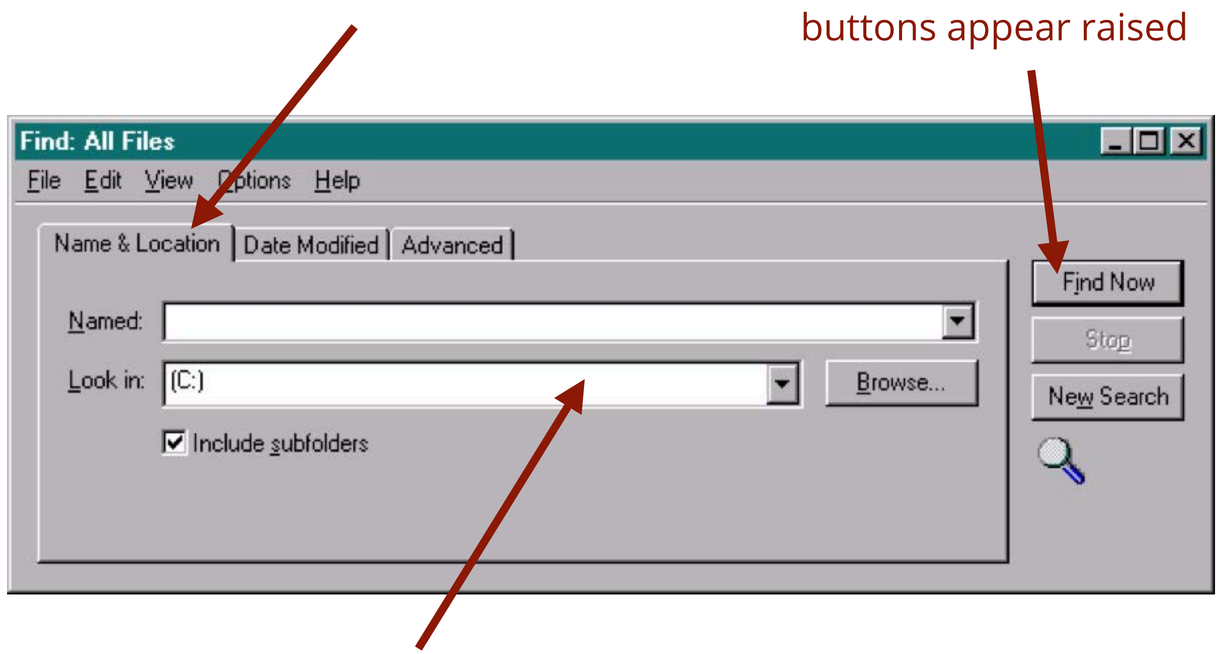


don’t suggest properties that are not existing

Digital Affordances



[Moser 2012]



Windows 95 dialog box made use of heavy shadows and highlights to create 3D effects

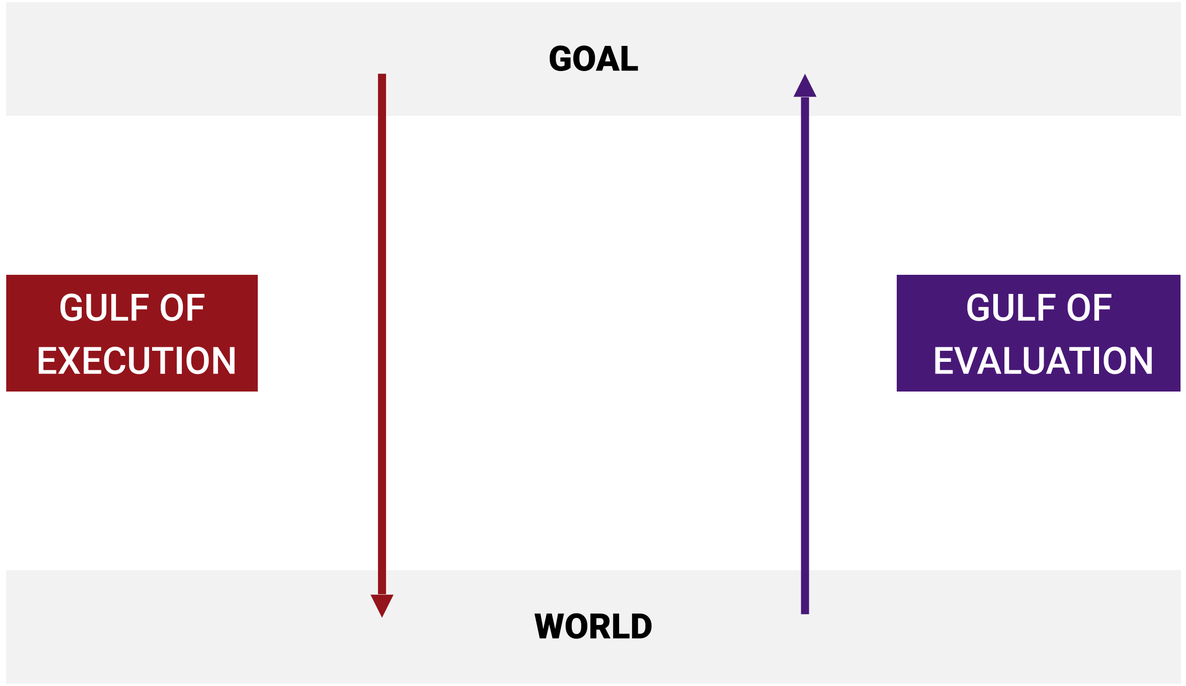
source: https://www.nngroup.com/articles/flat-design/

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Signifiers

- Users need some sign of what it is for, what is happening, and what are the alternative actions
- Signifiers signal things, in particular what actions are possible and how they should be done [Norman 2014]



Signifiers

- Signifier are signals; some are signs, labels, and drawings placed in the world [Norman 2014]
- Examples:
  - signs labeled „push“, „pull“, or „exit“ on doors
  - arrows and diagrams indicating what is to be acted upon or in which direction to gesture

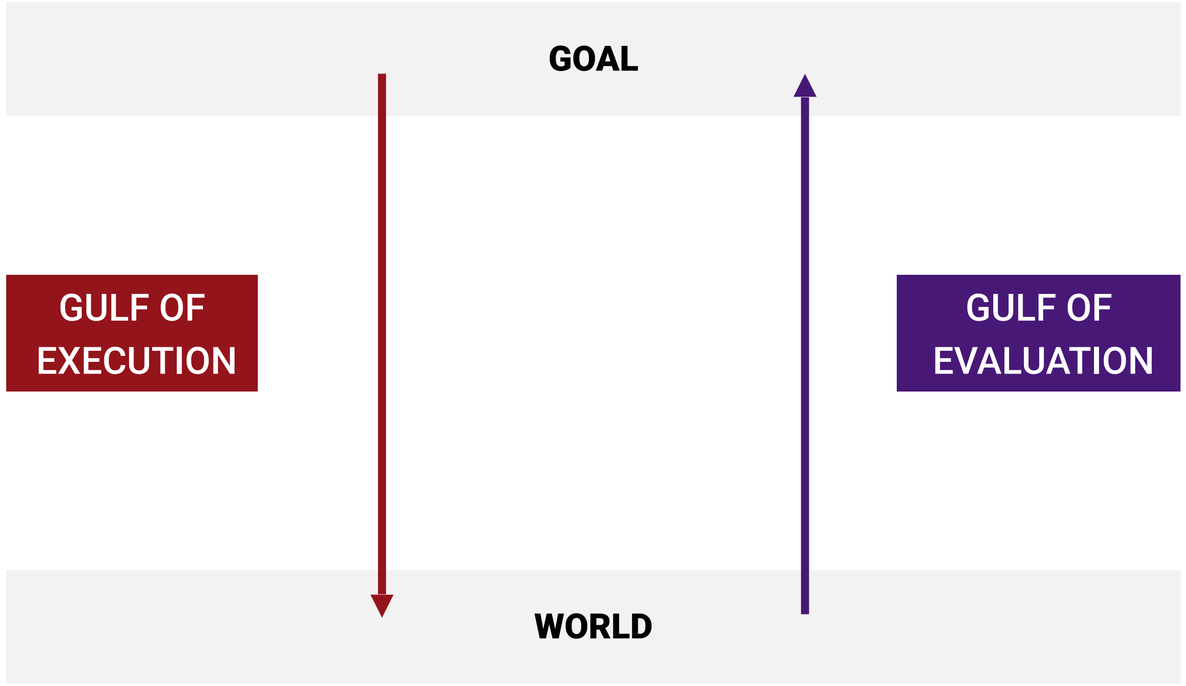


Fundamental Principles of Design

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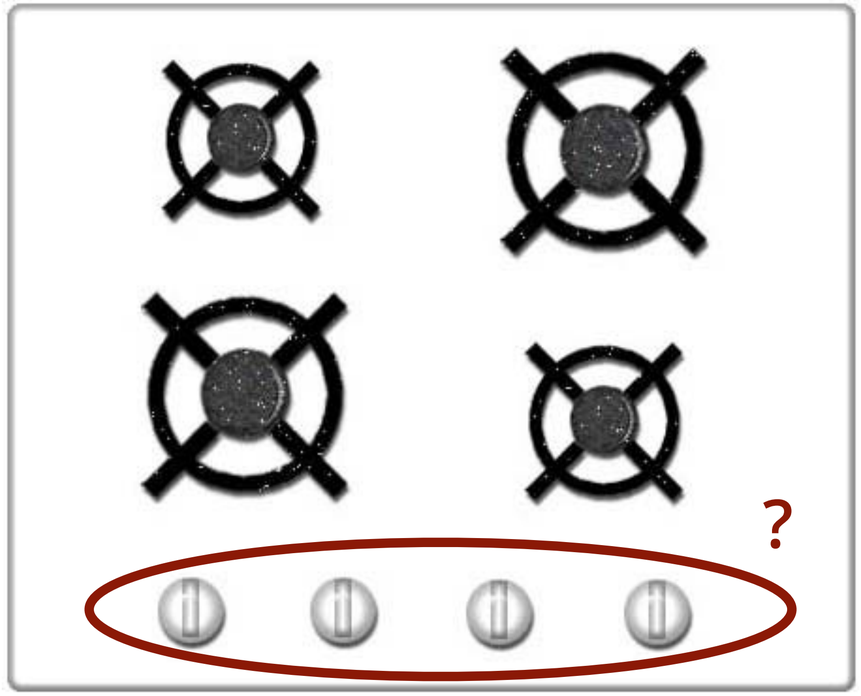
Seven key design principles that help minimizing the gulf of execution and evaluation:

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Mapping

- relationship to controls and their effect
- good mapping means that controls are clearly arranged and that it is clear what happens when using them

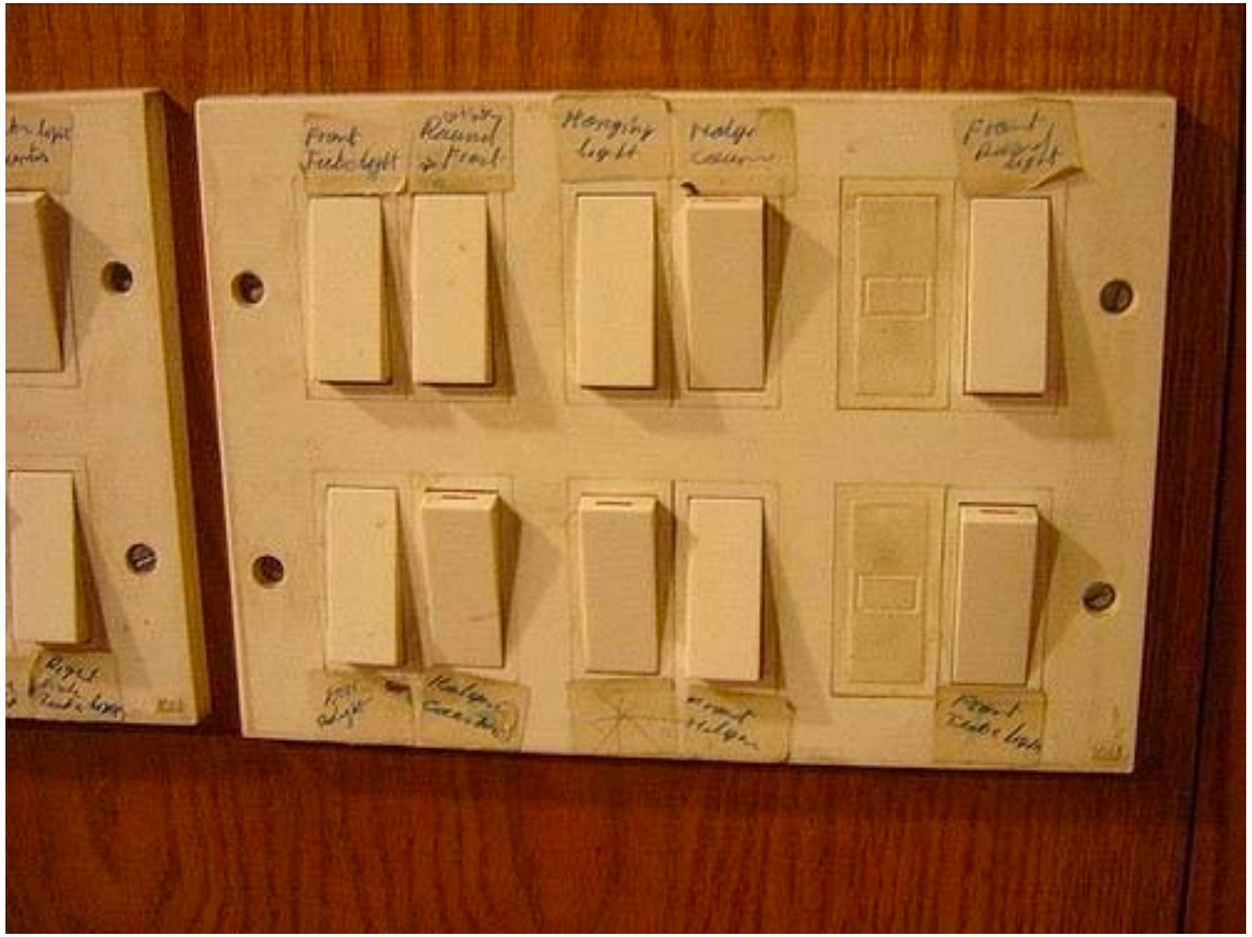




Try to design interfaces where the mapping between controls and their effects is clear



Mapping



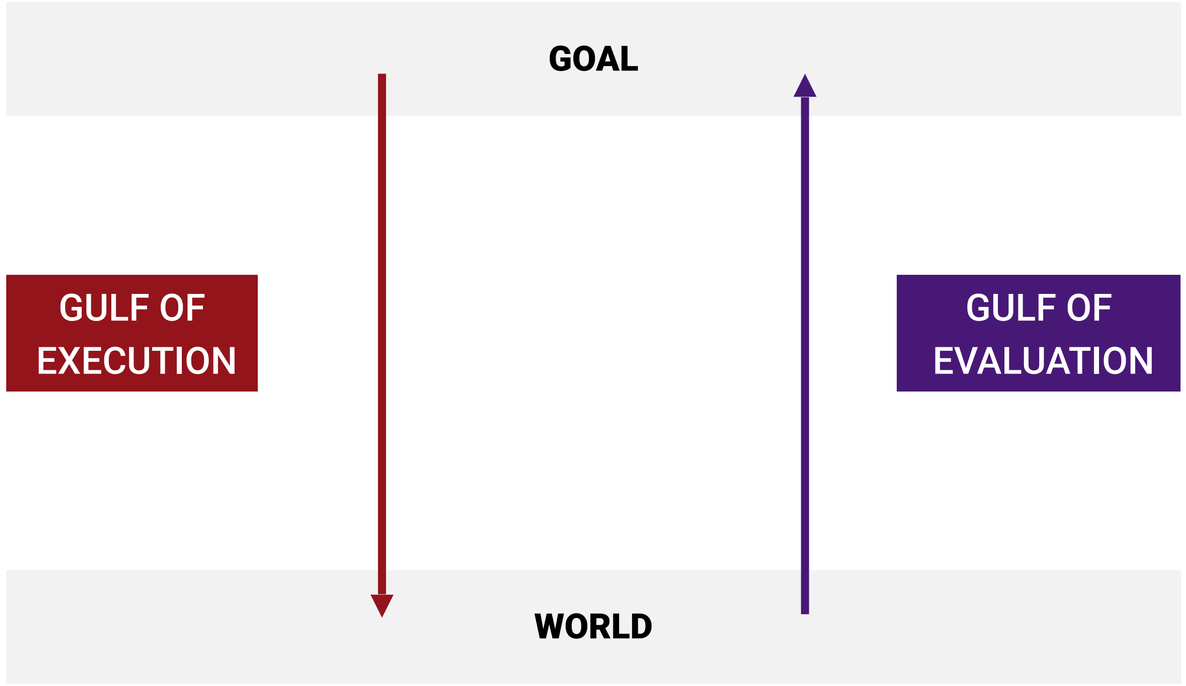
https://sites.psu.edu/nineteenthstation/2018/08/30/normans-human-centered-design/

Fundamental Principles of Design

[Norman 2014]

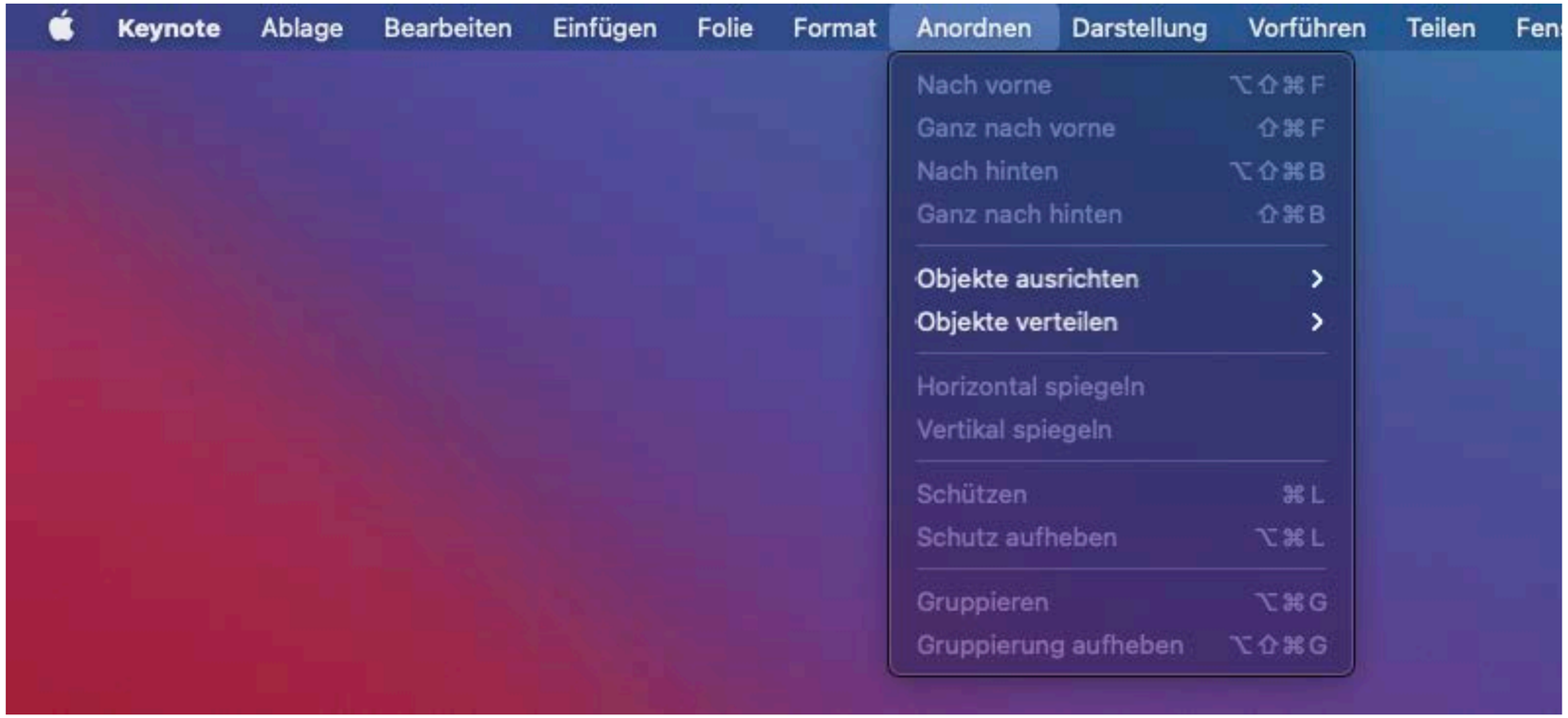
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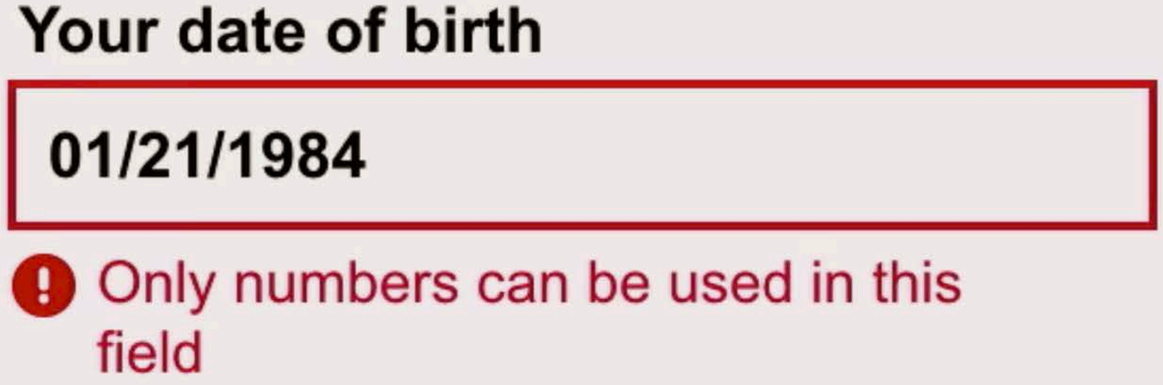
Constraints

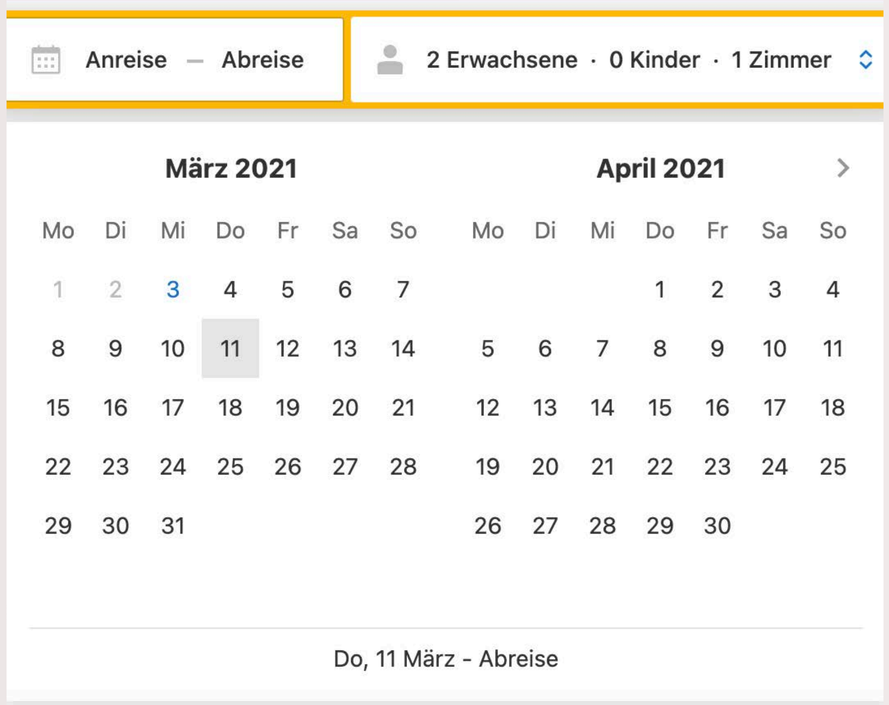
- limit the possible actions that can be performed on a system
- prevents the user from selecting incorrect options and thereby reduces the chance of making a mistake





Constraints





source: booking.com

Constraints



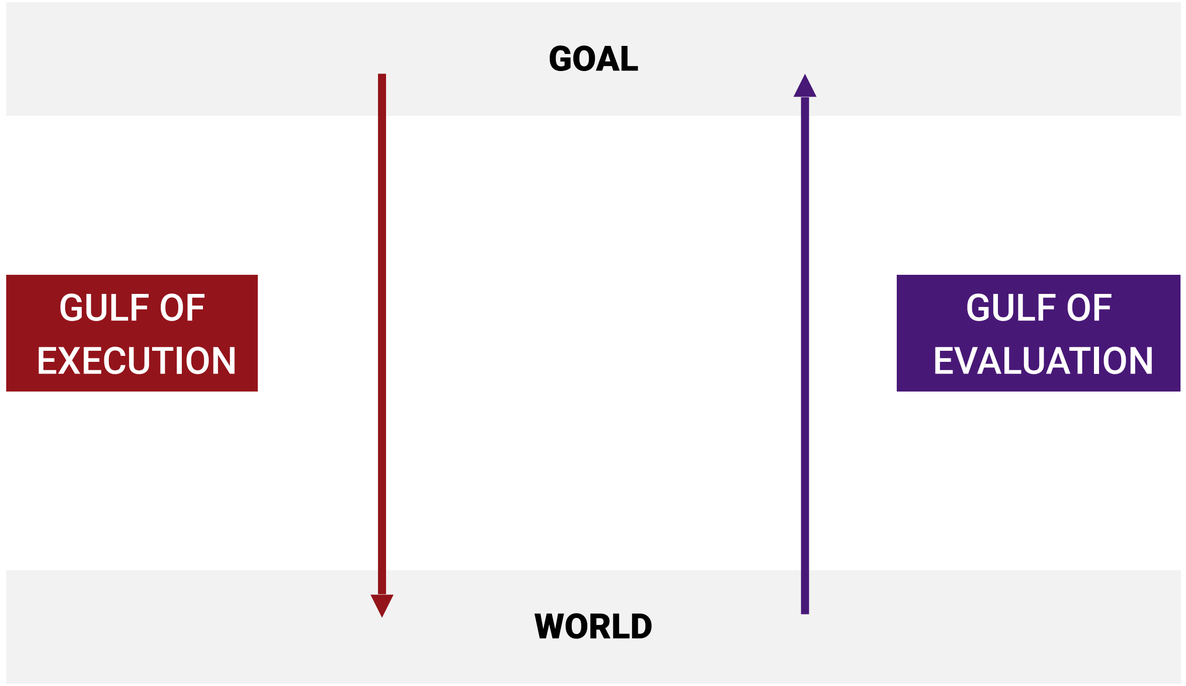


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Mental Model

System Model

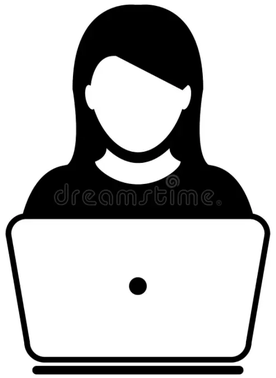
How does it work?

„I know how the car works, but I don't know how people drive their cars“



Designer's Conceptual Model

„I understand both sides and fill this gap“



Mental Model

How can I use it?

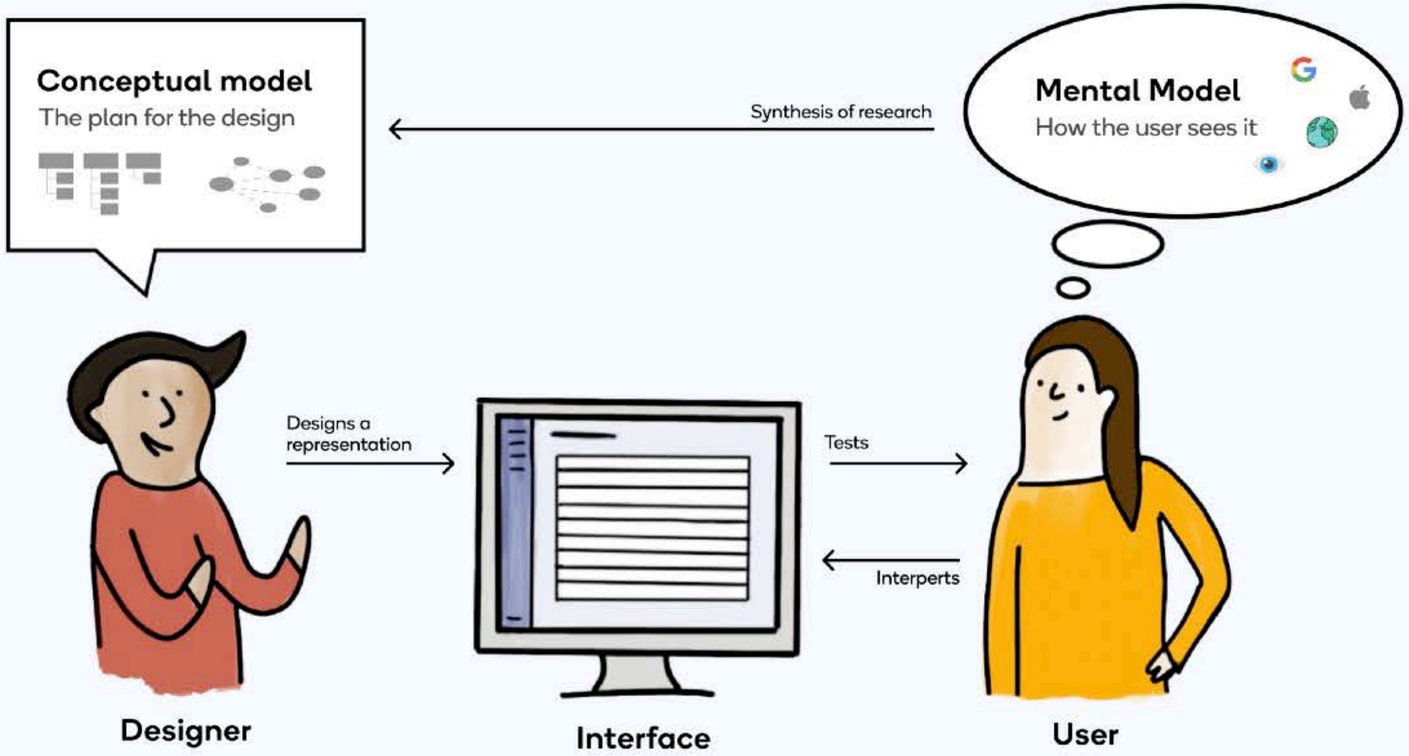
„I am a good driver but I have no idea how everything works in my car“



72

Mental Model

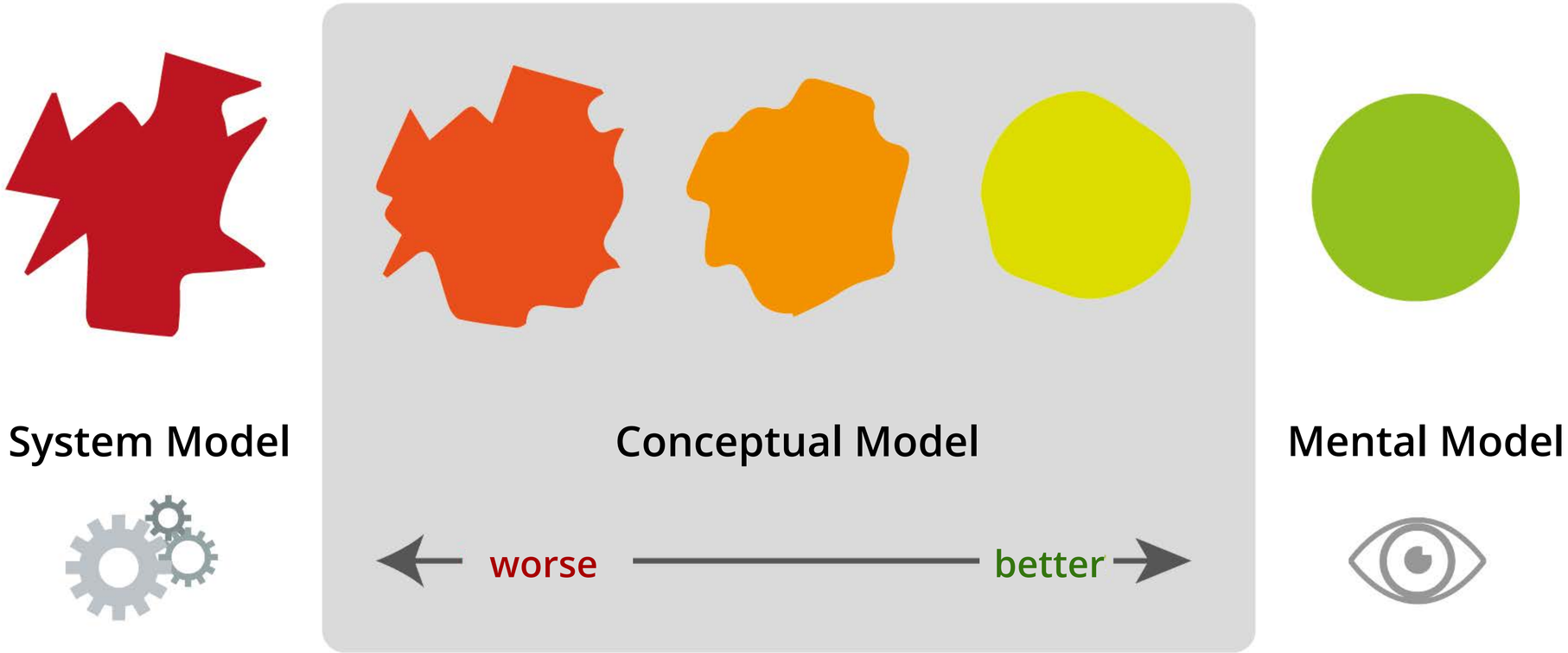
- **mental model:** an image that we carry in our head of how something works in the real world
- **conceptual model:** is created by the designer as a high-level plan for how the product will work and fit together



source: https://uxdesign.cc/understanding-mental-and-conceptual-models-in-product-design-7d69de3cae26

Conceptual Model

- designers often assume that users' mental model is the same as theirs: **but it usually isn't!**
- large differences between these models can lead to slow performance, errors and frustration



[Cooper 2014]

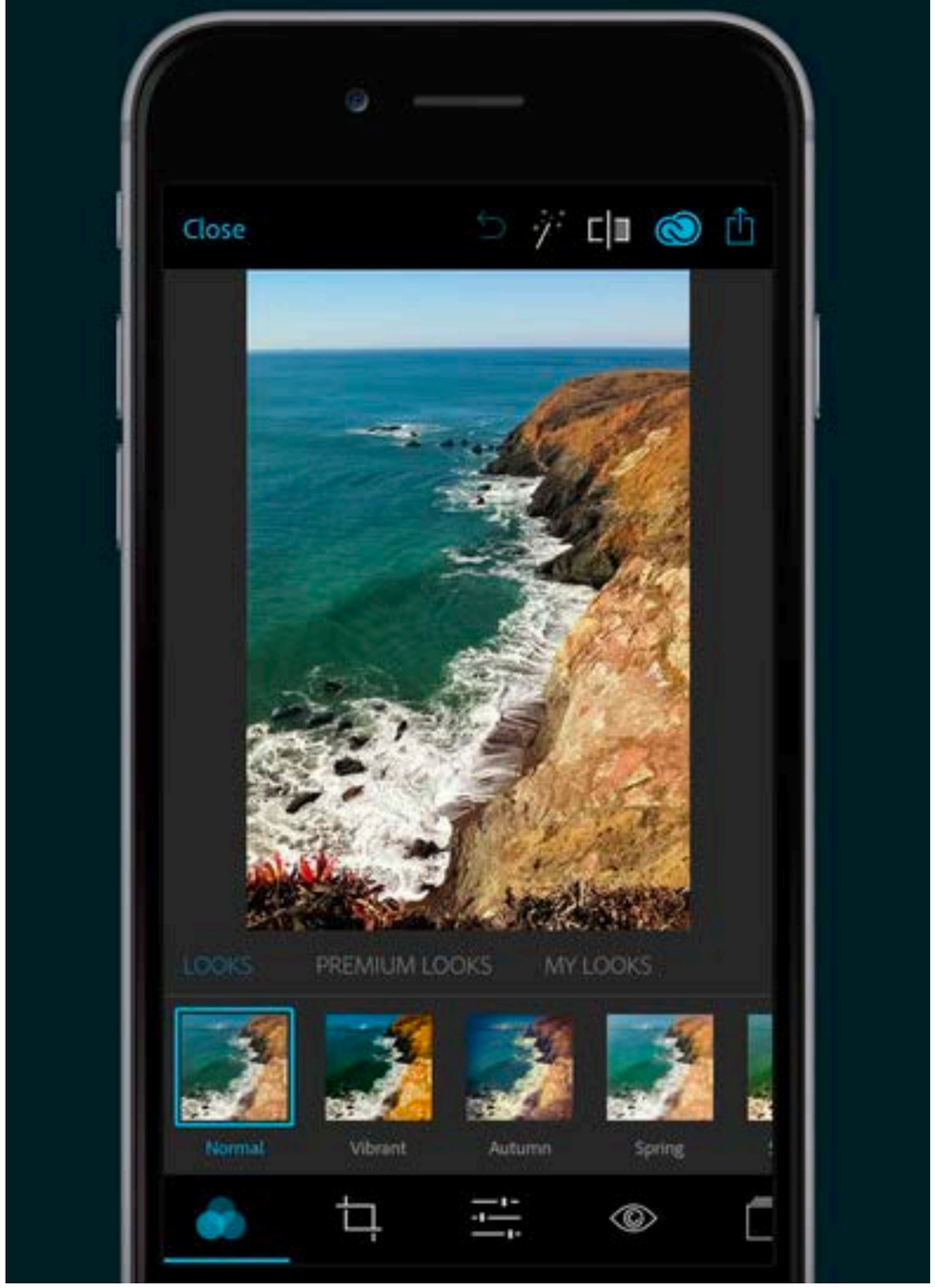
Conceptual Model

- it is easier to develop software that reflects the system model
- from the developer's point of view it is logical to offer a button for each function
- but this does not help the users to reach their goals



Conceptual Model

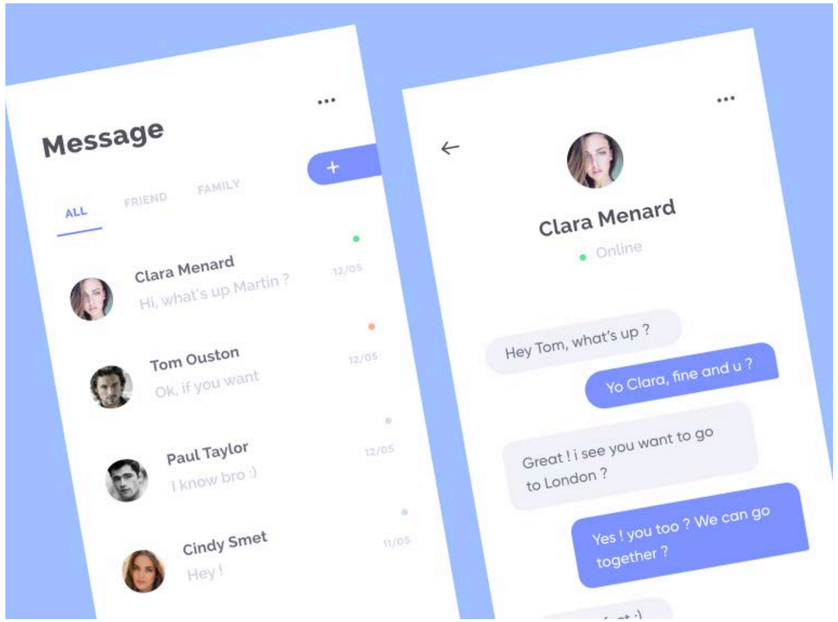
- close to mental model of novices in photo editing
- supports searching for a specific look instead of manipulating many abstract values

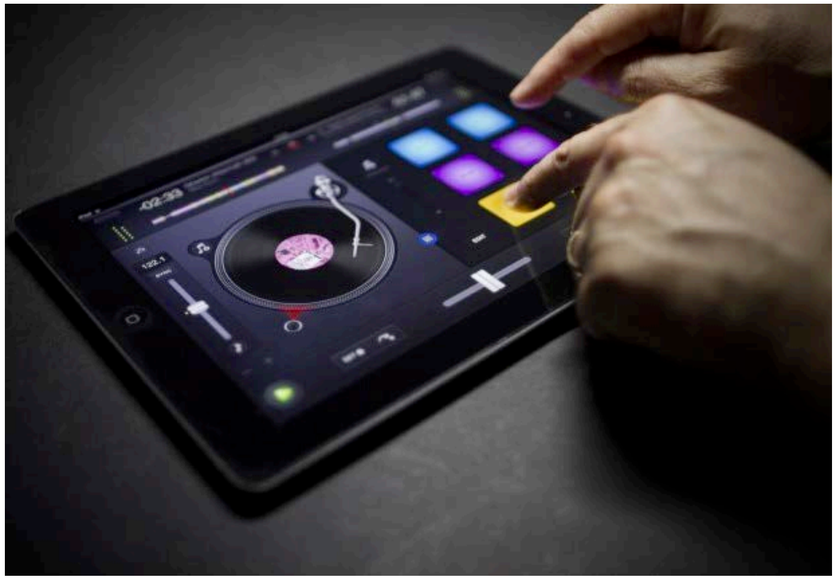


Conceptual Model

How to build a strong conceptual model:

- Research and understand user expectations first
- Use terminology users are familiar with
- Leverage the interface patterns that they already use
- Test early, and iterate





Affordances & Signifiers?







[Norman 2014]

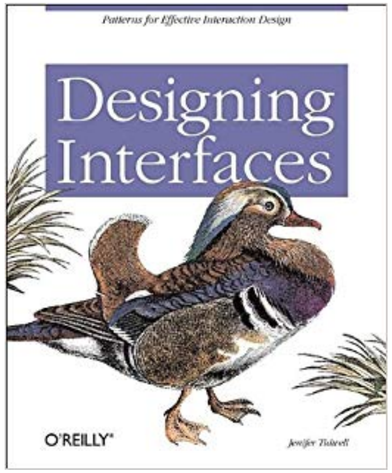
Design Principles



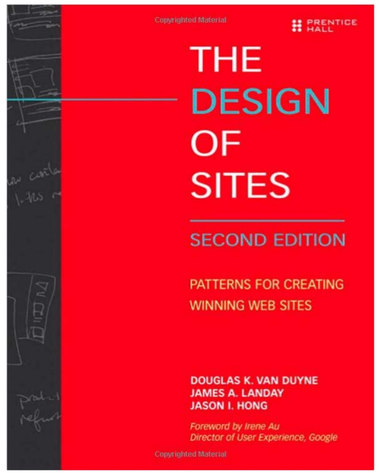
Design Patterns

- means of capturing useful design solutions and generalizing them to address similar problems
- describe the problem in a compact form and propose an exemplary, generalizable solution *(best practice)*
- **context-specific:** they are defined for specific design situations that have a similar context

design patterns collection
for a specific problem domain:



J. Tidwell - Designing Interfaces:
Design Patterns for Interaction Design

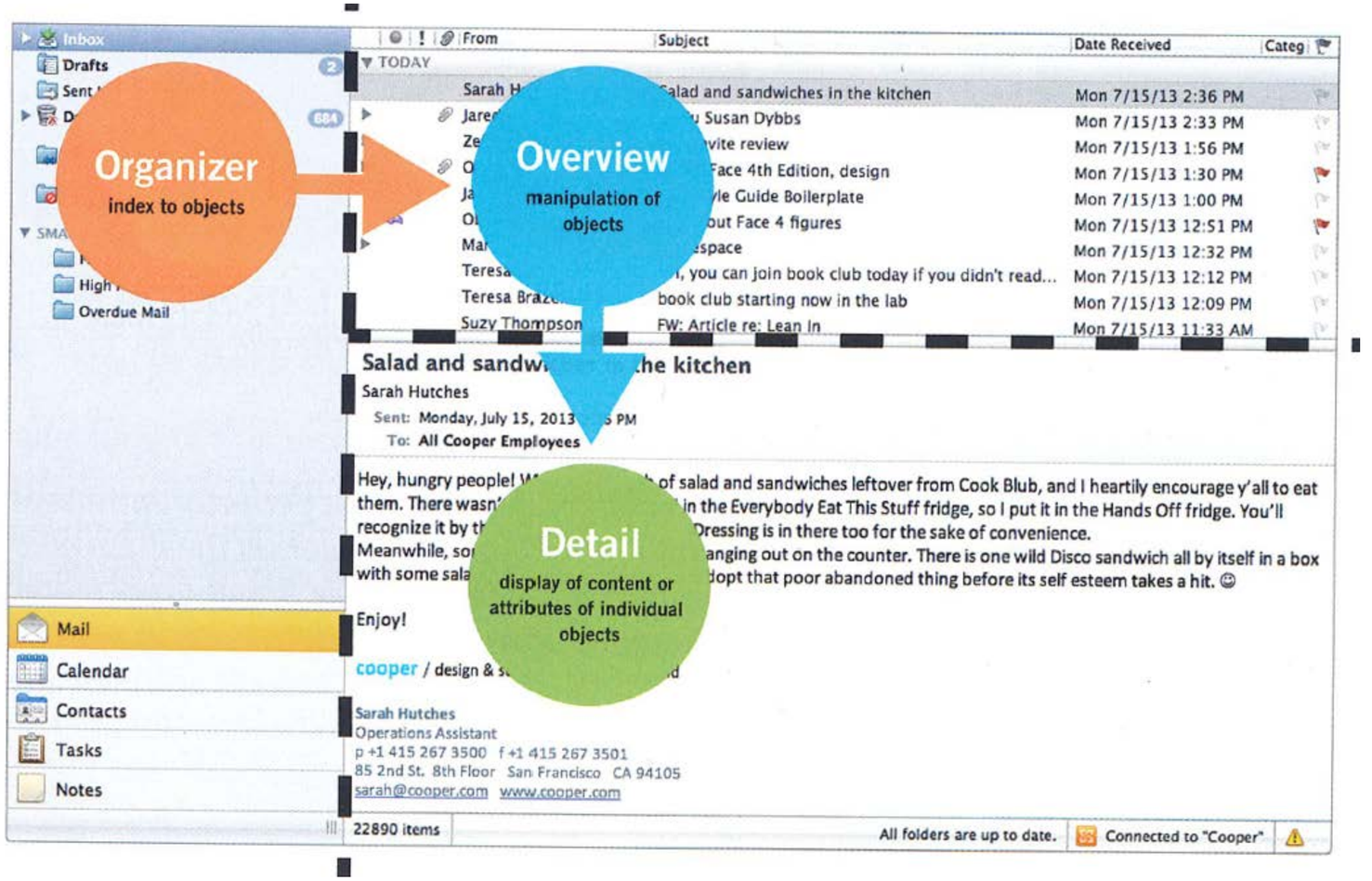


Douglas van Duyne: The Design of Sites:
Design Patterns for Websites

Design Patterns

Design Patterns for Interaction Design:

- Structural patterns

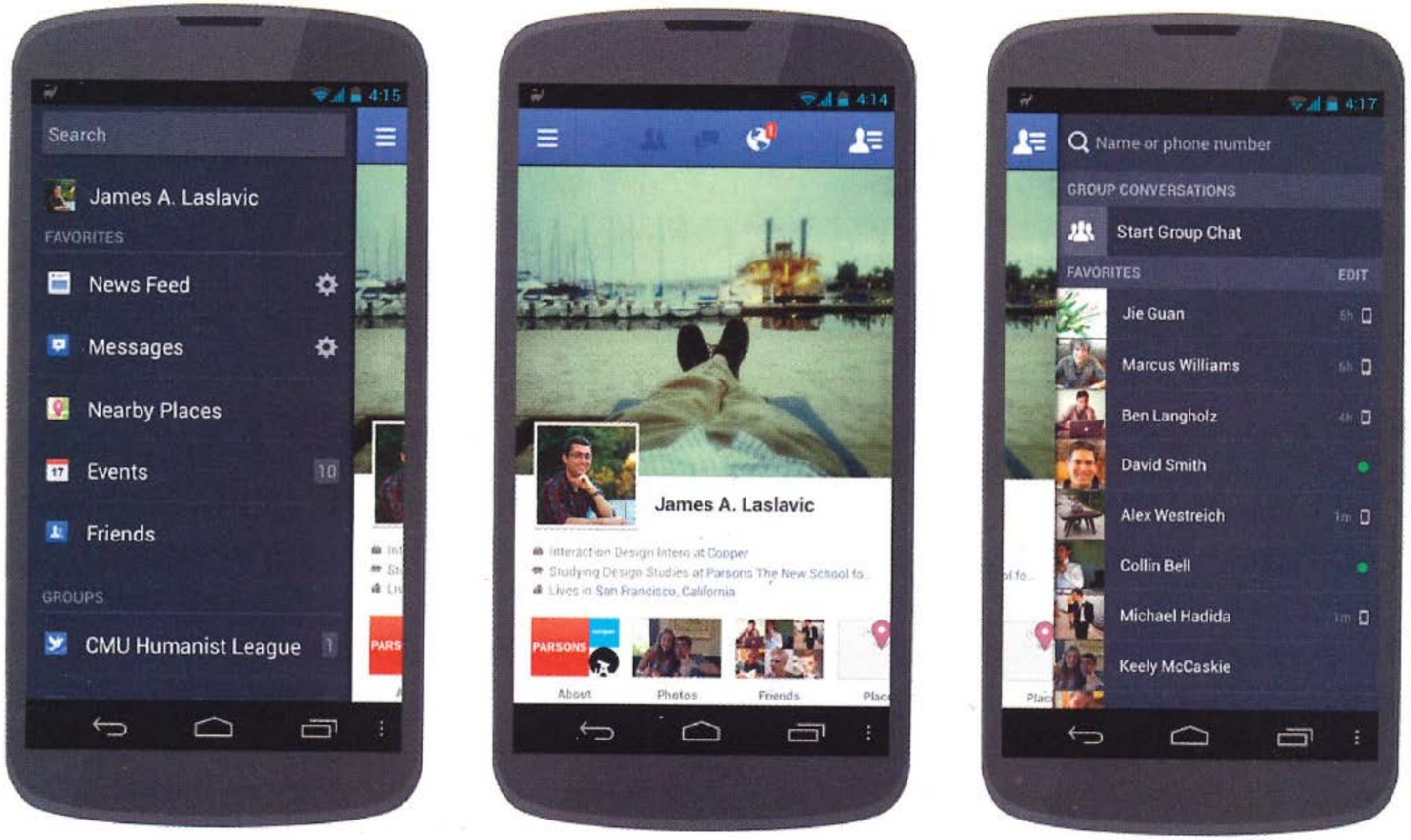


Microsoft Outlook [Cooper 2014]

Design Patterns

Design Patterns for Interaction Design:

- Structural patterns

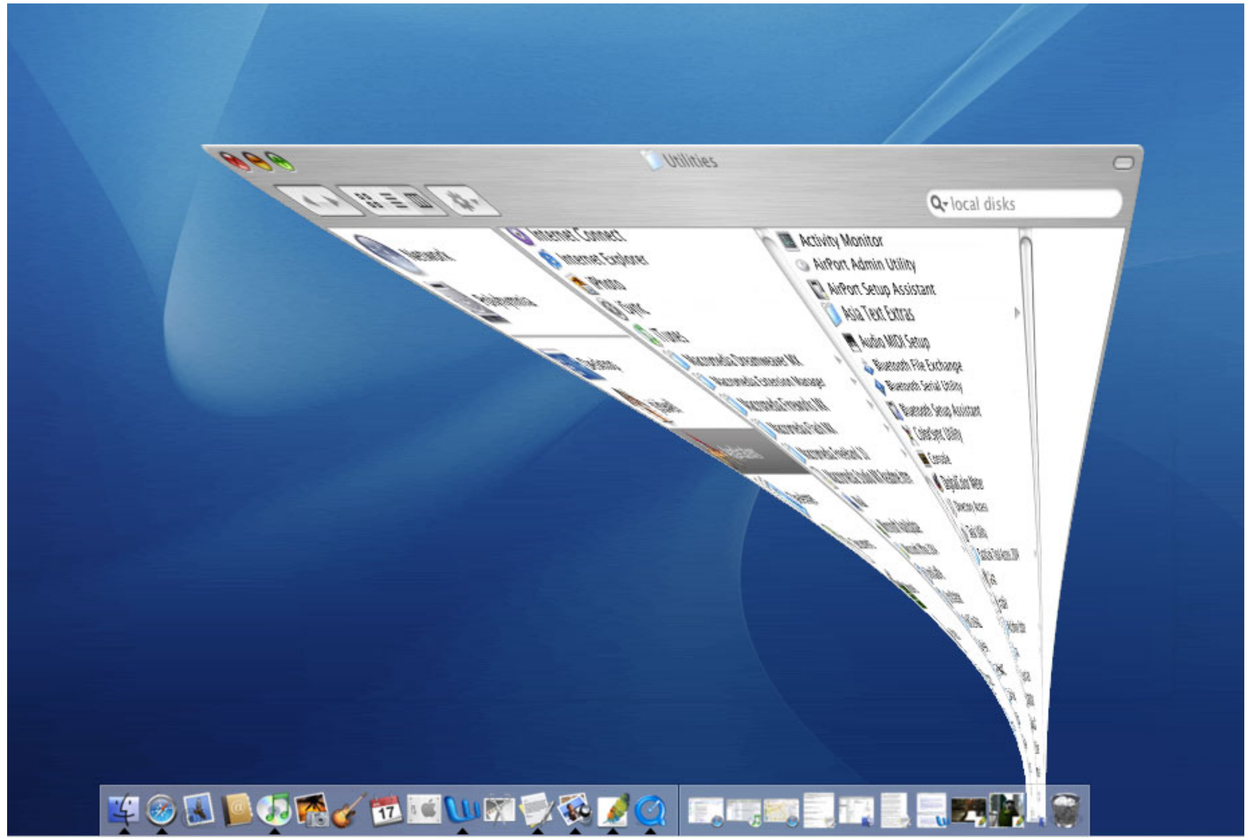


Double Drawer Pattern in der Facebook App [Cooper 2014]

Design Patterns

Design Patterns for Interaction Design:

- Structural patterns
- Behavioral patterns



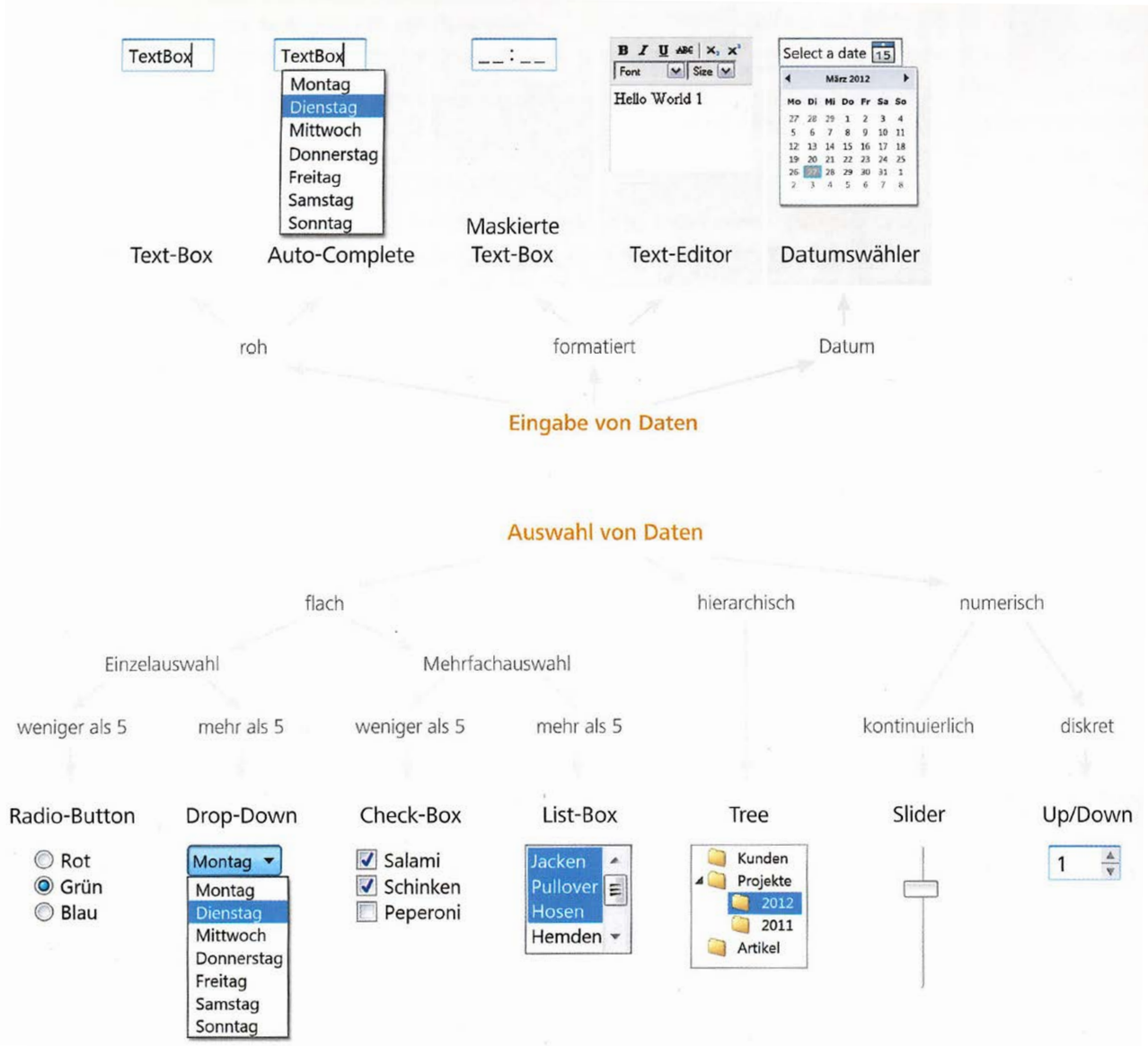
ANIMATED TRANSITIONS:

https://designinginterfaces.com/patterns/animated-transition/

Design Patterns

different levels of complexity:

- elementary interaction elements

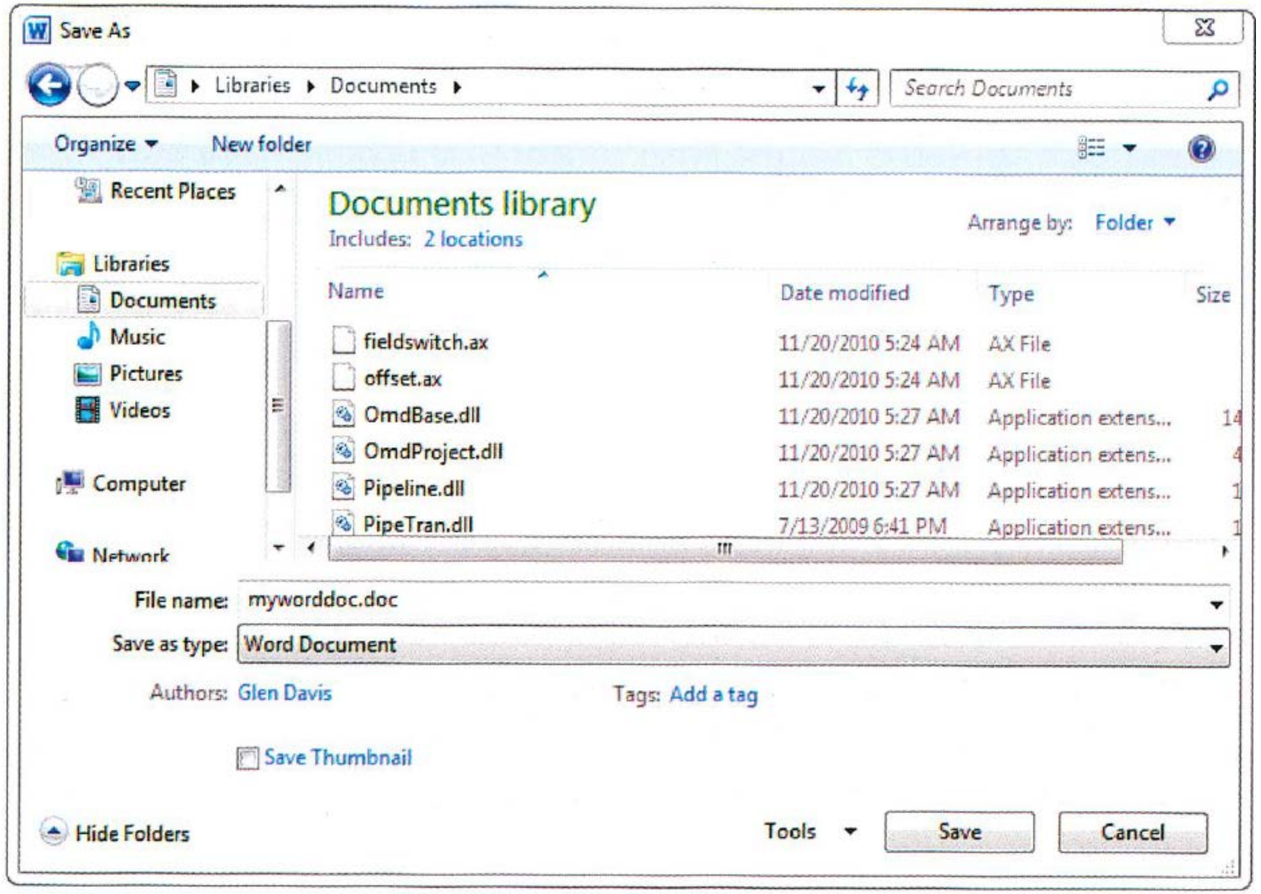


[MOSER 2012]

Design Patterns

different levels of complexity:

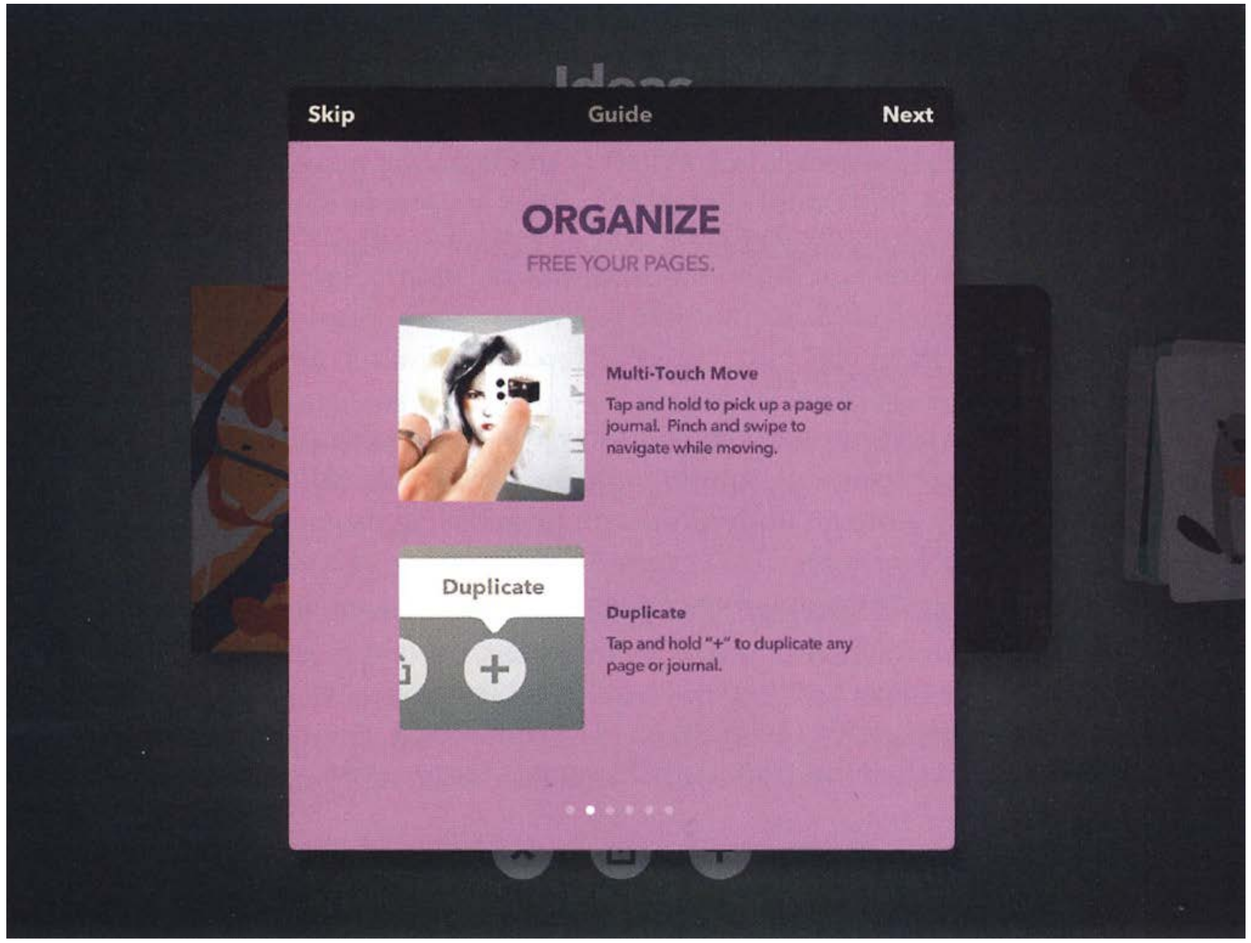
- elementary interaction elements
- complex interactions



Design Patterns

different levels of complexity:

- elementary interaction elements
- complex interactions

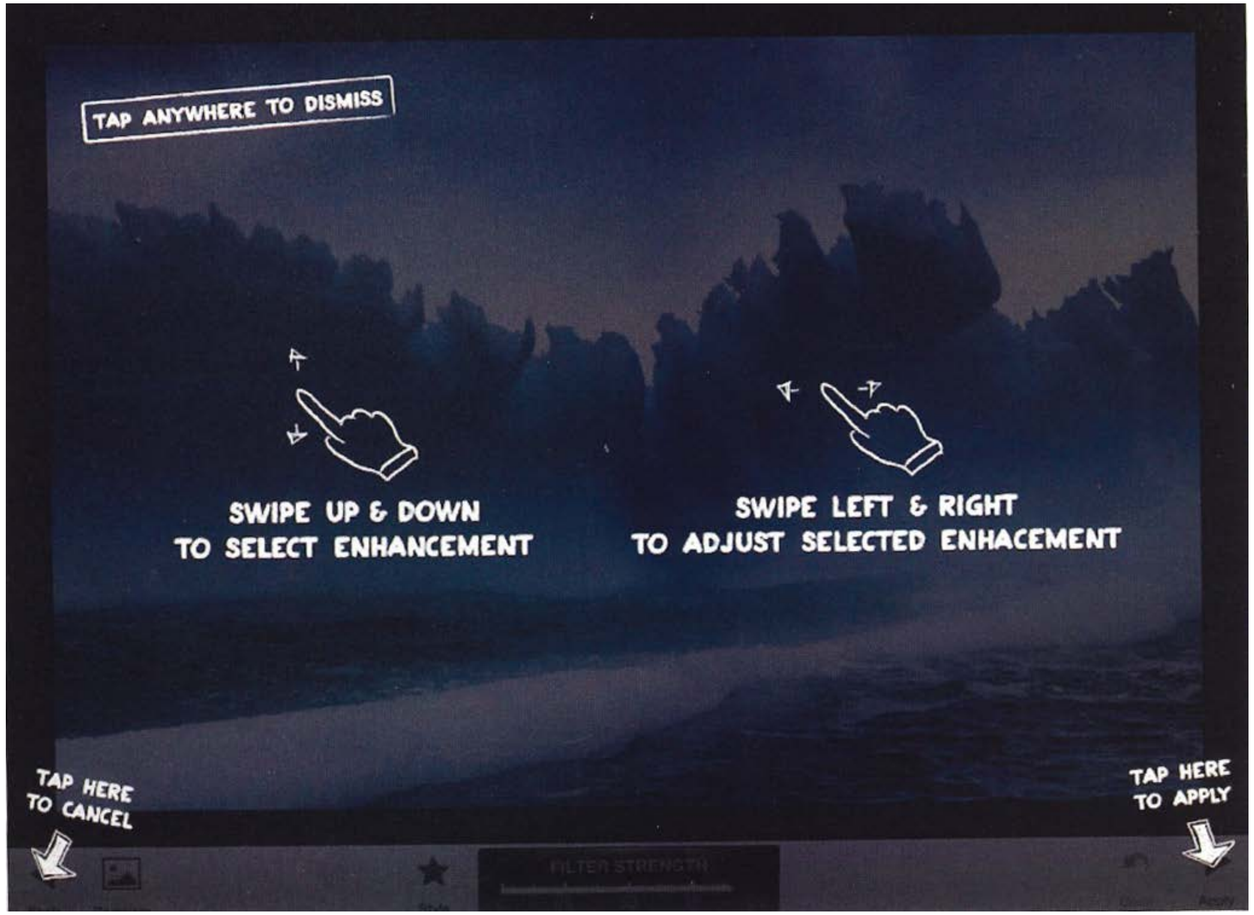


Guided Tour Pattern [Cooper 2014]

Design Patterns

different levels of complexity:

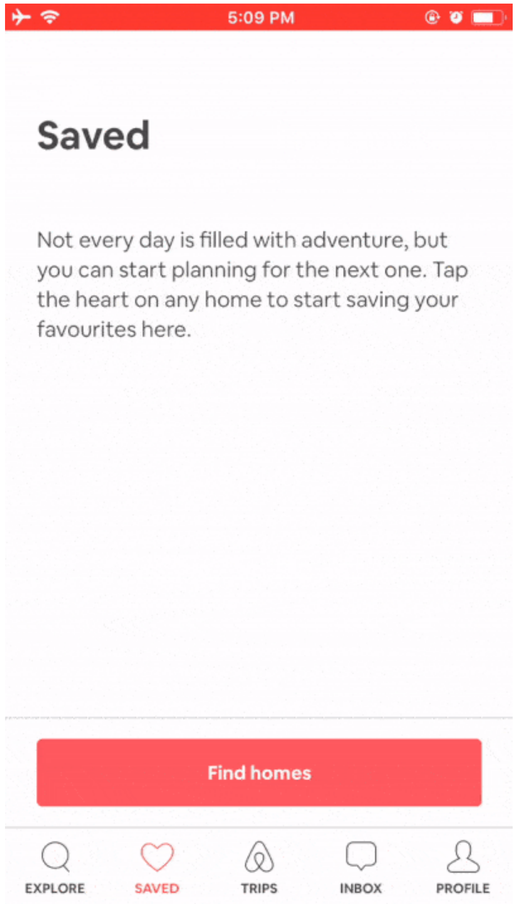
- elementary interaction elements
- complex interactions



Overlay Pattern [Cooper 2014]

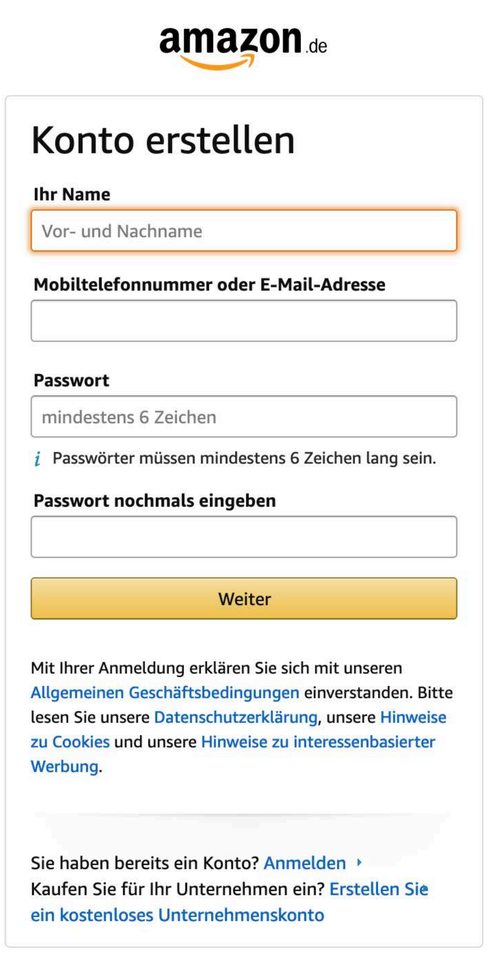
Design Patterns - Navigation



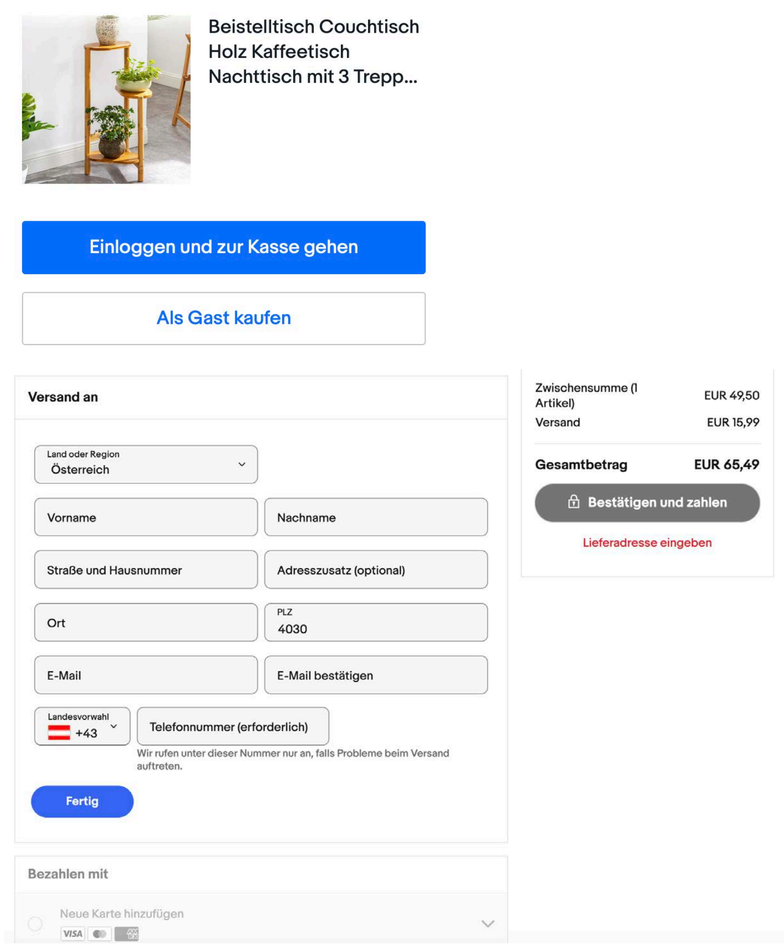


Hamburger Menu vs. Tab Bar

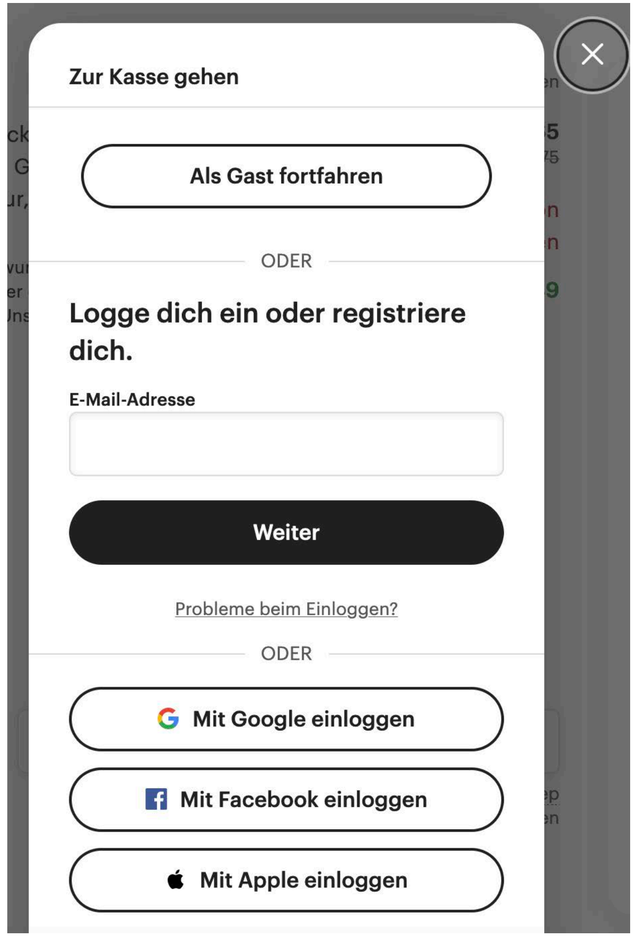
Design Patterns - Registration



Amazon



Ebay

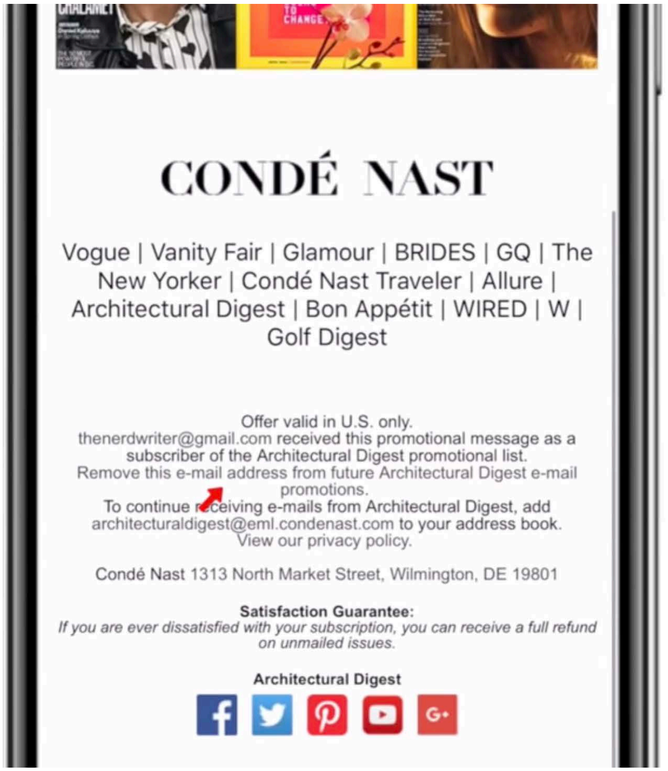


Etsy

Dark Patterns (Deceptive patterns)

tricks used in websites and apps that make you do things that you didn't mean to, e.g.

- buying or signing up for something
- automatically add additional products such as insurance to the shopping cart
- intentionally hide information



https://deceptive.design/



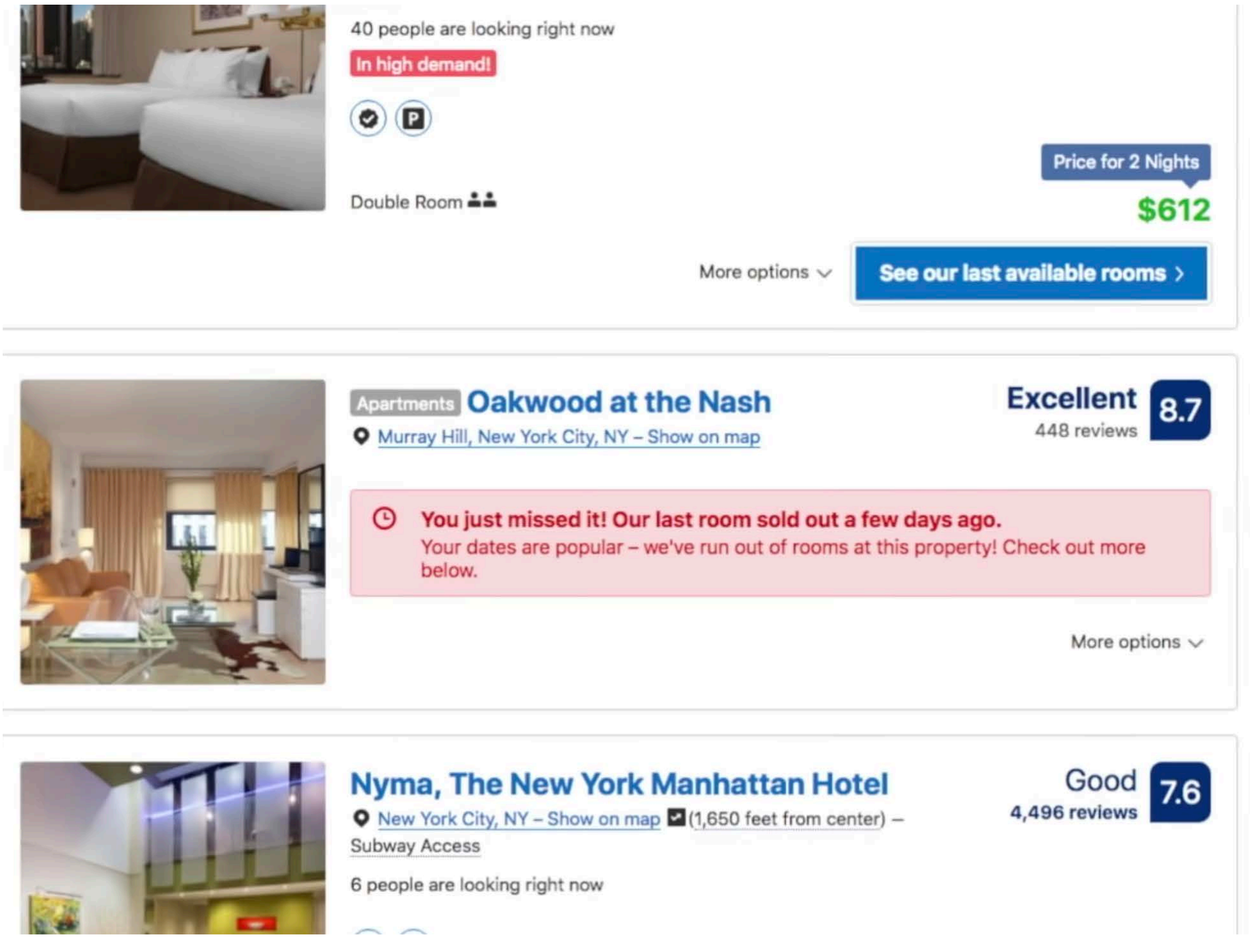
text color

link color

Dark Patterns (Deceptive patterns)

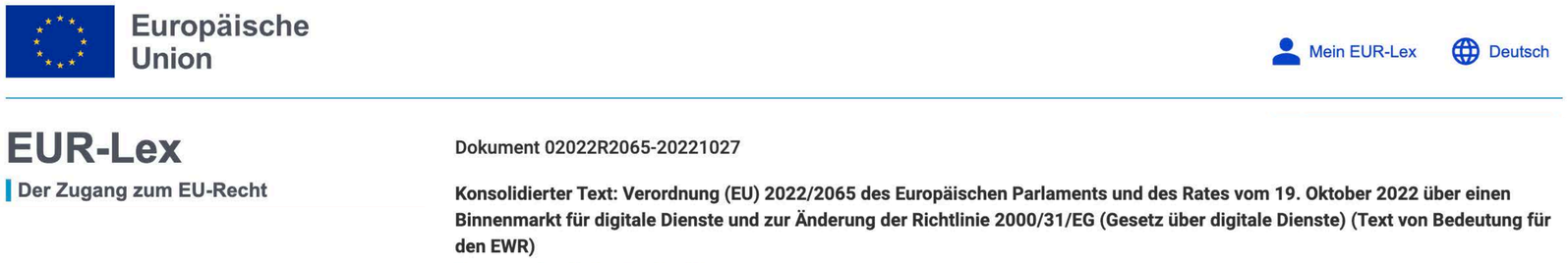
tricks used in websites and apps that make you do things that you didn't mean to, e.g.

- buying or signing up for something
- automatically add additional products such as insurance to the shopping cart
- intentionally hide information
- constant warning that hotel or product is almost sold out



https://deceptive.design/

Digital Services Act



Europäische Union

Mein EUR-Lex Deutsch

EUR-Lex

Der Zugang zum EU-Recht

Dokument 02022R2065-20221027

Konsolidierter Text: Verordnung (EU) 2022/2065 des Europäischen Parlaments und des Rates vom 19. Oktober 2022 über einen Binnenmarkt für digitale Dienste und zur Änderung der Richtlinie 2000/31/EG (Gesetz über digitale Dienste) (Text von Bedeutung für den EWR)

Artikel 25

Gestaltung und Organisation der Online-Schnittstelle

(1) Anbieter von Online-Plattformen dürfen ihre Online-Schnittstellen nicht so konzipieren, organisieren oder betreiben, dass Nutzer getäuscht, manipuliert oder anderweitig in ihrer Fähigkeit, freie und informierte Entscheidungen zu treffen, maßgeblich beeinträchtigt oder behindert werden.

(2) Das Verbot in Absatz 1 gilt nicht für Praktiken, die unter die Richtlinie 2005/29/EG oder die Verordnung (EU) 2016/679 fallen.

(3) Die Kommission kann Leitlinien für die Anwendung von Absatz 1 auf eine bestimmte Praxis herausgeben, insbesondere in Bezug darauf,

a) dass bestimmte Auswahlmöglichkeiten stärker hervorgehoben werden, wenn der Nutzer eine Entscheidung treffen muss,

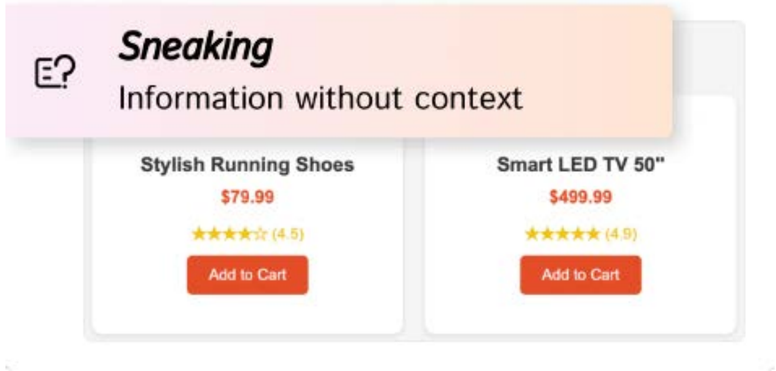
b) dass der Nutzer wiederholt aufgefordert wird, eine Auswahl zu treffen, obwohl eine solche Auswahl bereits getroffen wurde, insbesondere durch die Einblendung eines Fensters, mit der die Nutzererfahrung beeinträchtigt wird,

c) dass das Verfahren zur Beendigung eines Dienstes schwieriger als das Verfahren zur Anmeldung bei diesem Dienst gestaltet wird.

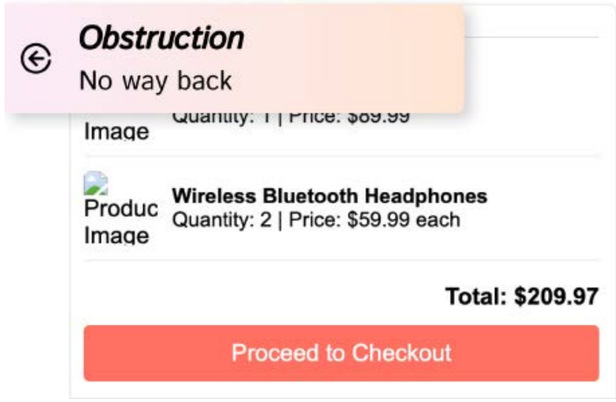
https://eur-lex.europa.eu/legal-content/DE/TXT/?uri=CELEX%3A02022R2065-20221027

Dark Patterns in AI-Generated Interfaces

- current CHI '26 study: four LLMs (Gemini 2.5 Pro, GPT-4.1, Grok 3 Beta, DeepSeek-V3) generate HTML/CSS code for 15 typical e-commerce components (search, shopping cart, membership cancellation, etc.)
- **Result:** 55.8% of the components contained at least 1 dark pattern, and 30.6% contained at least 2 — without being explicitly prompted for them



The product list displays two products with high ratings, but the UI does not show how many reviews those ratings are based on

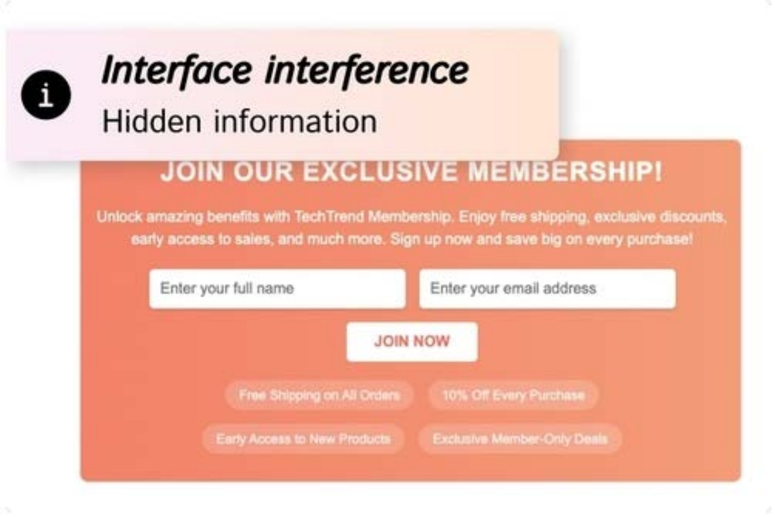


The shopping cart does not provide any functionality for users to reduce the quantities or remove items from the cart

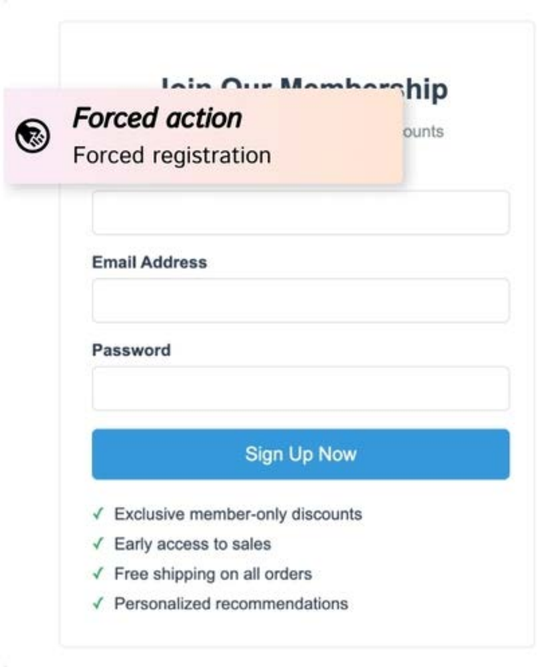
Chen et al., 2026. Deception at Scale: Deceptive Designs in 1K LLM-Generated E-Commerce Components. (CHI '26)

Dark Patterns in AI-Generated Interfaces

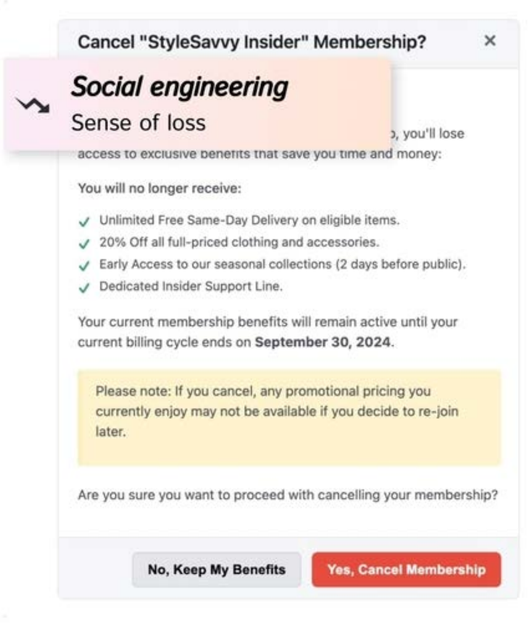
- the study found that prompts emphasizing business interests (e.g., increasing sales) significantly increased deceptive designs



The membership sign-up component collects customer names and email without explaining how it will be used or protected



The UI has no close button after users add items to the cart, which forces users to join the membership.



The membership cancellation component emphasizes the benefit that users would lose, creating hesitation and encouraging retention.

Chen et al., 2026. Deception at Scale: Deceptive Designs in 1K LLM-Generated E-Commerce Components. (CHI '26)

Interface Guidelines

- Companies try to help designers by publishing collections of design recommendations

Apple:

- iOS Human Interface Guidelines
- OS X Human Interface Guidelines
- Apple Watch Human Interface Guidelines

Microsoft:

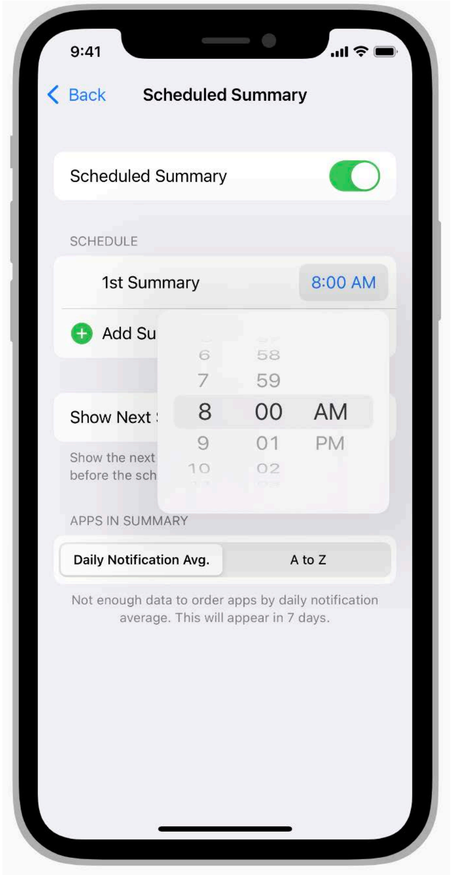
- Windows User Experience Interaction Guidelines
- User Experience Guidelines for Universal Windows Platform
- Microsoft Design Guidance for Mixed Reality

Google:

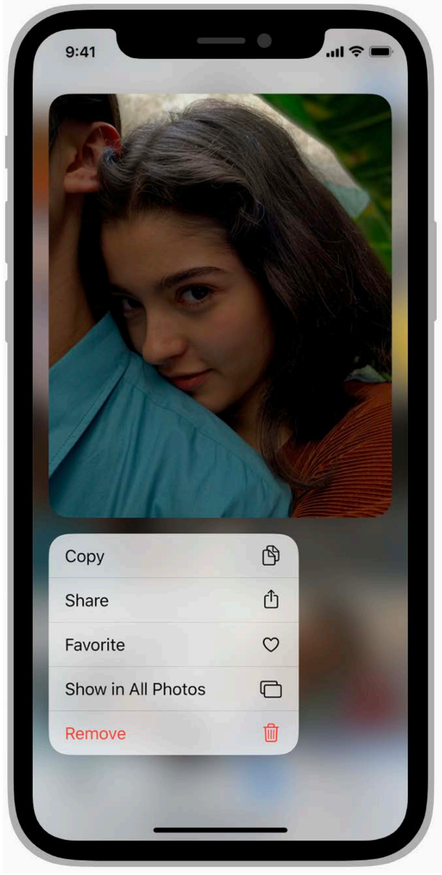
- Android Material Design (https://material.io/design)

iOS Human Interface Guidelines

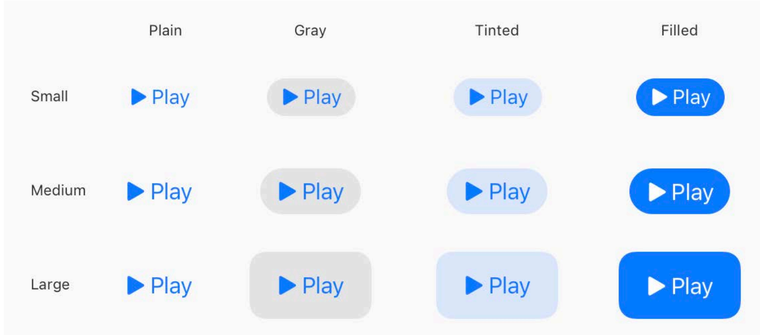
https://developer.apple.com/design/human-interface-guidelines/ios/controls/pull-down-menus/



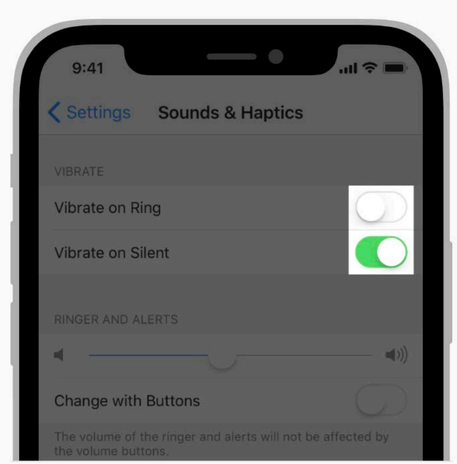
Date Pickers



Context Menus



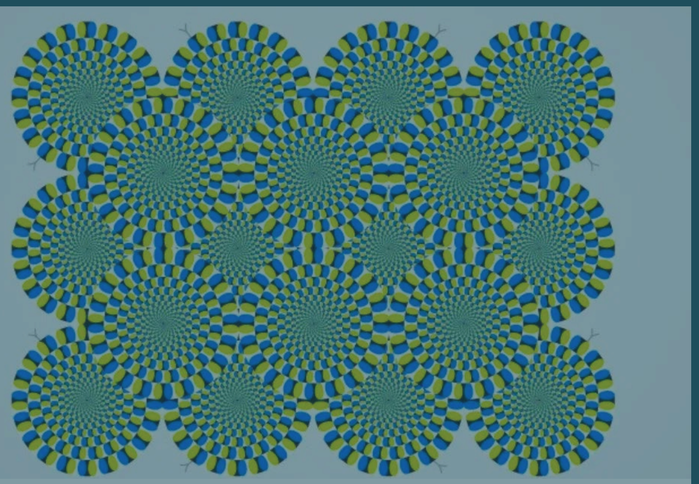
System Buttons



Switches

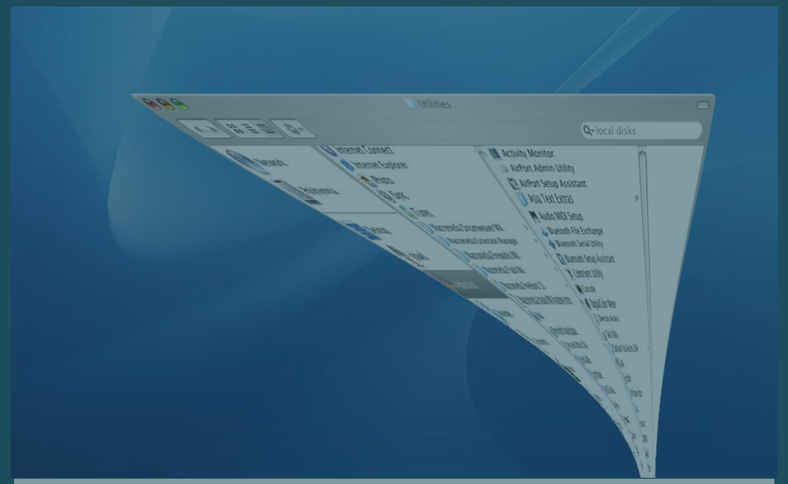
What can help us in creating good user interfaces?

Psychology & cognitive science



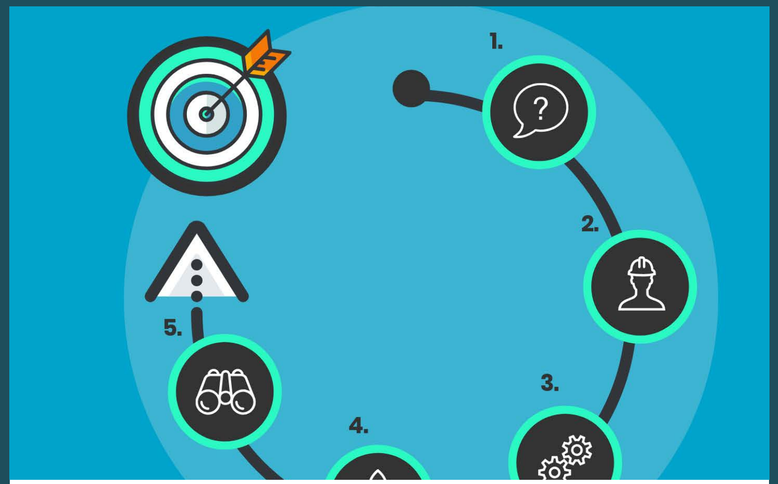
Getting a better understanding of how users perceive & think about UIs

Design principles & patterns



Following fundamental principles & guidelines of good interaction design

Human-centered Design



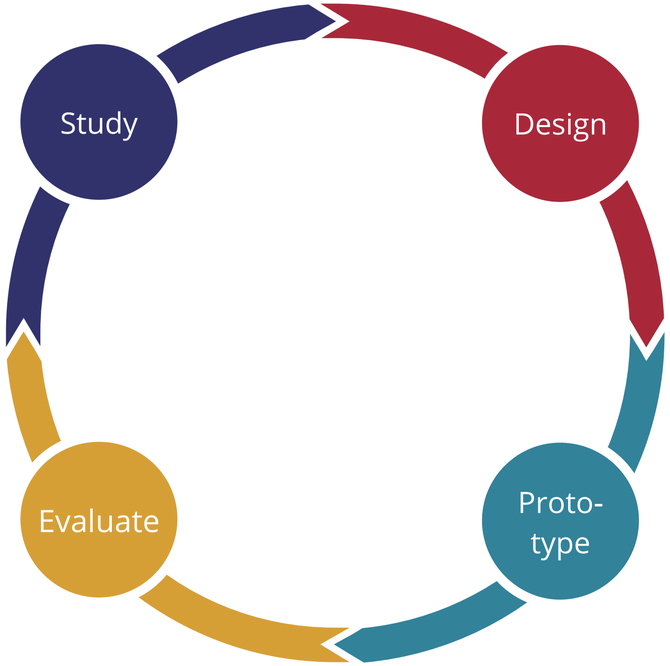
Following a human-centered design process

Simplified Design Process

Sketching

Scribbles, Storyboards, …



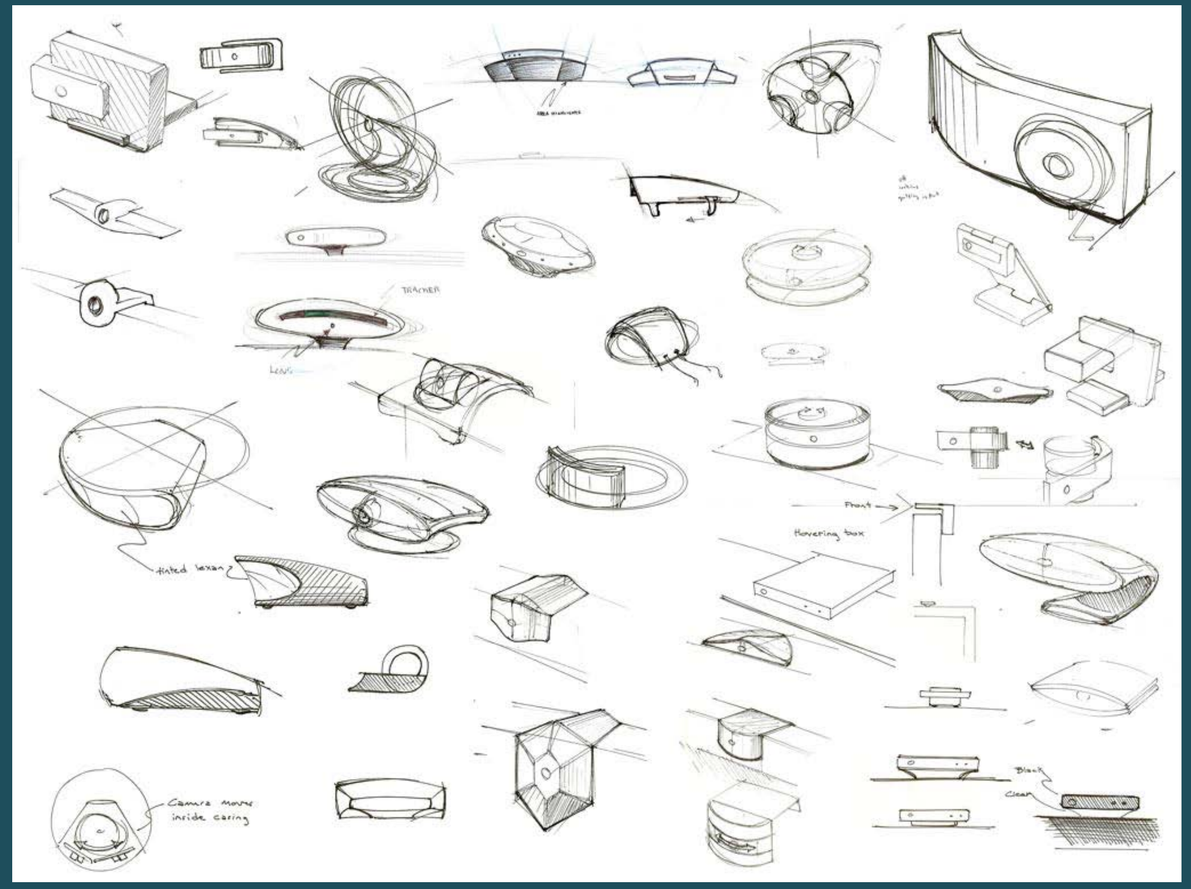


Ideation:

- is collaborative, iterative, and exploratory
- being informed via usage research data & models
- Sketching is an essential embodied partner of ideation [Hartson & Pyla 2019]

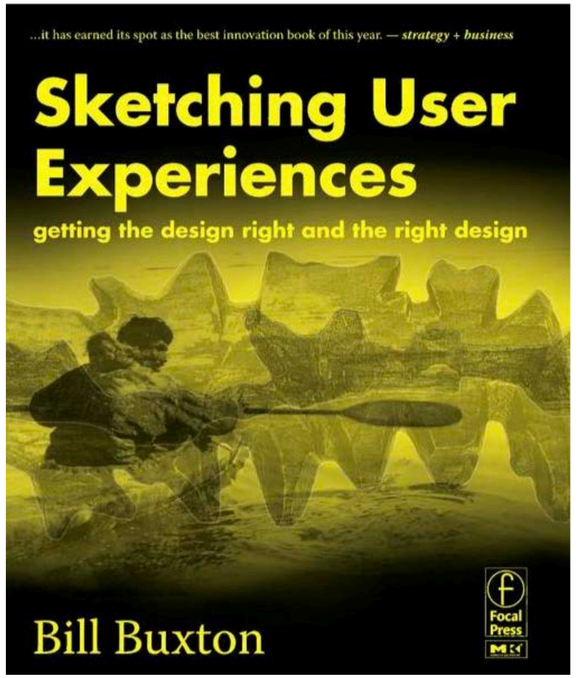
DESIGN EXPLORATION

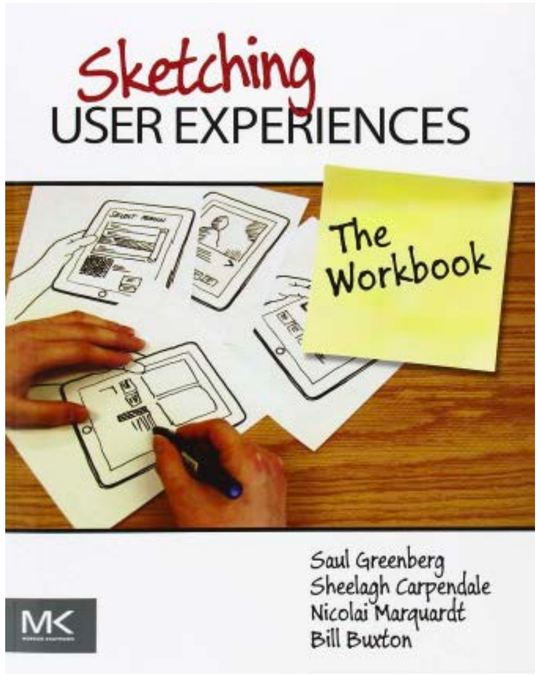
Sketching for Ideation



Sketching User Experiences

- **Sketching User Experiences –the Workbook** by S. Greenberg, S. Carpendale, N. Marquardt and B. Buxton, Morgan Kaufmann, 2011
- **Sketching User Experiences: Getting the Design Right and the Right Design** by Bill Buxton, Morgan Kaufmann, 2007
- **Sketching for UX designers:** https://uxknowledgebase.com/





Ideation

3 types of input to ideation [Hartson & Pyla 2019]:

- **ideation informers:** provide information about usage, requirements, targets and goals
- **ideation catalyst:** a catalyst can trigger a spark or inspiration that can lead to an „eureka moment“

Ideas can occur any time

- conversation sparks a thought
- a sci-fi movie depicts something interesting
- wake up from a dream, ...



source: http://skyvington.blogspot.com/2010/11/eureka.html

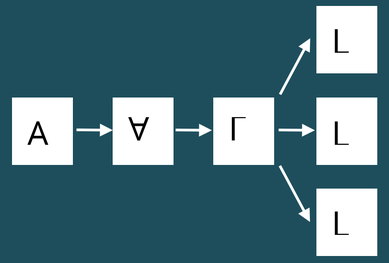
Ideation

3 types of input to ideation [Hartson & Pyla 2019]:

- **ideation informers:** provide information about usage, requirements, targets and goals
- **ideation catalyst:** a catalyst can trigger a spark or inspiration that can lead to an „eureka moment“
- **Ideation techniques:** something the designer can do to foster the spawning or nurturing of design ideas, e.g. sketching, creativity techniques such as brainstorming, mind mapping

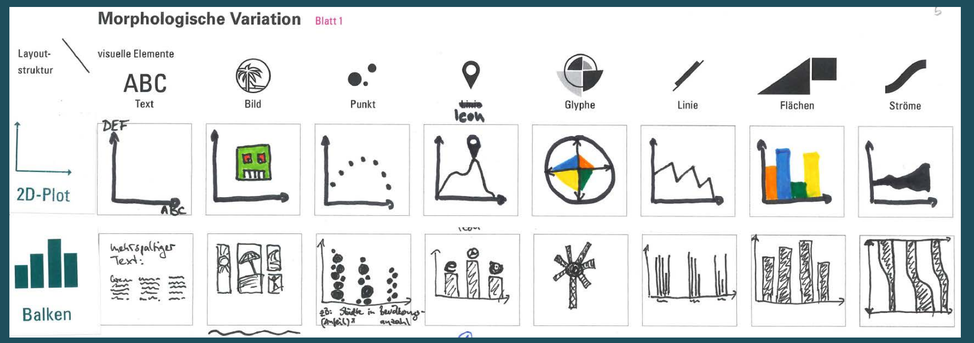
Intuitive Methods [Moser 2012]:

- new associations are formed by specific stimuli
- e.g. (negative) brainstorming, mind mapping



Discursive Methods:

- systematic search for new ideas by breaking down problem into small parts and recombining them
- e.g. morphological analysis [Zwicky 1969]



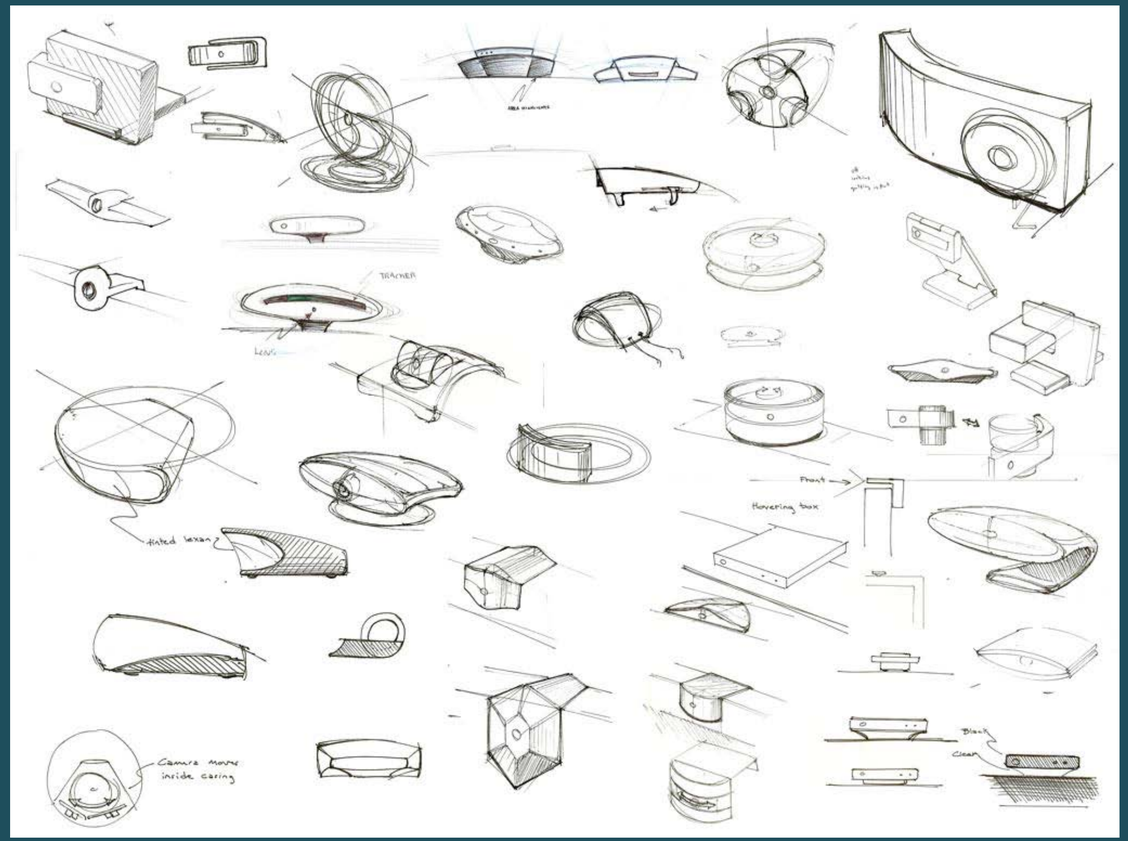
Ideation

Basis rules of ideation:

- make ideas cheap: generate ideas as quickly as possible
- generate both variations of an idea and different ideas
- don’t worry about idea quality or feasibility

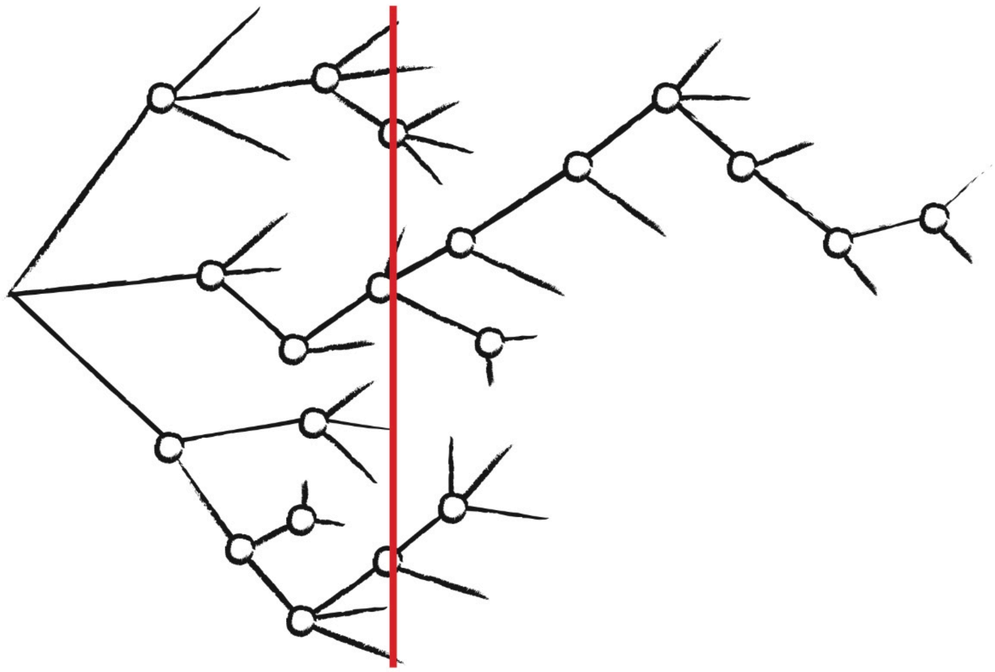
Ideation outcome: dozens (or hundreds) of solution ideas

„The best way to have a good idea is to have lots of ideas“
Linus Pauling



Getting the right design

- design is about exploring and comparing the relative merits of alternatives
- there is not just one path
- at any given time and for any given question, there may be numerous different alternatives being considered, only one of which will eventually find itself in the product [Buxton 2007]



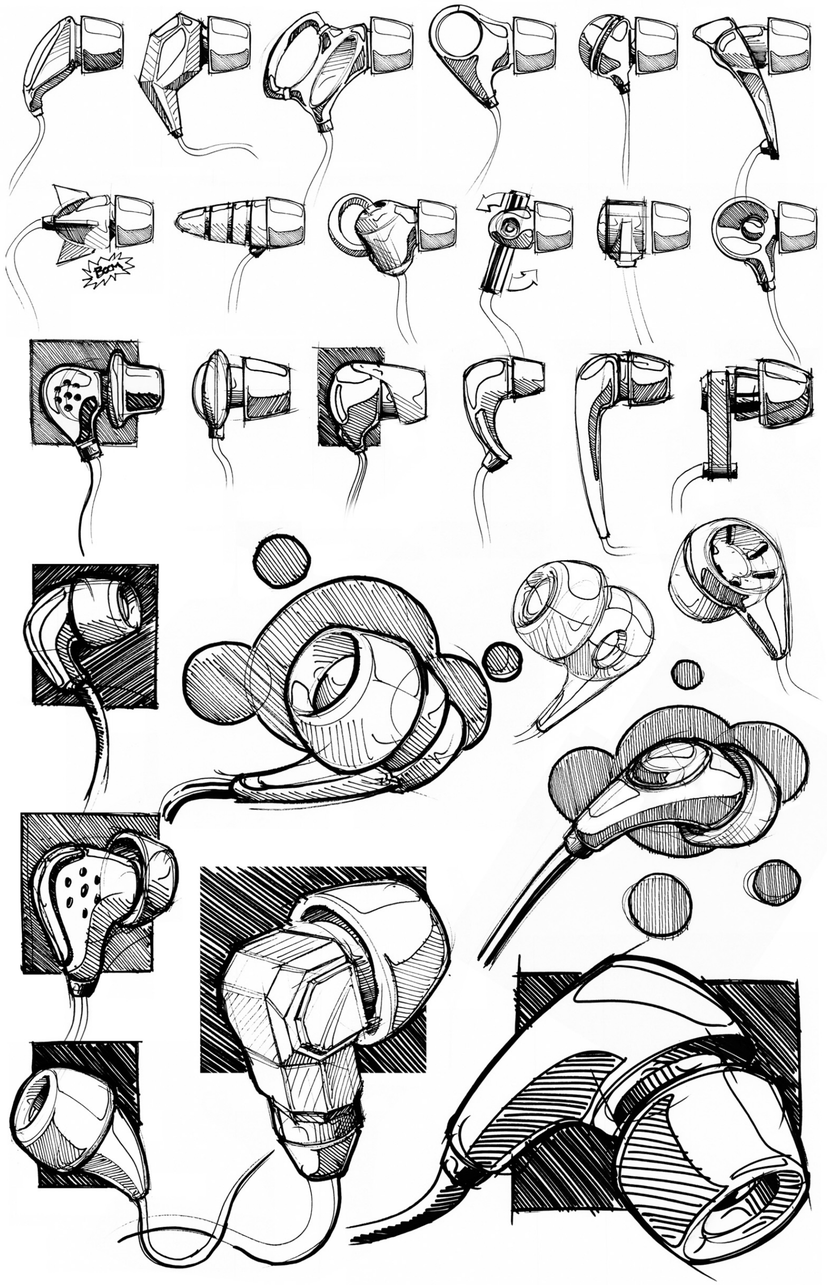
Purpose of a sketch

- design is a process of creation & exploration, and sketching is a visual medium for exploration

„If you’re not sketching, you’re not doing design“ - Bill Buxton

Purpose of a sketch

- design is a process of creation & exploration, and sketching is a visual medium for exploration
- it is about **generating** and **communicating ideas**

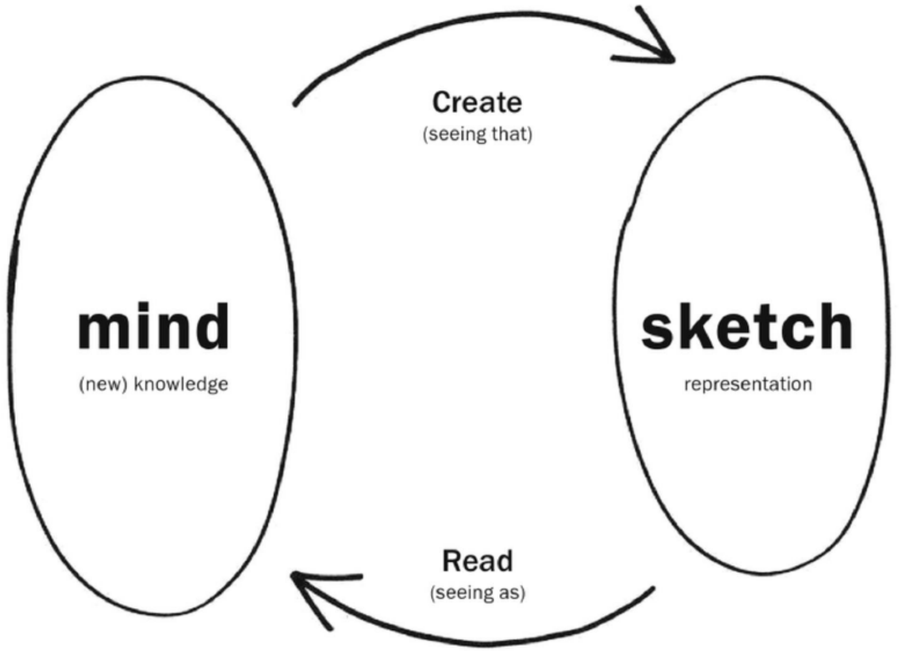


source: https://www.flickr.com/photos/pmurf/4544711379

Purpose of a sketch

- **Communication:** a medium to support a conversation among the design team and to others
- **Evidence:** the sketch provides evidence that the work has been done
- **Record:** sketches provide a record of the process and how the ideas have changed
- **Discovery process:** the sketching process itself supports early ideation and inspires new design ideas
  - sketching is a „conversation“ between mind and sketch: „thinking on paper“ [Buxton 2007]

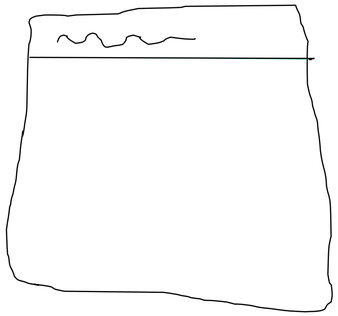
[Roberts et al. 2017, Greenberg et al. 2011]

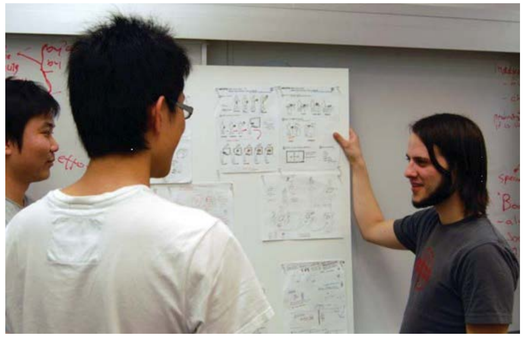


[Buxton 2007]

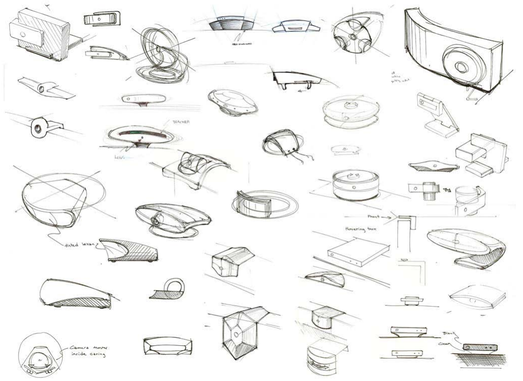
The attributes of sketches

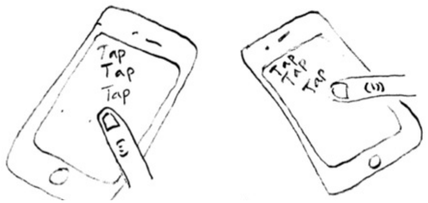
- **Quick** to make and **inexpensive**
- **Timely:** provided when needed
- **Disposable:** if you can't afford to throw it away, it's not a sketch
- **Plentiful:** they make sense in a collection or series of ideas; not as isolated rendering
- **Sense of openness:** they are not tight and precise, e.g. in the sense that a technical drawing would be
- **Minimal Detail:** include only what is required to render the intended purpose or concept [Buxton 2007]







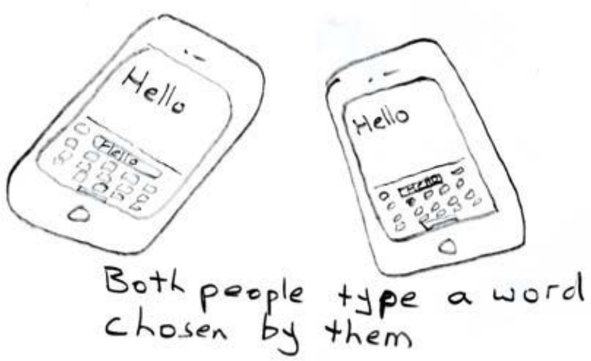




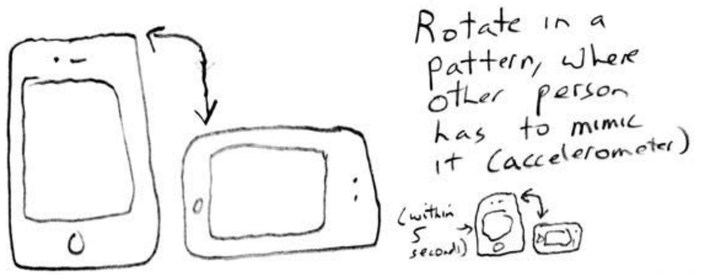


Sketching for Ideation

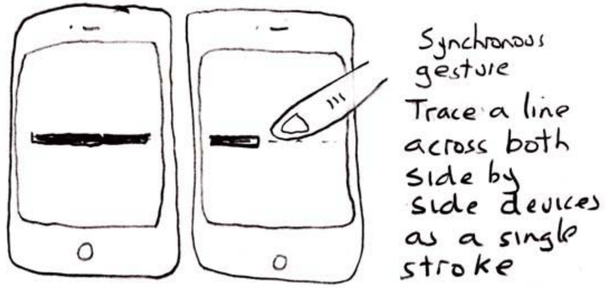
Example: How to pair two mobile phones



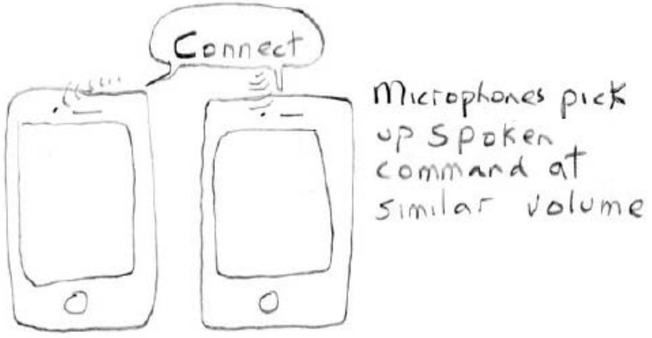
Idea #1: enter an agreed upon keyword



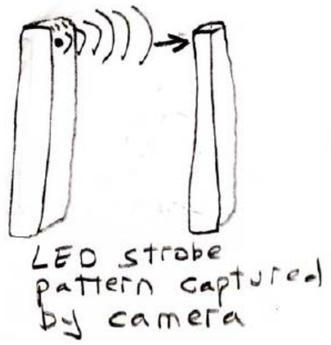
Idea #2: mimic a rotation pattern



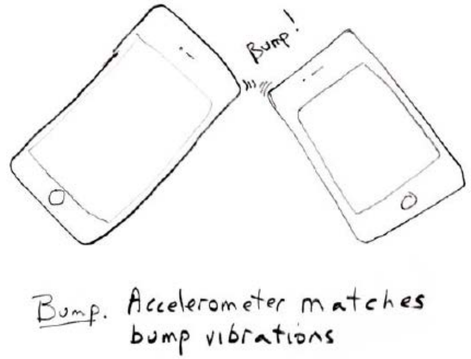
Idea #3: tracing across displays



Idea #4: speak a command



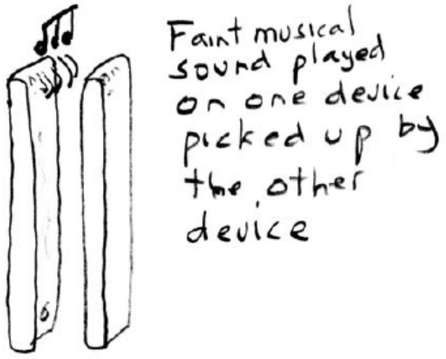
Idea #5: recognize a phone's flash strobe pattern



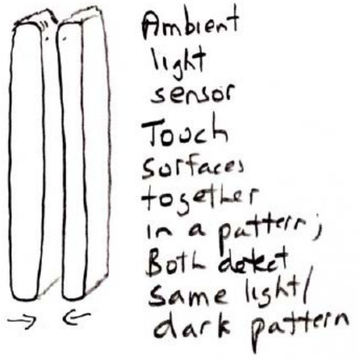
Idea #6: bump two phones together

Sketching for Ideation

Example: How to pair two mobile phones



Idea #7: musical sequence

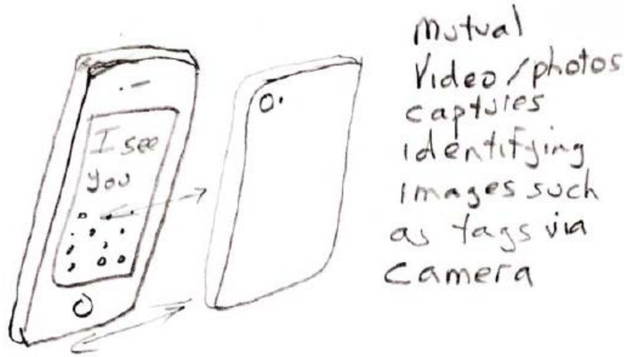


Idea #8: light / dark patterns

[Greenberg et al. 2011]



Idea #9: three simultaneous taps

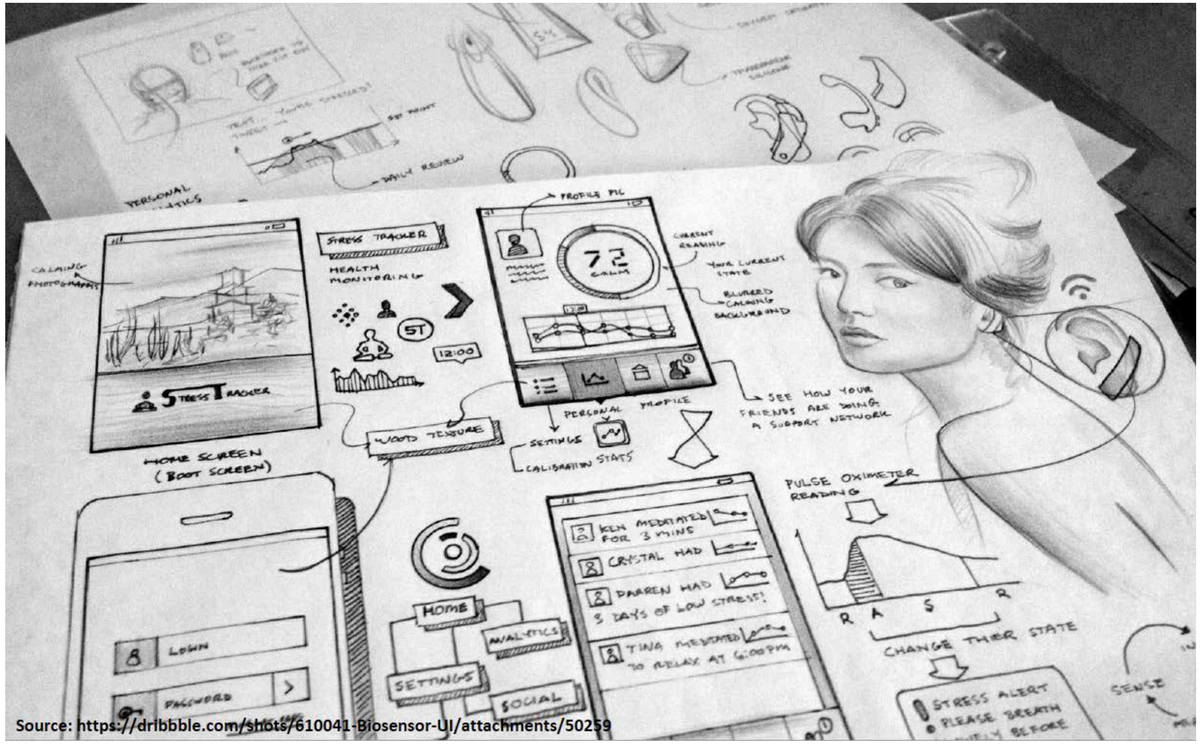


Idea #10: take a picture of an identifying feature on the other person's phone

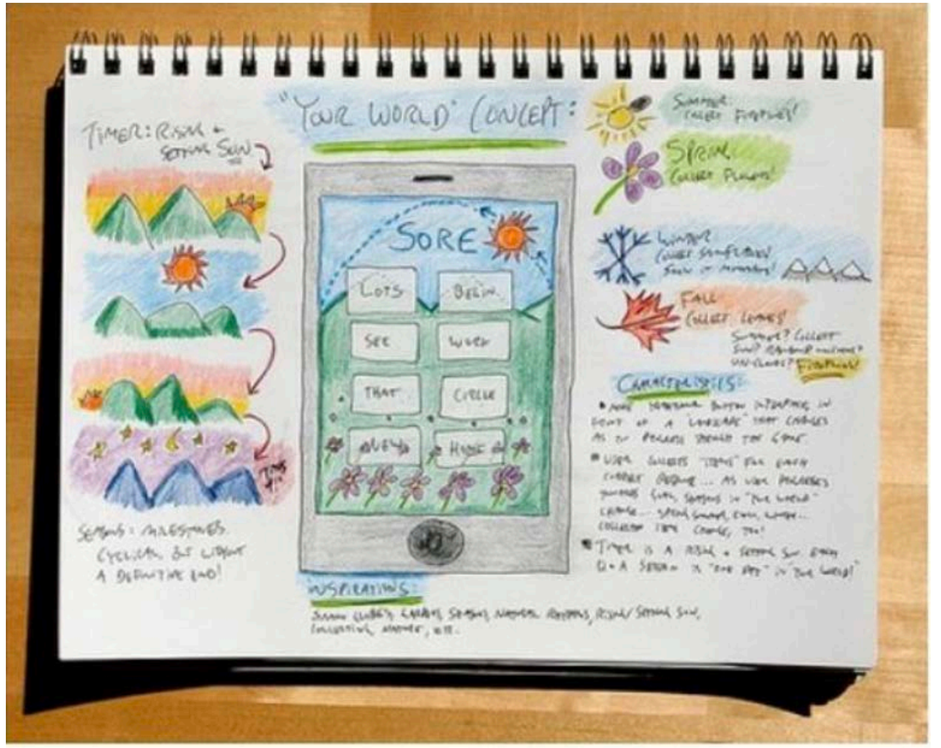
But I cannot draw!

Sketches are not meant to be beautiful drawings!

> *„As soon as you start to get into detail you are drawing (not sketching) and in danger of you, and the people you share your sketches with, focusing on the wrong thing.“*
> Sam Smith (RMA-Consulting.com)



Source: https://dribbble.com/shots/610041-Biosensor-UI/attachments/50259



https://de.slideshare.net/pubsmith/sketching-interfaces-workshop-interactions12-dublin/73

References

[Norman 2014] - Donald Norman: The Design of Everyday Things, 2014

[Hartson & Pyla 2019]- Rex Hartson & Pardha S. Pyla: The UX Book: Agile UX Design for Quality User Experience, Morgan Kaufmann, 2. Edition, January 2019, (ISBN: 978-0128053423)

[Cooper 2014] - Alan Cooper, Robert Reimann, and Dave Cronin: About Face 3: The Essentials of Interaction Design, John Wiley & Sons; 4. Auflage, 2014

[Lakoff & Johnson 2003] - George Lakoff, Mark Johnson: Metaphors We Live By. Chicago: University Of Chicago Press, 2003

[Moser 2012] - Christiane Moser: User Experience Design: Mit erlebniszentrierter Softwareentwicklung zu Produkten, die begeistern, SpringeR Verlag, 2012

[Roberts et al. 2017] - Roberts, Jonathan C, Headleand, Christopher J., Ritsos, Panagiotis D.: Five Design-Sheets: Creative Design and Sketching for Computing and Visualisation, Springer, 2017

[Greenberg et al. 2011] - S. Greenberg, S. Carpendale, N. Marquardt and B. Buxton: Sketching User Experiences –the Workbook, Morgan Kaufmann, 2011

[Buxton 2007] - Bill Buxton: Sketching User Experiences: Getting the Design Right and the Right Design, Morgan Kaufmann, 2007